

>> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<

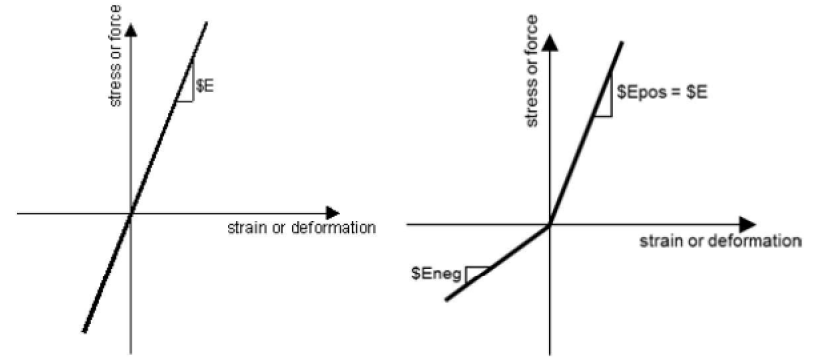
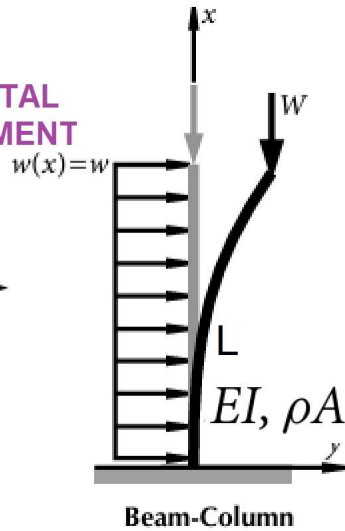
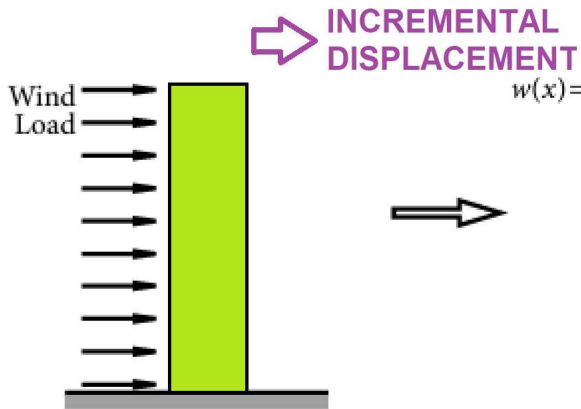
**COMPREHENSIVE NONLINEAR
SEISMIC ASSESSMENT OF A STEEL
COLUMN: AN OPENSEES
FRAMEWORK FOR STATIC
PUSHOVER, CYCLIC DEGRADATION,
STATIC TIME-HISTORY AND
DYNAMIC TIME-HISTORY ANALYSIS**

THIS PROGRAM WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)

20
21
19
m
7

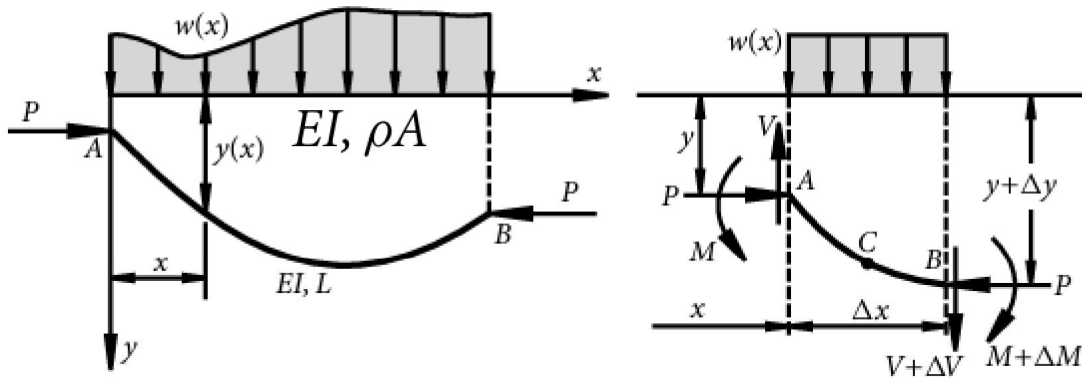
2
3
1
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1

$$W(x, t) = P(t) = W_0 e^{-0.05 \bar{\omega} t} \sin(\bar{\omega} t)$$

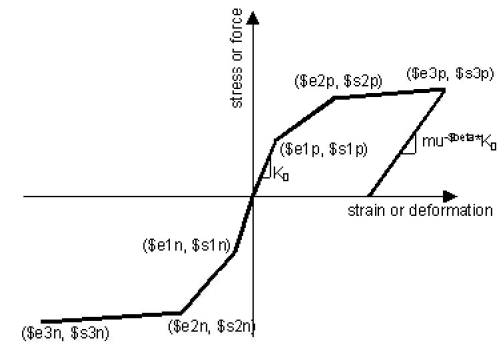


ELEMENT ELASTIC MATERIAL

$$W(x, t) = P(t) = W_0 e^{-0.05 \bar{\omega} t} \sin(\bar{\omega} t)$$



Beam-column.



ELEMENT INELASTIC MATERIAL

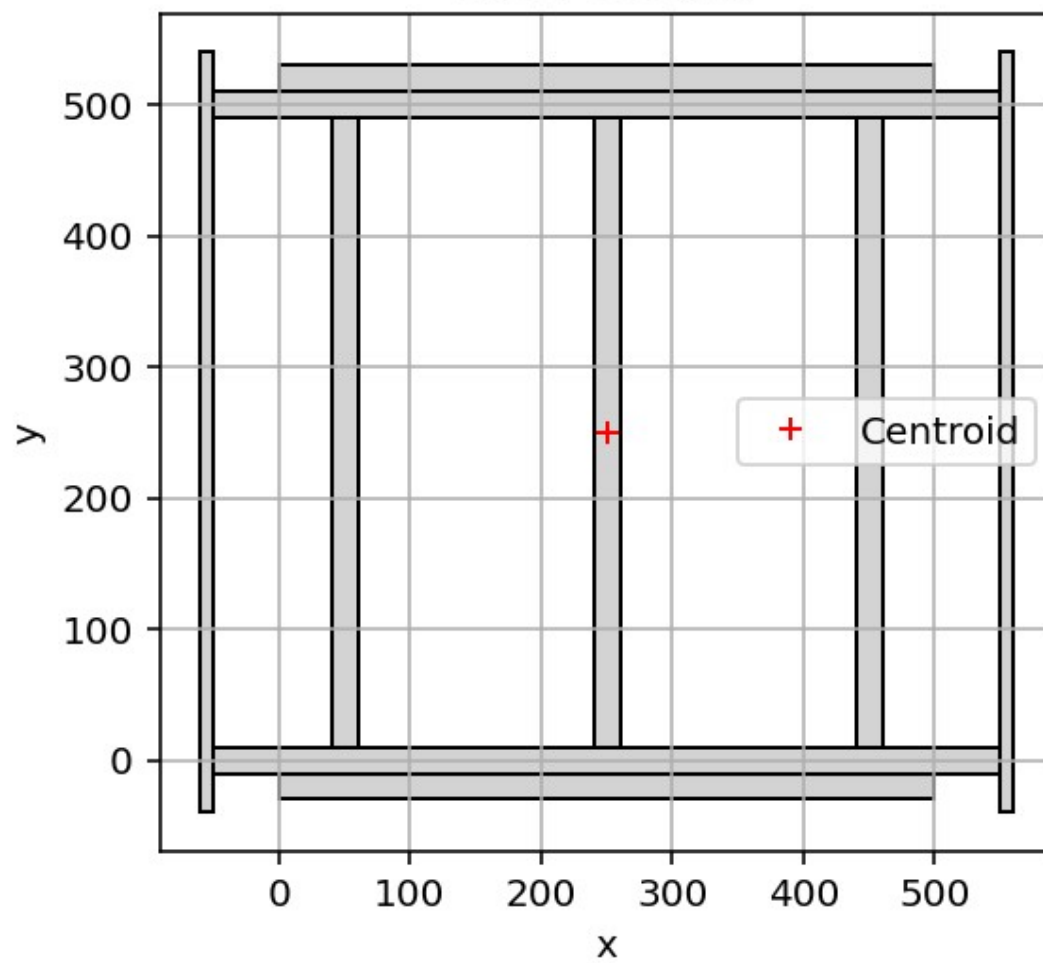
$$W(x,t)=P(t)=W_0e^{-0.05\bar{\omega}t}sin(\bar{\omega}t)$$

$$m\ddot{u}+c\dot{u}+ku=P(t)$$

$$Structure\;Ductility\;Damage\;Index=\frac{\Delta_d-\Delta_y}{\Delta_u-\Delta_y}$$

$$\rho A \frac{\partial^2 v}{\partial t^2} + EI \frac{\partial^4 v}{\partial x^4} = w(x,t).$$

Steel Section



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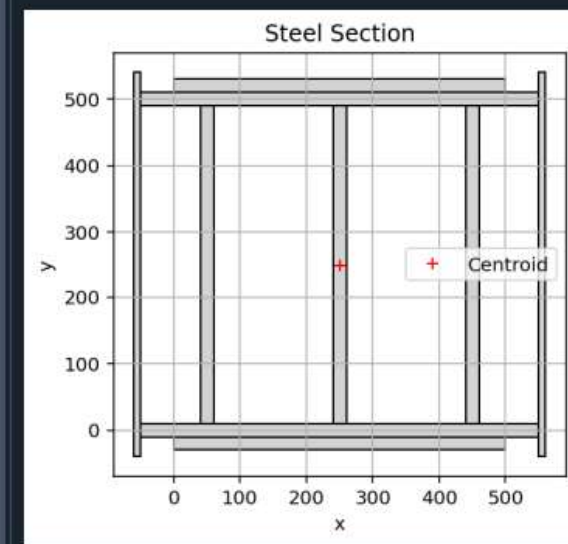
W(x,t)_COLUMN_AGGR...N_ELEMENT_8_ANA.py x
 AGGREGATED_AXIAL_F...ION_ELEMENT_FUN.py x

```

1 #####
2 #
3 # >> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<
4 # COMPREHENSIVE NONLINEAR SEISMIC ASSESSMENT OF A STEEL COLUMN : AN OPENSEES
5 # FRAMEWORK FOR STATIC PUSHOVER, CYCLIC DEGRADATION, STATIC TIME-HISTORY AND DYNAMIC TIME-HISTORY ANALYSIS
6 #
7 #
8 # W(x,t) = W0 sin(wt)
9 # W(x,t) = W0 exp(-0.05wt) sin(wt)
10 #
11 # -----
12 # ASSESSMENT OF DUCTILITY DAMAGE INDICES FOR STRUCTURAL ELEMENTS AND SYSTEMS AND EVALUATION
13 # OF ENERGY DISSIPATION CAPACITY INDEX
14 # -----
15 # THIS PYTHON SCRIPT IS WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)
16 # EMAIL: salar.d.ghashghaei@gmail.com
17 #####
18 """
19 Nonlinear Seismic Performance Assessment of a steel column:
20 An OpenSeesPy Framework for Material and Geometric Nonlinearity Under Static, Cyclic, and Earthquake Loading
21
22 This OpenSeesPy script performs rigorous nonlinear static and dynamic analysis of a steel column
23 for performance-based earthquake engineering. The 2D model incorporates both material nonlinearity
24 (elastic-perfectly plastic or hysteretic steel with strain hardening)
25 and geometric nonlinearity to capture P-delta effects
26 and large displacements-essential for collapse assessment.
27
28 Eight analysis protocols are implemented:
29 (1) [PERIOD] : Structural Period
30 (2) [STATIC] : Gravity load analysis establishing dead load state
31 (3) [PUSHOVER] : Displacement-controlled pushover generating full capacity curves
32 and plastic mechanism identification
33 (4) [CYCLIC_DISPLACEMENT] : Symmetric cyclic displacement protocol capturing hysteresis,
34 pinching behavior, and energy dissipation degradation
35 (5) [STATIC_EXTERNAL_TIME-DEPENDENT_LOADING] : Static Analysis of External time-dependent Loading P(t)
36 (6) [DYNAMIC_EXTERNAL_TIME-DEPENDENT_LOADING] : Dynamic Analysis of External time-dependent Loading P(t)
37 (7) [FREE_VIBRATION] : Free-vibration with initial conditions extracting damping ratios
    
```

..._INDEX\37_COLUMN_AGGREGATED_AXIAL_FLEXURAL_SECTION_8_ANA+

Steel Section
 62 %



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W(x,t)_COLUMN_AGGREGATED_AXIAL_FLEXURAL_SECTION_ELEMENT_FUN.py

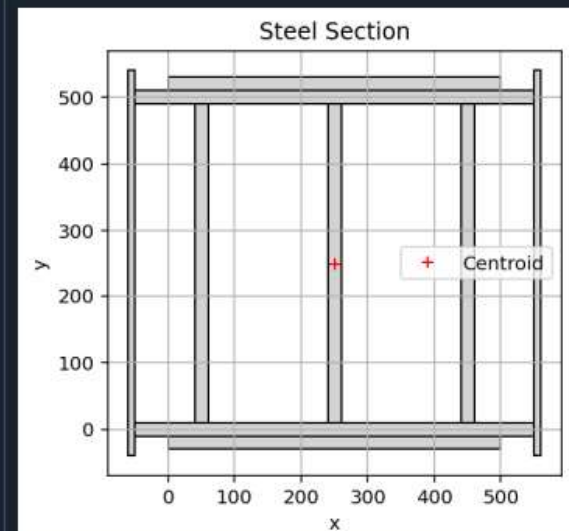
```

1 def AGGREGATED_AXIAL_FLEXURAL_SECTION_ELEMENT_FUN(ELE_TAG,
2           NODE_I, NODE_J,
3           SECTION_Es, SECTION_Iz, SECTION_Area,
4           DENSITY_STEEL, STEEL_TYPE):
5     # THIS PYTHON SCRIPT WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)
6     import openseespy.opensees as ops
7
8     ColMatTagFlex = ELE_TAG + 1000
9     ColMatTagAxial = ELE_TAG + 2000
10    ColMatTagShear = ELE_TAG + 3000
11    ColSecTag = ELE_TAG + 4000
12
13    EI = SECTION_Es * SECTION_Iz
14    EA = SECTION_Es * SECTION_Area
15
16    if STEEL_TYPE == 'ELASTIC':
17        # Define Nonlinear Sectional Moment-Curvature
18        FiY = 0.000001474 # [1/mm] Yield Curvature of element
19        MY = EI * FiY # [N.mm] Yield Moment of element
20        MU = 1.36 * MY # [N.mm] Ultimate Moment of element
21        FiSU = 0.000541 # [1/mm] Ultimate Curvature of element
22        EIColCrack = MY / FiY # [N.mm^2] Elastic Flextural Rigidity
23        # MOMENT-CURVATURE RELATION FOR SECTION
24        #ops.uniaxialMaterial('Elastic', ColMatTagFlex, EIColCrack) # TESNSION AND COMPRESS
25        ops.uniaxialMaterial('Elastic', ColMatTagFlex, EIColCrack, 0.0, 0.5*EIColCrack) # TESNSION AND COMPRESS
26        # SHEAR-SHEAR STRAIN RELATION FOR SECTION
27        # Define Nonlinear Sectional Shear-shear Strain
28        ViY = 0.0009474 # [mm/mm] Yield Shear-strain of element
29        VY = EA * ViY # [N] Yield Shear of element
30        VU = 1.1818 * VY # [N] Ultimate Shear of element
31        ViSU = 0.0541 # [mm/mm] Ultimate Shear-strain of element
32        EIColShear = VY / ViY # [N.mm^2] Elastic Shear Rigidity
33        #ops.uniaxialMaterial('Elastic', ColMatTagShear, EIColShear) # TESNSION AND COMPRESS
34        ops.uniaxialMaterial('Elastic', ColMatTagShear, EIColShear, 0.0, 0.5*EIColShear) # TESNSION AND COMPRESS

```

..._INDEX\37_COLUMN_AGGREGATED_AXIAL_FLEXURAL_SECTION_8_ANA+

62 %

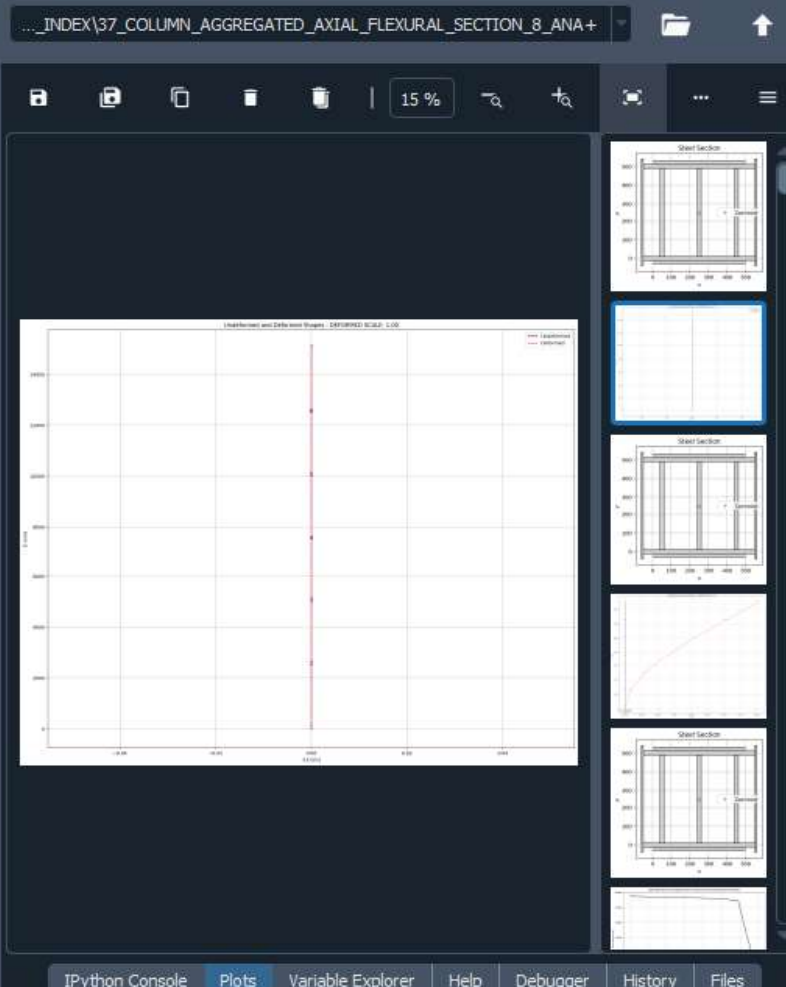


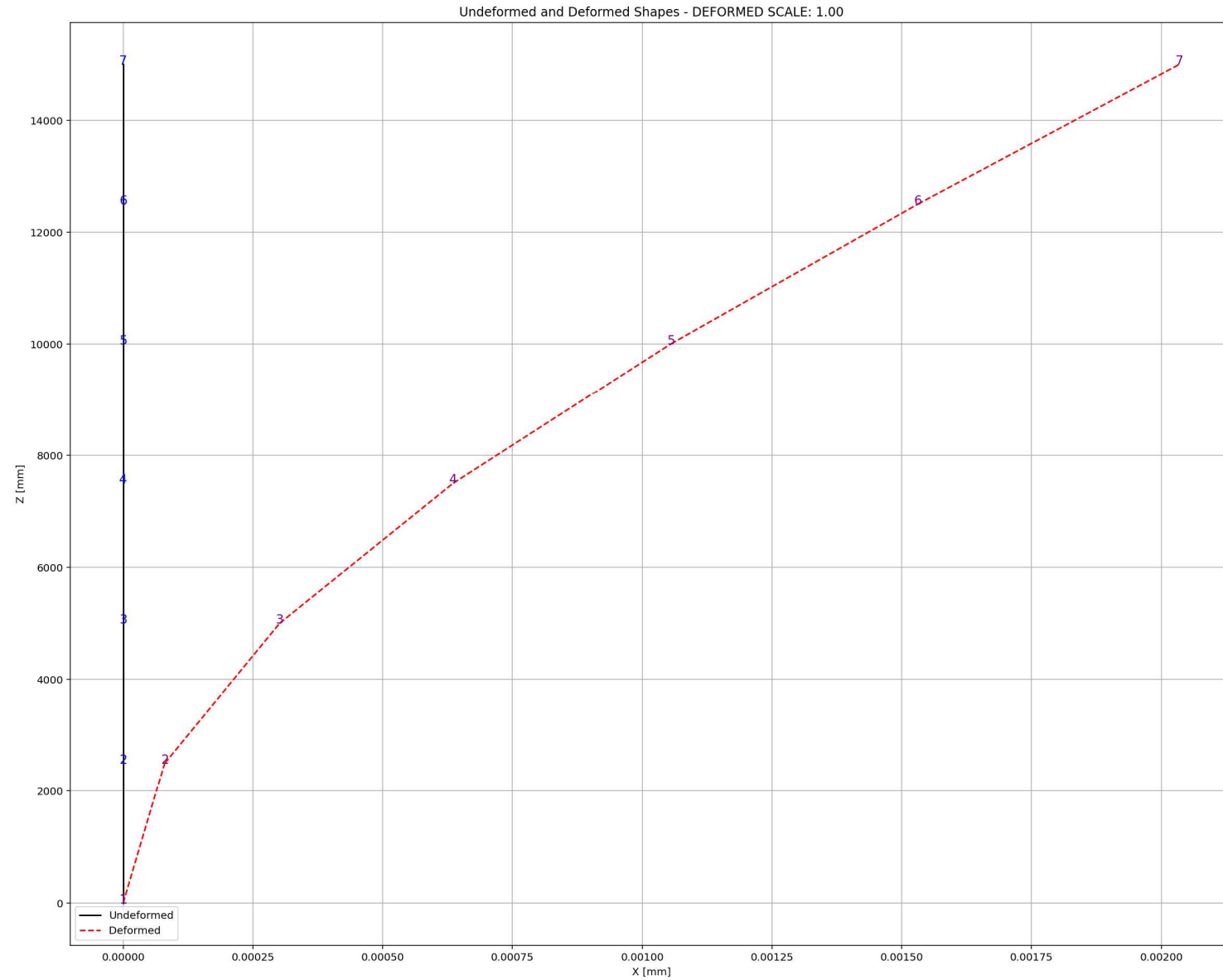
STATIC ANALYSIS

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```

W(x,t)_COLUMN_AGGR...N_ELEMENT_8_ANA.py X AGGREGATED_AXIAL_F...ION_ELEMENT_FUN.py X
950 #####
951 # PERIOD ANALYSIS
952 #ELE_TYPE = 'elasticBeamColumn' # MATERIAL LINEARITY
953 ELE_TYPE = 'nonlinearBeamColumn' # MATERIAL AND GEOMETRIC NONLINEARITIES
954 MAT_TYPE = 'INELASTIC' # 'ELASTIC' OR 'INELASTIC'
955 ANAL_TYPE = 'PERIOD'
956
957 DATA = AGGREGATED_AXIAL_FLEXURAL_COLUMN(LENGTH, MAT_TYPE, ANAL_TYPE, TOTAL_MASS)
958 (PERIOD_MIN_X, PERIOD_MAX_X) = DATA
959 print('Structure First Period: ', PERIOD_MIN_X)
960 print('Structure Second Period: ', PERIOD_MAX_X)
961
962 S01.PLOT_2D FRAME(deformed_scale=1.0)
963 #####
964 # STATIC ANALYSIS
965 MAT_TYPE = 'INELASTIC' # 'ELASTIC' OR 'INELASTIC'
966 ANAL_TYPE = 'STATIC'
967
968 DATA = AGGREGATED_AXIAL_FLEXURAL_COLUMN(LENGTH, MAT_TYPE, ANAL_TYPE, TOTAL_MASS)
969 (reactionX, reactionY, reactionZ, disp_mid,
970 ele_axialforce, ele_shearforce, ele_momentforce,
971 node_displacementsX, node_displacementsY,
972 dispX, dispY) = DATA
973
974 S01.PLOT_2D FRAME(deformed_scale=1.0)
975 #####
976 # PUSHOVER ANALYSIS (STATIC TIME-HISTORY ANALYSIS)
977 MAT_TYPE = 'INELASTIC' # 'ELASTIC' OR 'INELASTIC'
978 ANAL_TYPE = 'PUSHOVER'
979
980 DATA = AGGREGATED_AXIAL_FLEXURAL_COLUMN(LENGTH, MAT_TYPE, ANAL_TYPE, TOTAL_MASS)
981 (reactionX_PUSH, reactionY_PUSH, reactionZ_PUSH, disp_mid_PUSH,
982 ele_axialforce_PUSH, ele_shearforce_PUSH, ele_momentforce_PUSH,
983 node_displacementsX_PUSH, node_displacementsY_PUSH,
  
```





PUSHOVER ANALYSIS

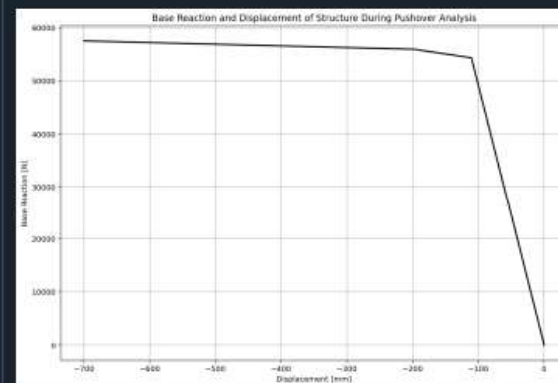
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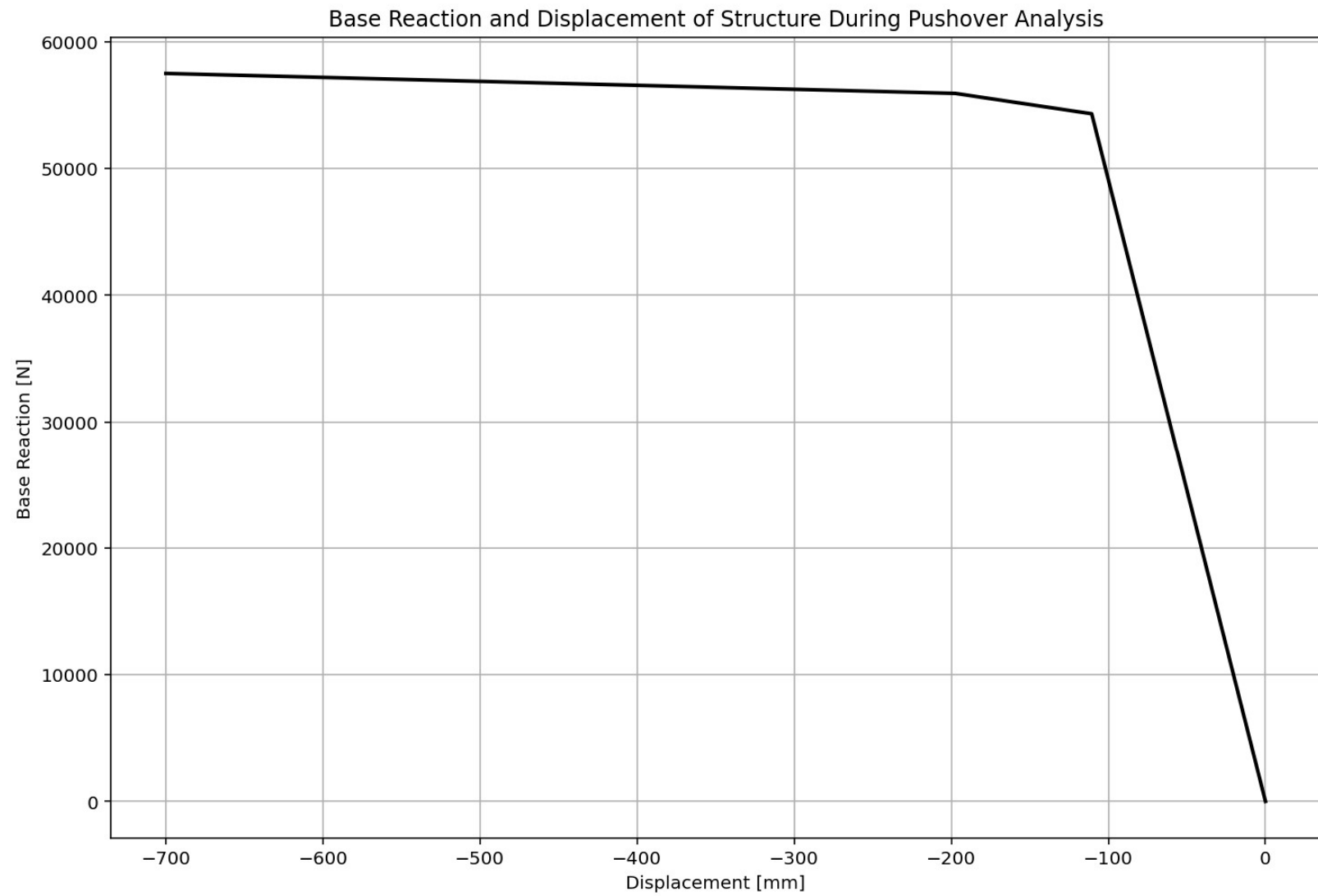
W(x,t)_COLUMN_AGGR...N_ELEMENT_8_ANA.py X AGGREGATED_AXIAL_F...ION_ELEMENT_FUN.py X

```
976 # PUSHOVER ANALYSIS (STATIC TIME-HISTORY ANALYSIS)
977 MAT_TYPE = 'INELASTIC' # 'ELASTIC' OR 'INELASTIC'
978 ANAL_TYPE = 'PUSHOVER'
979
980 DATA = AGGREGATED_AXIAL_FLEXURAL_COLUMN(LENGTH, MAT_TYPE, ANAL_TYPE, TOTAL_MASS)
981 (reactionX_PUSH, reactionY_PUSH, reactionZ_PUSH, disp_mid_PUSH,
982  ele_axialforce_PUSH, ele_shearforce_PUSH, ele_momentforce_PUSH,
983  node_displacementsX_PUSH, node_displacementsY_PUSH,
984  dispX_PUSH, dispY_PUSH,
985  PERIOD_MIN_PUSH, PERIOD_MAX_PUSH) = DATA
986
987
988 XDATA = disp_mid_PUSH
989 YDATA = reactionX_PUSH
990 XLABEL = 'Displacement [mm]'
991 YLABEL = 'Base Reaction [N]'
992 TITLE = 'Base Reaction and Displacement of Structure During Pushover Analysis'
993 COLOR = 'black'
994 SEMIOLOGY = False
995 PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMIOLOGY)
996
997 DATA = S07.BILINEAR_CURVE(np.abs(disp_mid_PUSH), np.abs(reactionX_PUSH), SLOPE_NODE=10)
998 (X_PUSH, Y_PUSH, Elastic_ST, Plastic_ST, Tangent_ST, Ductility_Rito, Over_Strength_Factor) = DATA
999
1000 # PLOT STRUCTURAL PERIOD DURING THE ANALYSIS
1001 plt.figure(0, figsize=(12, 8))
1002 plt.plot(disp_mid_PUSH, PERIOD_MIN_PUSH, linewidth=3)
1003 plt.plot(disp_mid_PUSH, PERIOD_MAX_PUSH, linewidth=3)
1004 plt.title('Period of Structure During Pushover Analysis')
1005 plt.ylabel('Structural Period [s]')
1006 plt.xlabel('Displacement [mm]')
1007 #plt.semilogy()
1008 plt.grid()
1009 plt.legend(['f'PERIOD - MIN VALUES: Min: {np.min(PERIOD MIN PUSH):.3f} (s) - Mean: {np.mean(PERIOD MIN I
```

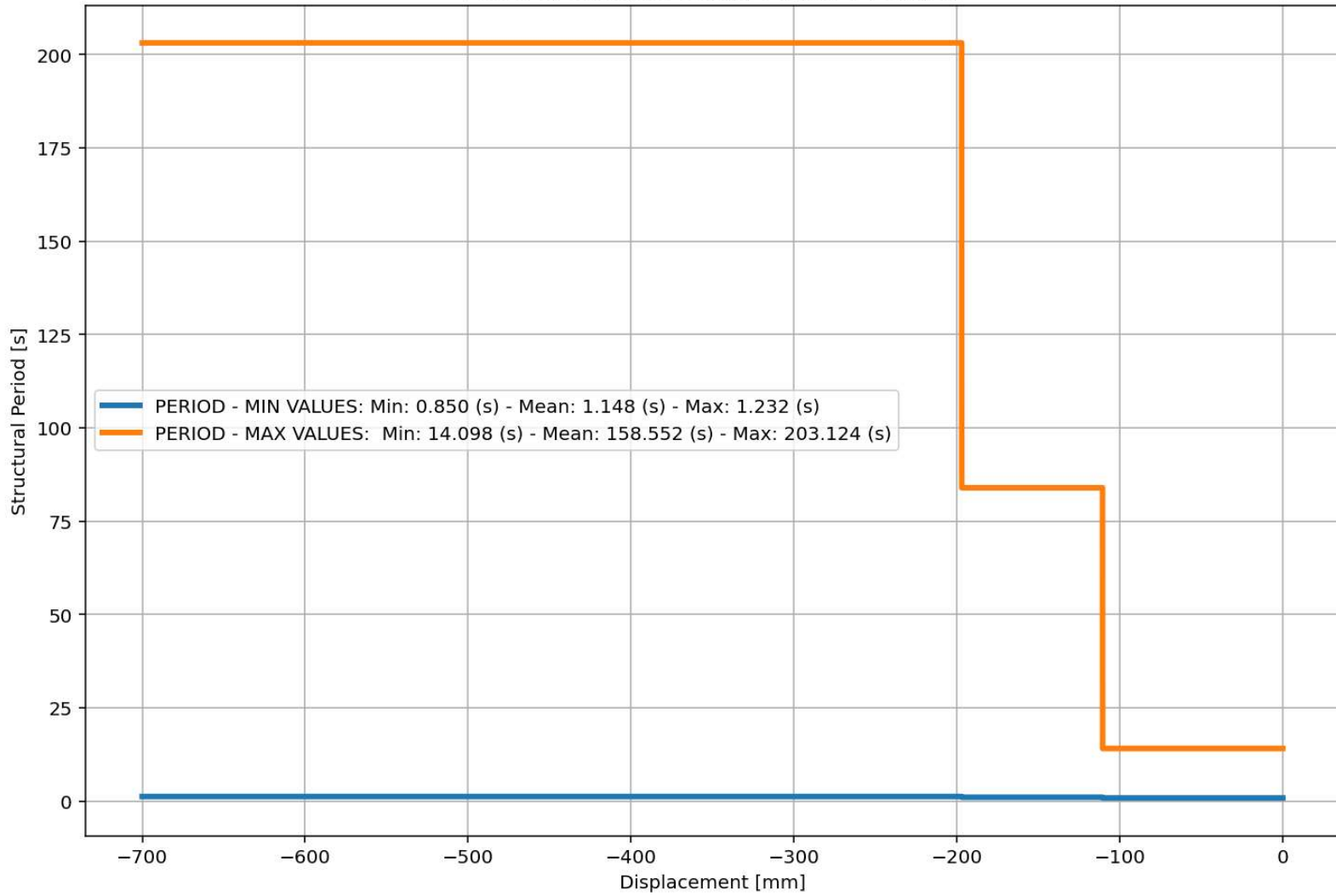
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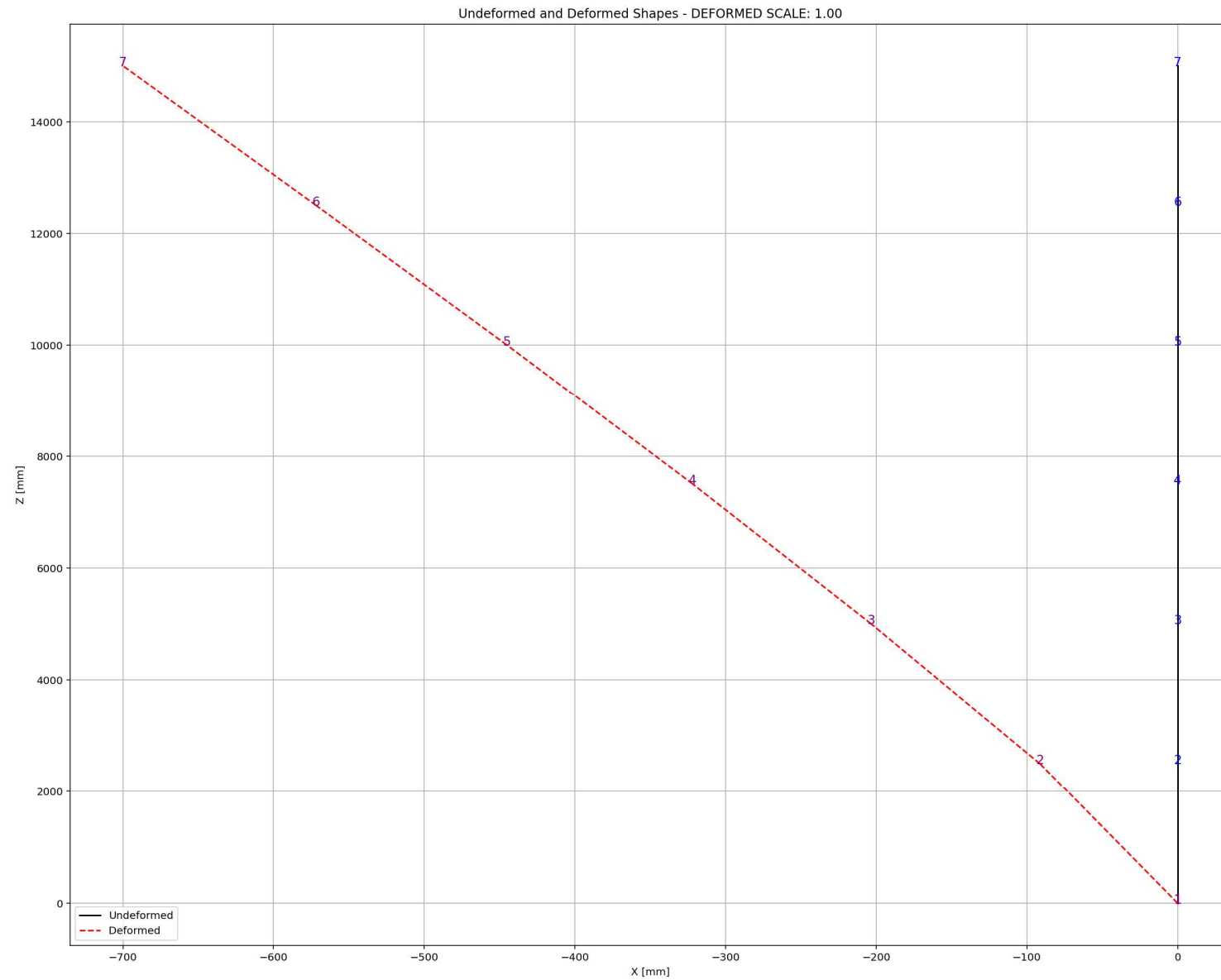
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Period of Structure During Pushover Analysis





CYCLIC DISPLACEMENT ANALYSIS

Spyder (Python 3.12)

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W(x,t)_COLUMN_AGGR...N_ELEMENT_8_ANA.py X AGGREGATED_AXIAL_F...ION_ELEMENT_FUN.py X

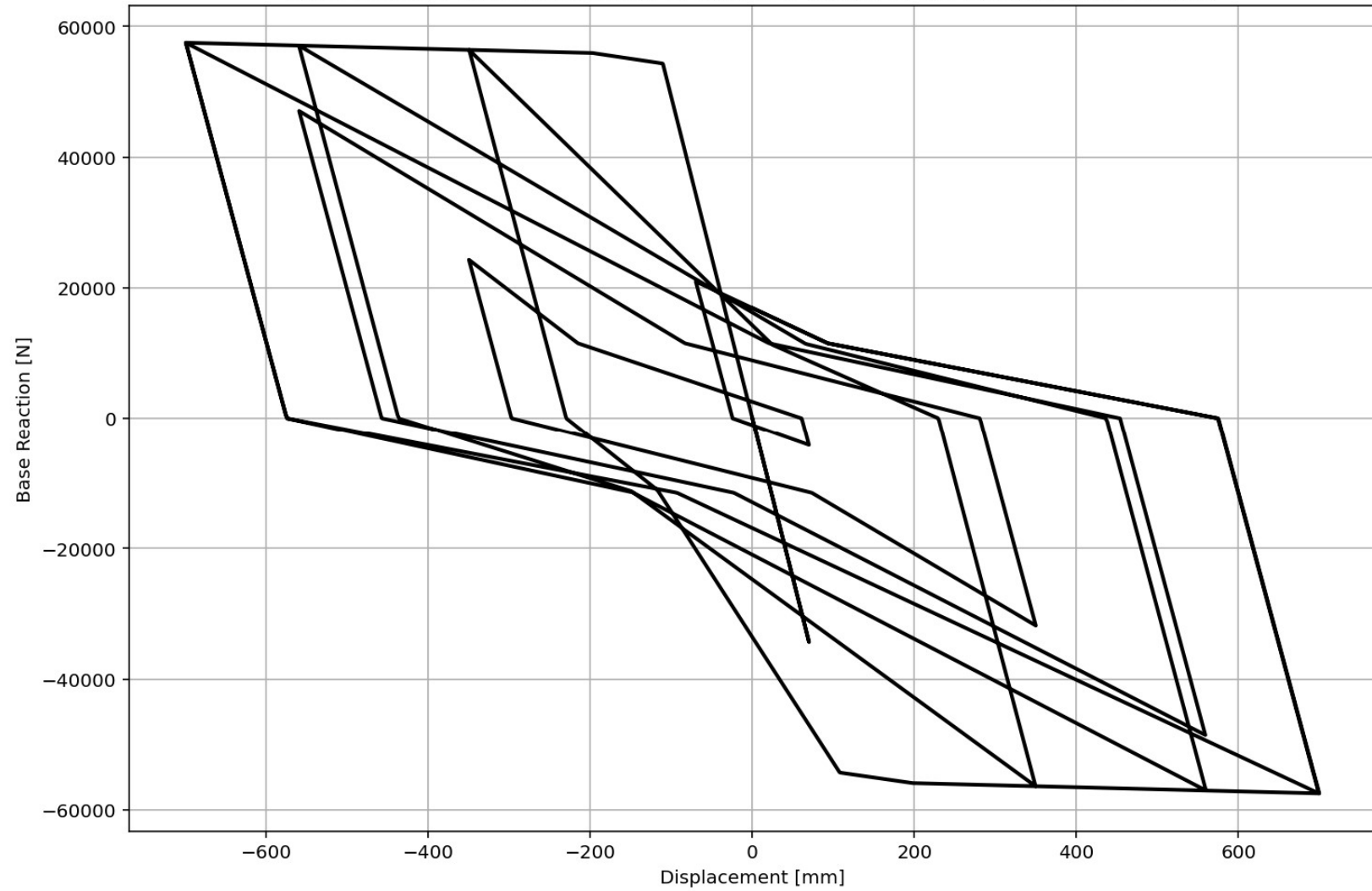
```
1016 # CYCLIC DISPLACEMENT ANALYSIS (STATIC TIME-HISTORY ANALYSIS)
1017 MAT_TYPE = 'INELASTIC' # 'ELASTIC' OR 'INELASTIC'
1018 ANAL_TYPE = 'CYCLIC_DISPLACEMENT'
1019
1020 DATA = AGGREGATED_AXIAL_FLEXURAL_COLUMN(LENGTH, MAT_TYPE, ANAL_TYPE, TOTAL_MASS)
1021 (reactionX_CP, reactionY_CP, reactionZ_CP, disp_mid_CP,
1022  ele_axialforce_CP, ele_shearforce_CP, ele_momentforce_CP,
1023  node_displacementsX_CP, node_displacementsY_CP,
1024  dispX_CP, dispY_CP,
1025  PERIOD_MIN_CP, PERIOD_MAX_CP) = DATA
1026
1027
1028 XDATA = disp_mid_CP
1029 YDATA = reactionX_CP
1030 XLABEL = 'Displacement [mm]'
1031 YLABEL = 'Base Reaction [N]'
1032 TITLE = 'Base Reaction and Displacement of Structure During Cyclic-Displacement Analysis'
1033 COLOR = 'black'
1034 SEMILOGY = False
1035 PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMILOGY)
1036
1037 # PLOT STRUCTURAL PERIOD DURING THE ANALYSIS
1038 plt.figure(0, figsize=(12, 8))
1039 plt.plot(disp_mid_CP, PERIOD_MIN_CP, linewidth=3)
1040 plt.plot(disp_mid_CP, PERIOD_MAX_CP, linewidth=3)
1041 plt.title('Period of Structure During Cyclic Displacement Analysis')
1042 plt.ylabel('Structural Period [s]')
1043 plt.xlabel('Displacement [mm]')
1044 #plt.semilogy()
1045 plt.grid()
1046 plt.legend([f'PERIOD - MIN VALUES: Min: {np.min(PERIOD_MIN_CP):.3f} (s) - Mean: {np.mean(PERIOD_MIN_CP):.3f} (s)',
1047            f'PERIOD - MAX VALUES: Min: {np.min(PERIOD_MAX_CP):.3f} (s) - Mean: {np.mean(PERIOD_MAX_CP):.3f} (s)'],
1048            [])
1049 plt.show()
```

Base Reaction and Displacement of Structure During Cyclic Displacement Analysis

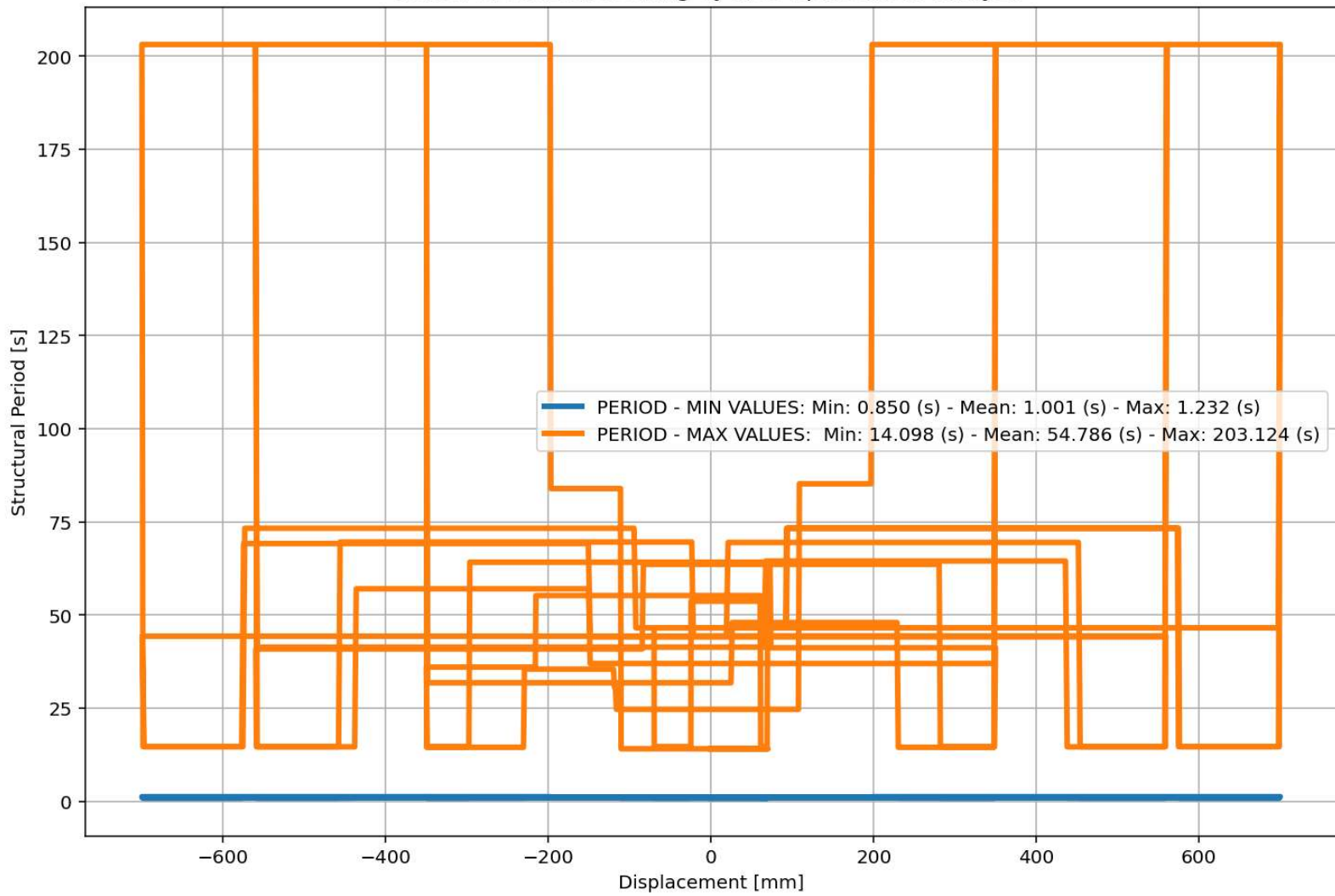
IPython Console Plots Variable Explorer Help Debugger History Files

File Tools Conda: anaconda3 (Python 3.12.7) ✓ LSP: Python Line 14 Col 49 UTF-8 CRLF RW Mem 47%

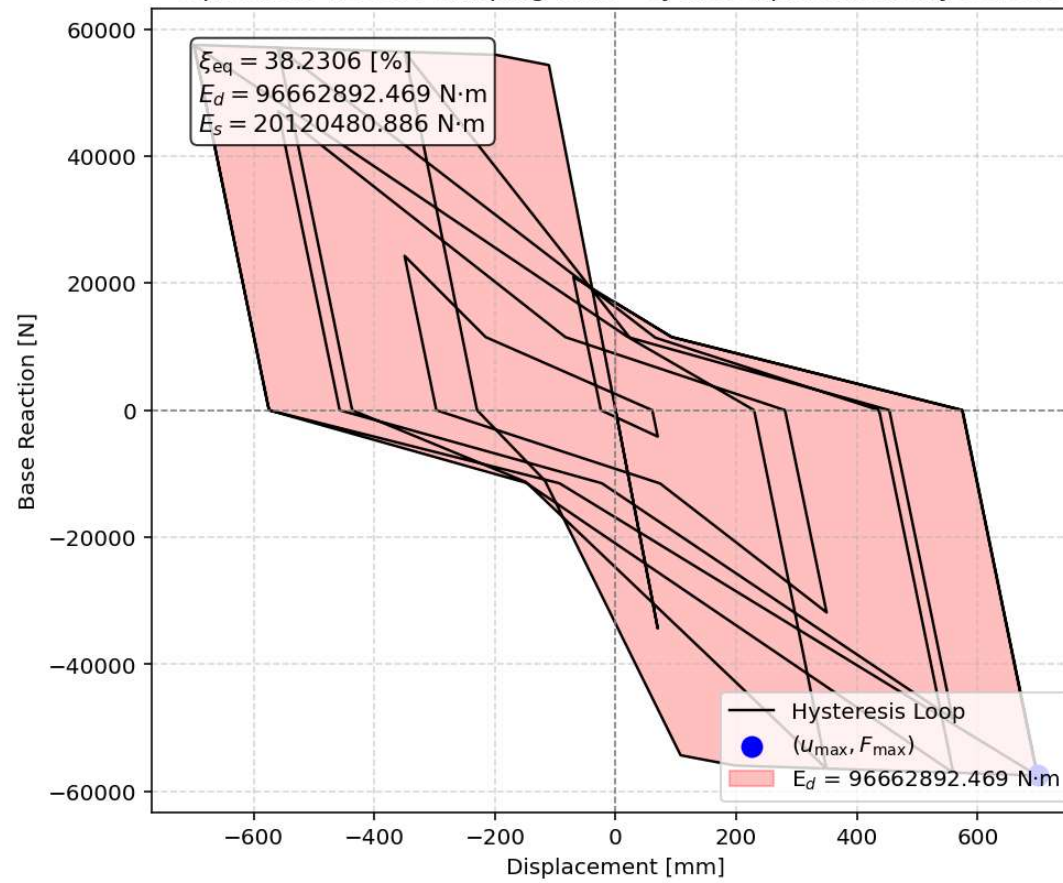
The graph plots Base Reaction [N] on the y-axis (ranging from -60,000 to 60,000) against Displacement [mm] on the x-axis (ranging from -600 to 600). Multiple black lines represent different load-displacement paths. Most curves start at high positive reaction values (up to 60,000 N) at negative displacements and show a sharp drop as displacement increases, eventually reaching a plateau or a second peak at positive displacements. This illustrates the complex nonlinear behavior of the column, including buckling and post-buckling stages.

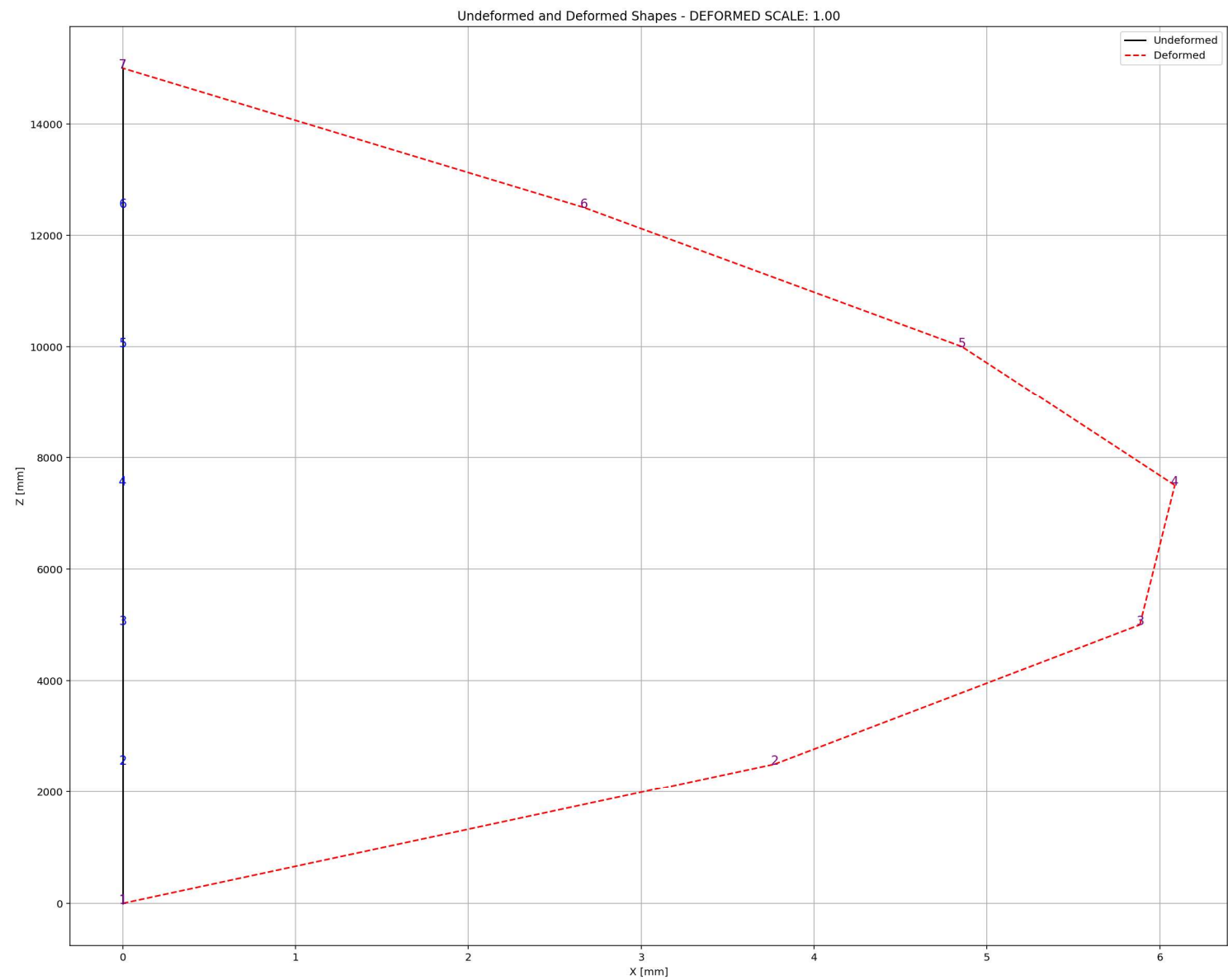


Period of Structure During Cyclic Displacement Analysis



Equivalent viscous damping ratio - Cyclic Displacement Hysteresis



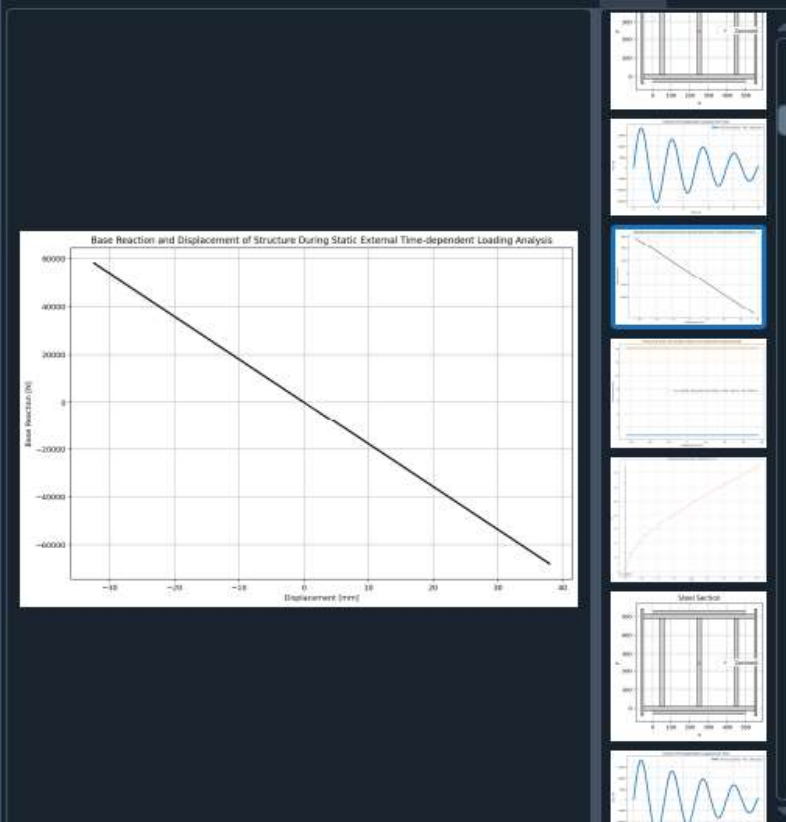


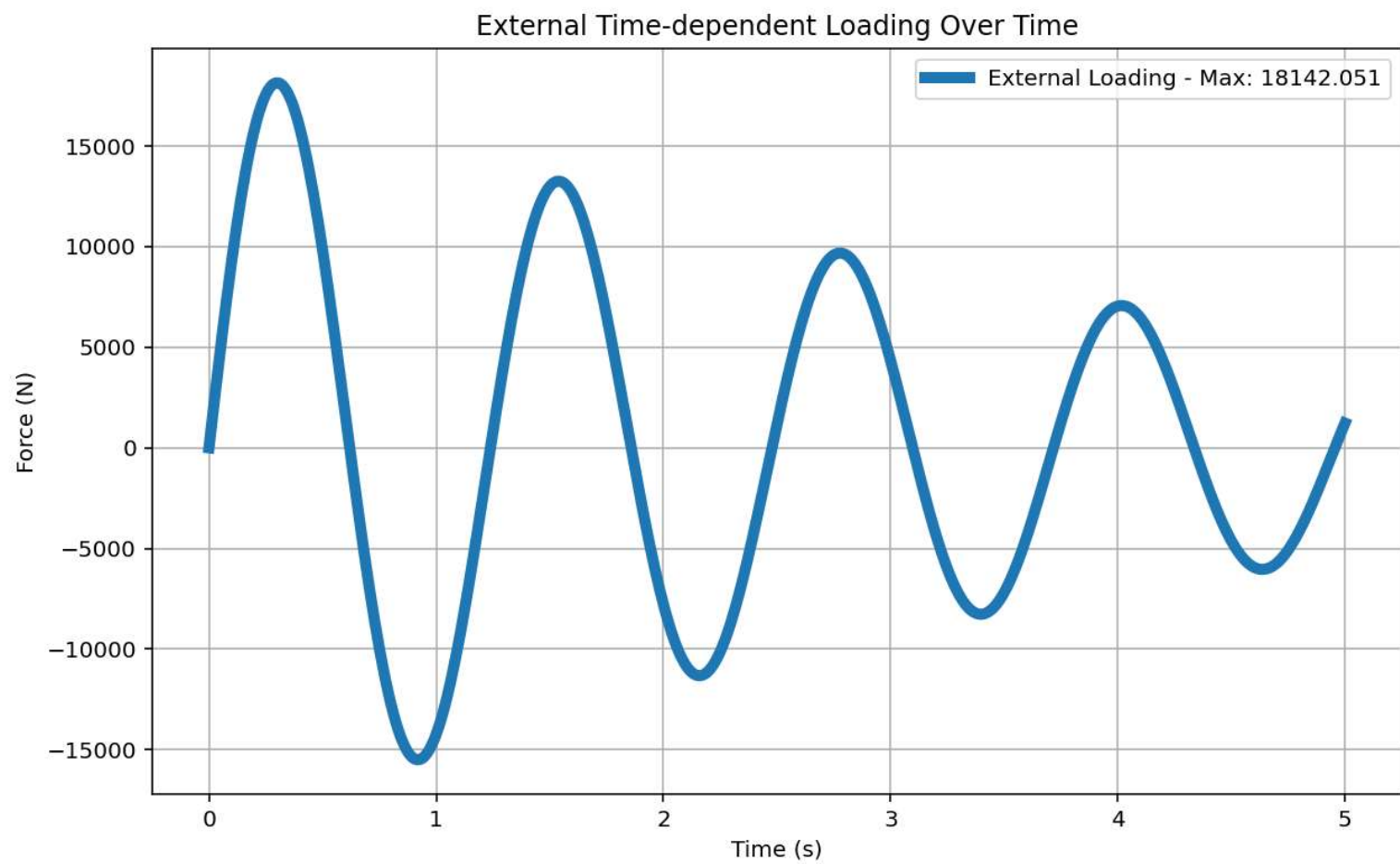
STATIC TIME-HISTORY WITH EXTERNAL TIME-DEPENDENT LOADING ANALYSIS

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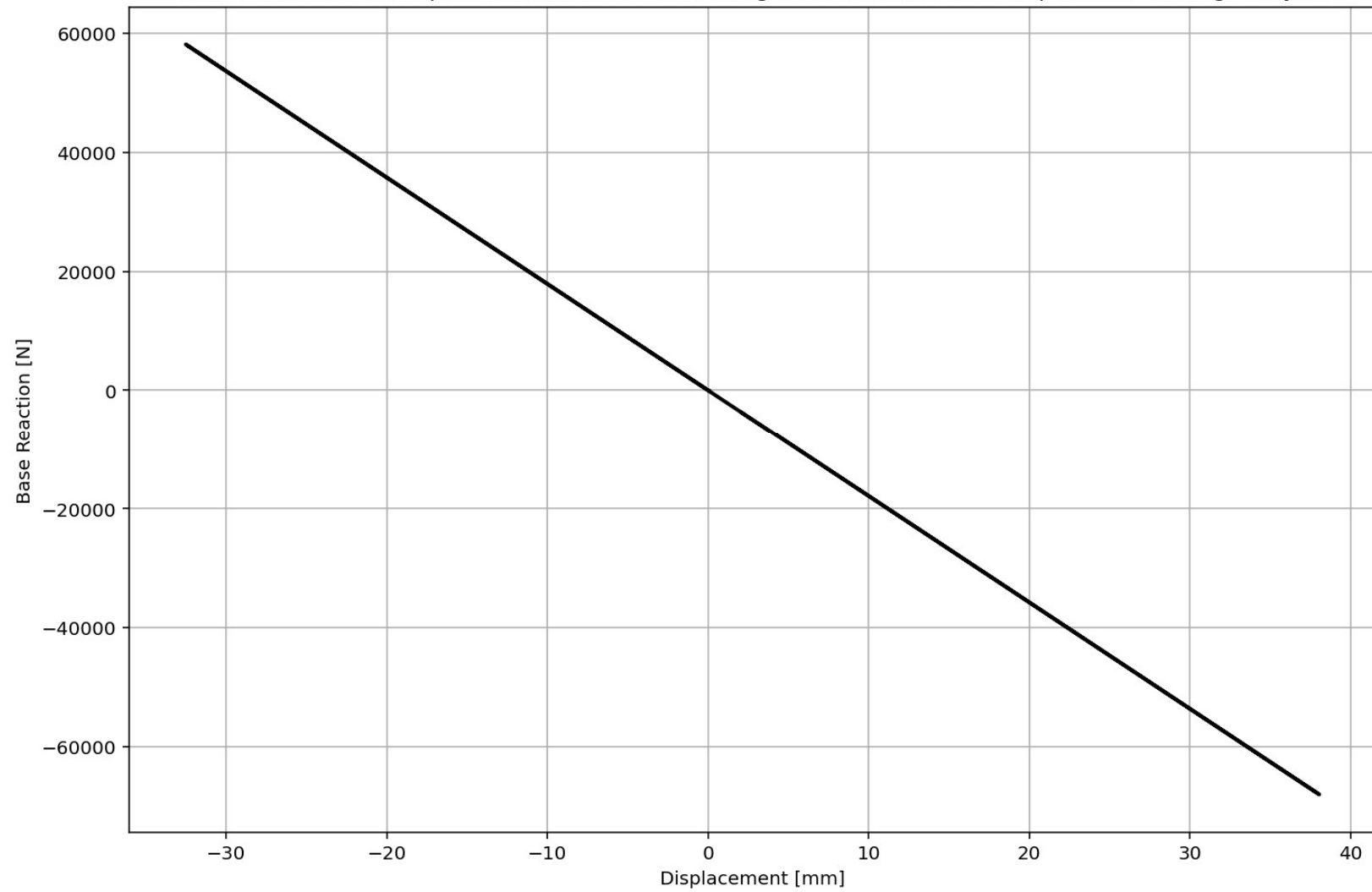
```
W(x,t)_COLUMN_AGR...N_ELEMENT_8_ANA.py X AGGREGATED_AXIAL_F...ION_ELEMENT_FUN.py X
1060 # EXTERNAL TIME-DEPENDENT LOADING ANALYSIS (STATIC TIME-HISTORY ANALYSIS)
1061 MAT_TYPE = 'INELASTIC' # 'ELASTIC' OR 'INELASTIC'
1062 ANAL_TYPE = 'STATIC_EXTERNAL_TIME-DEPENDENT_LOADING'
1063
1064 DATA = AGGREGATED_AXIAL_FLEXURAL_COLUMN(LENGTH, MAT_TYPE, ANAL_TYPE, TOTAL_MASS)
1065
1066 (reactionX_ETDLS, reactionY_ETDLS, reactionZ_ETDLS, disp_mid_ETDLS,
1067 ele_axialforce_ETDLS, ele_shearforce_ETDLS, ele_momentforce_ETDLS,
1068 node_displacementsX_ETDLS, node_displacementsY_ETDLS,
1069 dispX_ETDLS, dispY_ETDLS,
1070 PERIOD_MIN_ETDLS, PERIOD_MAX_ETDLS) = DATA
1071
1072
1073 XDATA = disp_mid_ETDLS
1074 YDATA = reactionX_ETDLS
1075 XLABEL = 'Displacement [mm]'
1076 YLABEL = 'Base Reaction [N]'
1077 TITLE = 'Base Reaction and Displacement of Structure During Static External Time-dependent Loading Anal
1078 COLOR = 'black'
1079 SEMIOLOGY = False
1080 PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMIOLOGY)
1081
1082 # PLOT STRUCTURAL PERIOD DURING THE ANALYSIS
1083 plt.figure(0, figsize=(12, 8))
1084 plt.plot(disp_mid_ETDLS, PERIOD_MIN_ETDLS, linewidth=3)
1085 plt.plot(disp_mid_ETDLS, PERIOD_MAX_ETDLS, linewidth=3)
1086 plt.title('Period of Structure During Static External Time-dependent Loading Analysis')
1087 plt.ylabel('Structural Period [s]')
1088 plt.xlabel('Displacement [mm]')
1089 #plt.semilogy()
1090 plt.grid()
1091 plt.legend([f'PERIOD - MIN VALUES: Min: {np.min(PERIOD_MIN_ETDLS):.3f} (s) - Mean: {np.mean(PERIOD_MIN_
1092 f'PERIOD - MAX VALUES: Min: {np.min(PERIOD_MAX_ETDLS):.3f} (s) - Mean: {np.mean(PERIOD_MA
1093 ])
```

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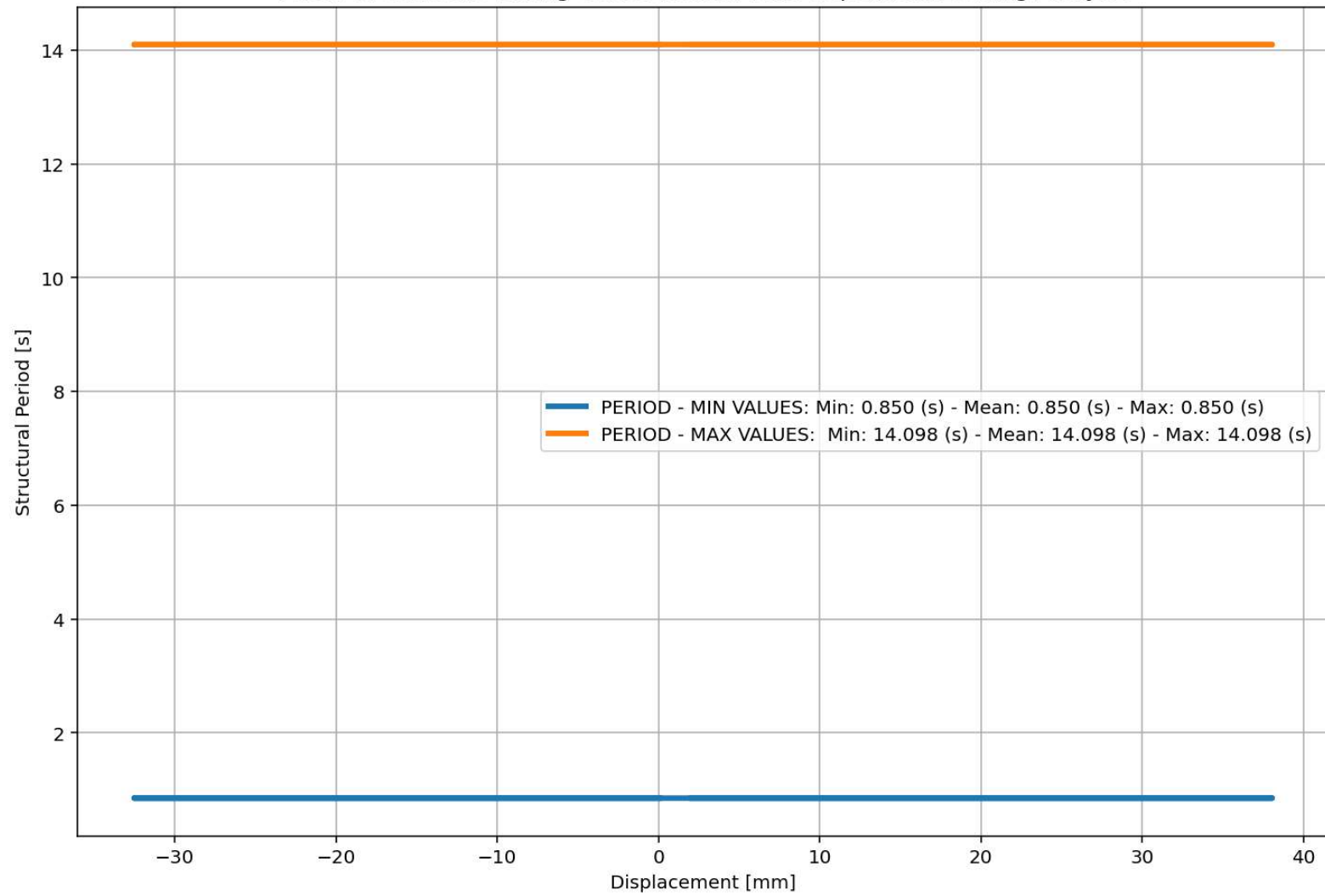


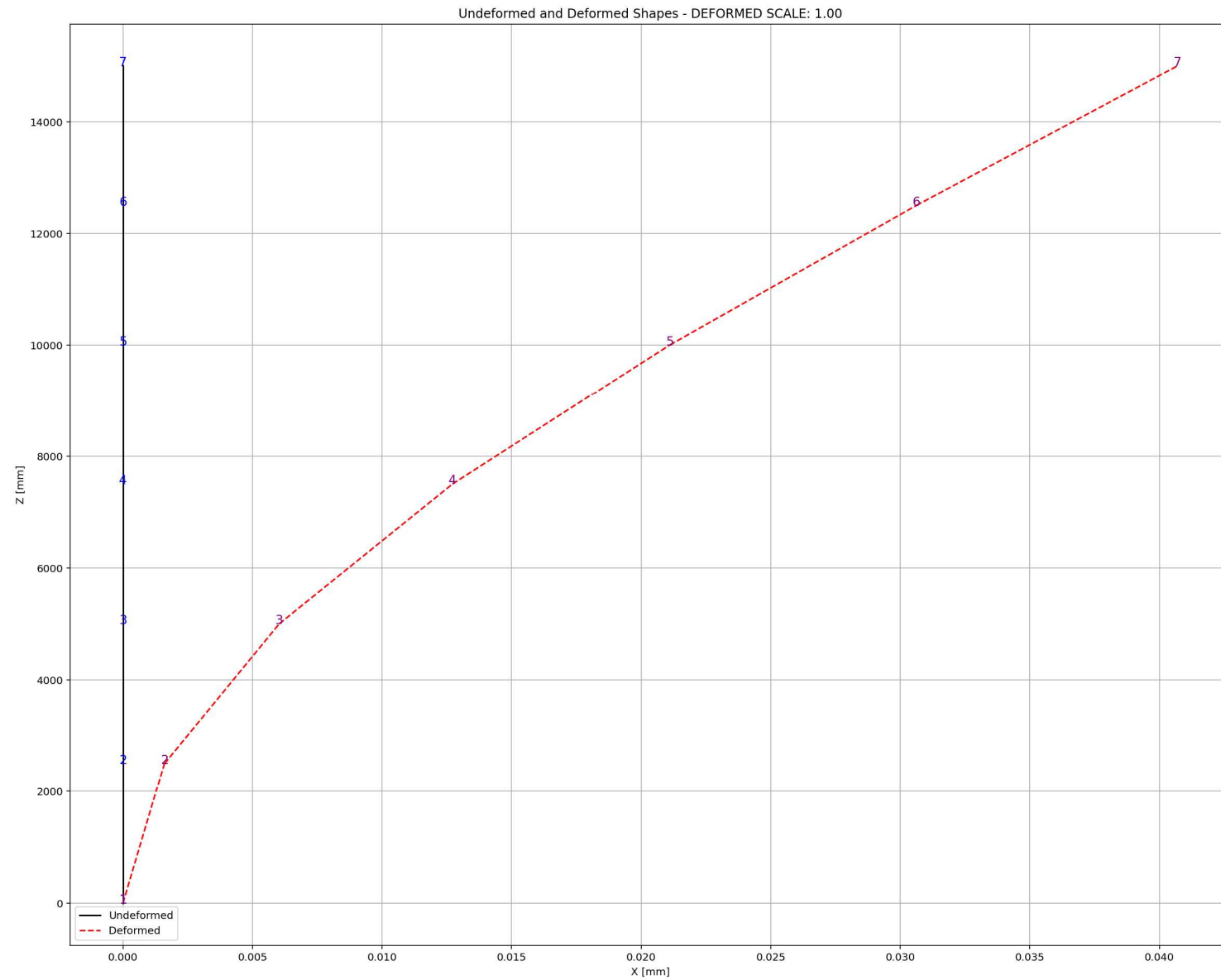


Base Reaction and Displacement of Structure During Static External Time-dependent Loading Analysis



Period of Structure During Static External Time-dependent Loading Analysis



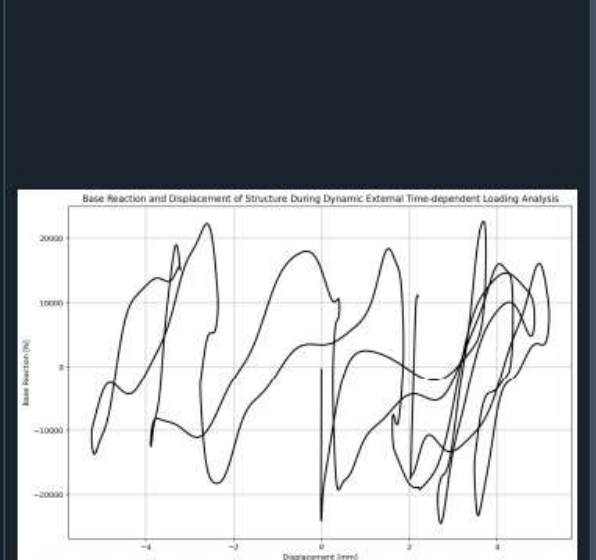


DYNAMIC TIME-HISTORY WITH EXTERNAL TIME-DEPENDENT LOADING ANALYSIS

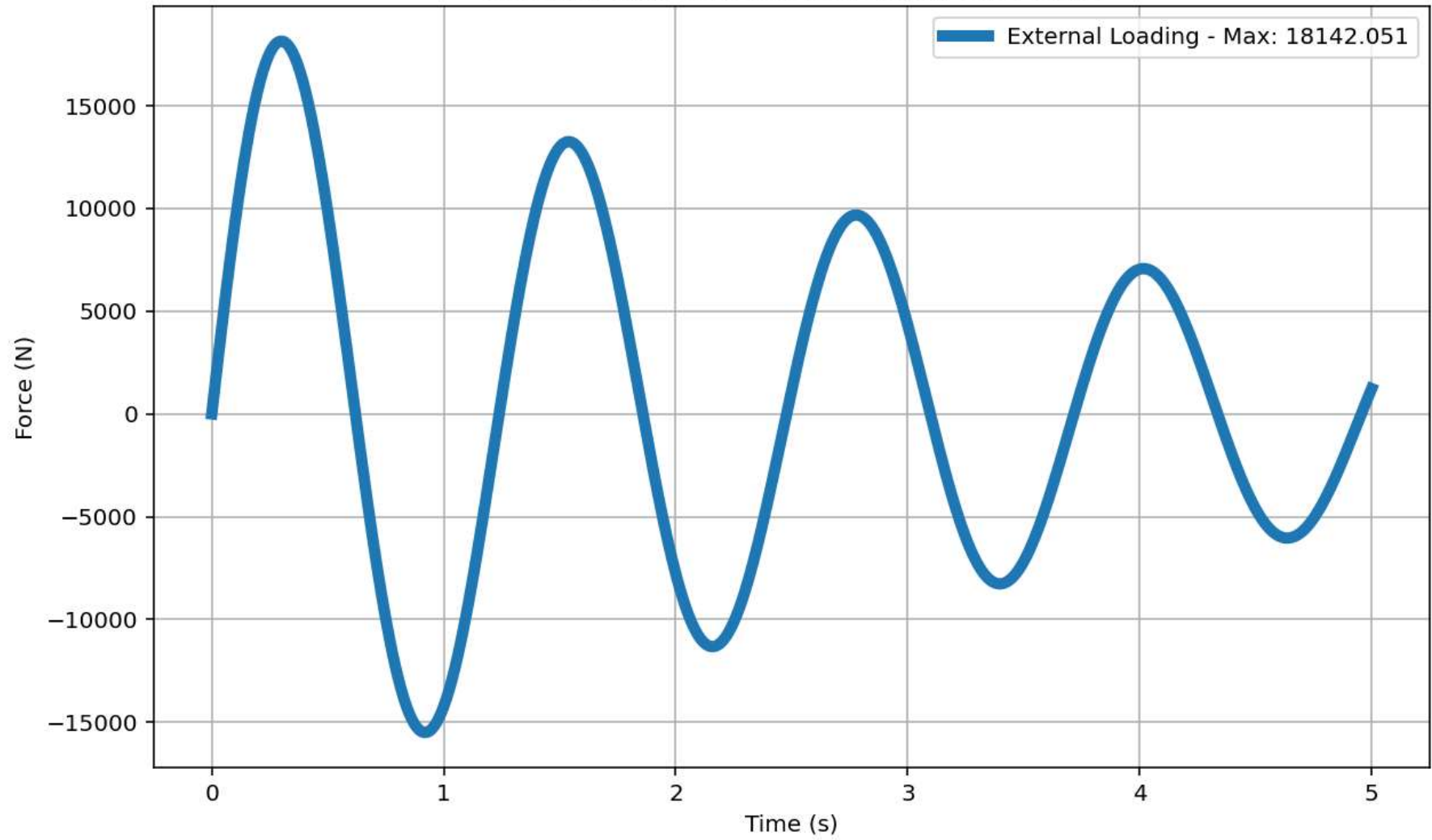
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1098 # EXTERNAL TIME-DEPENDENT LOADING ANALYSIS (DYNAMIC TIME-HISTORY ANALYSIS)
1099 MAT_TYPE = 'INELASTIC' # 'ELASTIC' OR 'INELASTIC'
1100 ANAL_TYPE = 'DYNAMIC_EXTERNAL_TIME-DEPENDENT_LOADING'
1101
1102 DATA = AGGREGATED_AXIAL_FLEXURAL_COLUMN(LENGTH, MAT_TYPE, ANAL_TYPE, TOTAL_MASS)
1103
1104 (time_ETDLD, reactionX_ETDLD, reactionY_ETDLD, reactionZ_ETDLD, disp_mid_ETDLD,
1105 ele_axialforce_ETDLD, ele_shearforce_ETDLD, ele_momentforce_ETDLD,
1106 node_displacementsX_ETDLD, node_displacementsY_ETDLD,
1107 dispX_ETDLD, dispY_ETDLD,
1108 veloX_ETDLD, veloY_ETDLD,
1109 accX_ETDLD, accY_ETDLD,
1110 stiffness_ETDLD, PERIOD_ETDLD, damping_ratio_ETDLD,
1111 PERIOD_MIN_ETDLD, PERIOD_MAX_ETDLD) = DATA
1112
1113
1114 XDATA = disp_mid_ETDLD
1115 YDATA = reactionX_ETDLD
1116 XLABEL = 'Displacement [mm]'
1117 YLABEL = 'Base Reaction [N]'
1118 TITLE = 'Base Reaction and Displacement of Structure During Dynamic External Time-dependent Loading Analysis'
1119 COLOR = 'black'
1120 SEMILOGY = False
1121 PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMILOGY)
1122
1123 PLOT_TIME_HISTORY(time_ETDLD, reactionX_ETDLD, disp_mid_ETDLD,
1124                   dispX_ETDLD, dispY_ETDLD,
1125                   veloX_ETDLD, veloY_ETDLD,
1126                   accX_ETDLD, accY_ETDLD)
1127
1128 # PLOT ELEMENTS AXIAL FORCE
1129 YLABEL = 'Element Axial Force [N]'
1130 TITLE = "elements force in X Dir. vs Time for Beam Element During Dynamic External Time-dependent Loading Analysis"
1131 PLOT_FORCES(time_ETDLD, ele_axialforce_ETDLD, YLABEL, TITLE) # Beam Element - ELEMENTS AXIAL FORCE

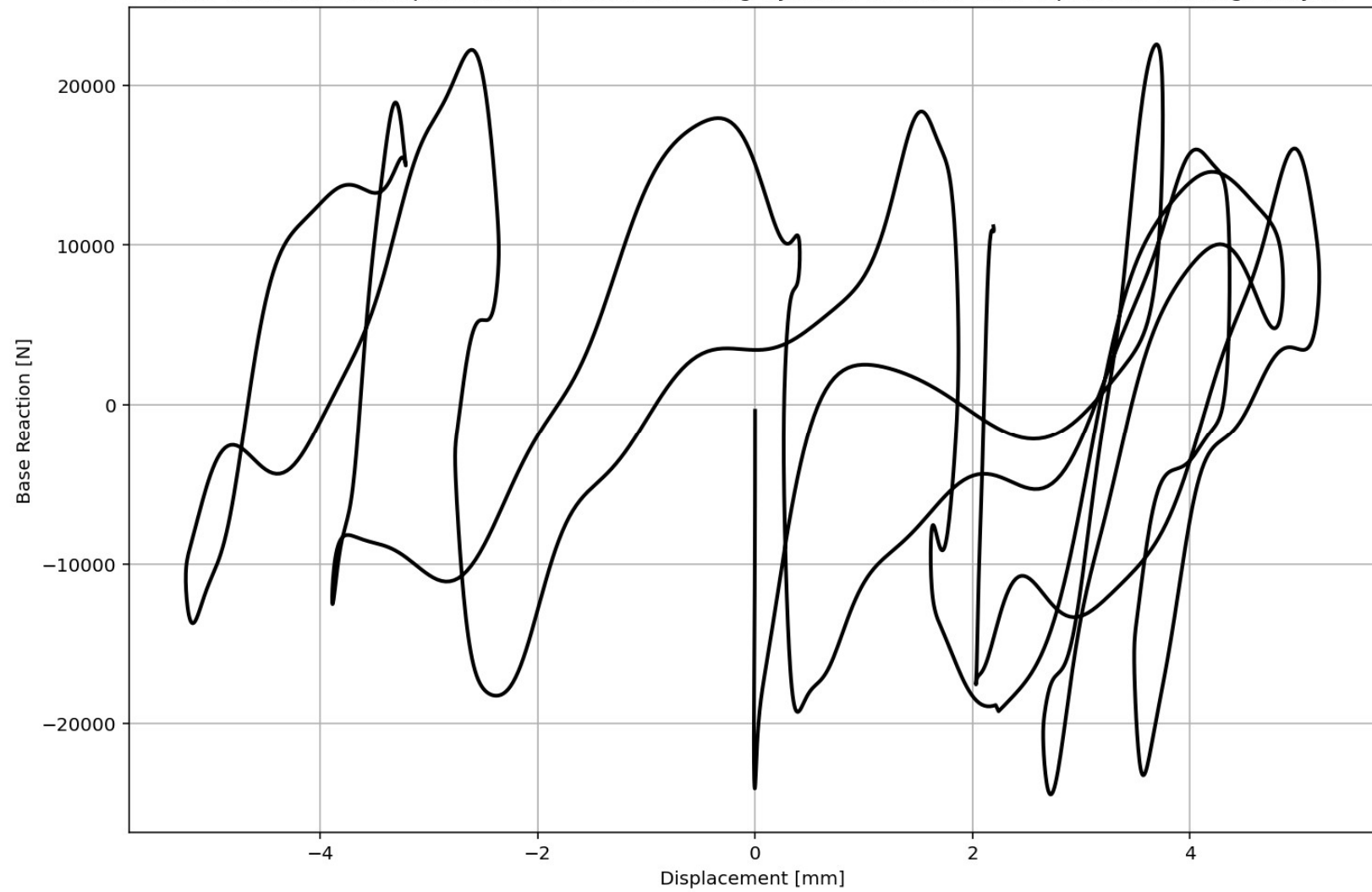
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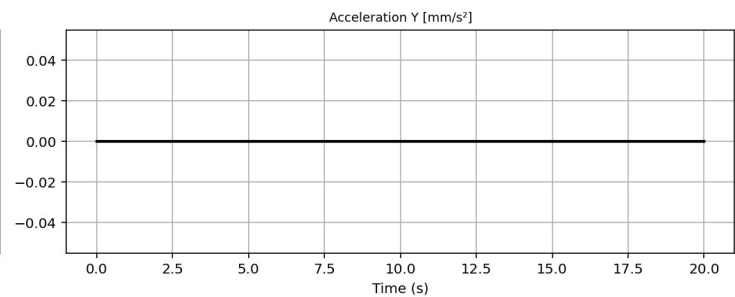
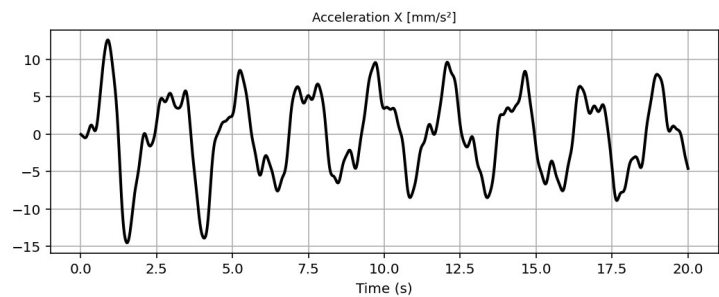
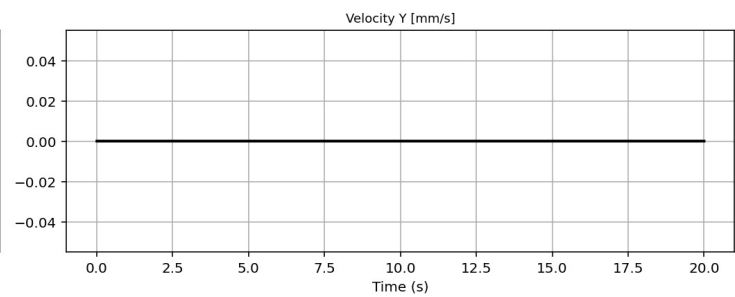
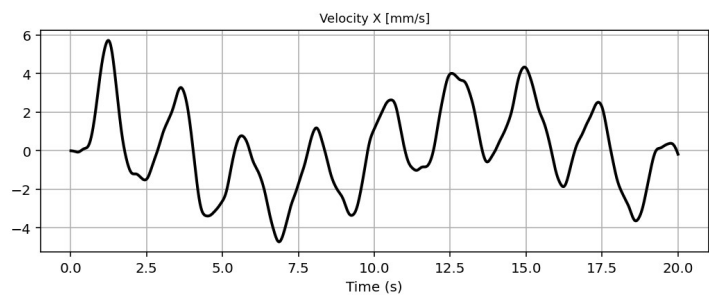
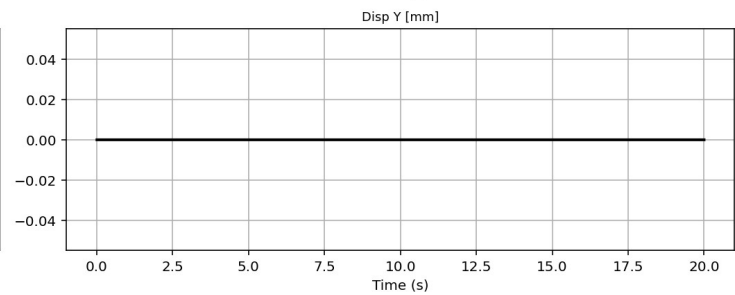
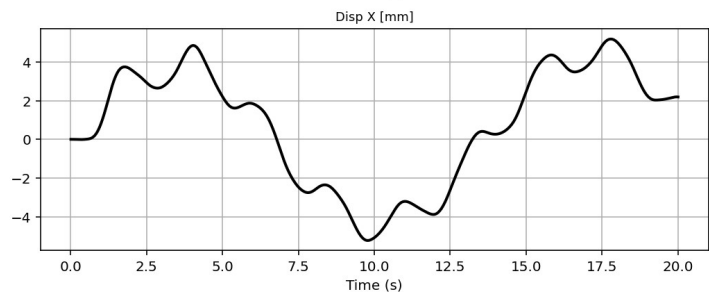
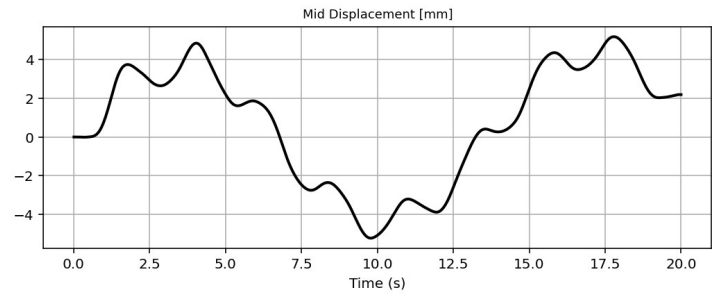
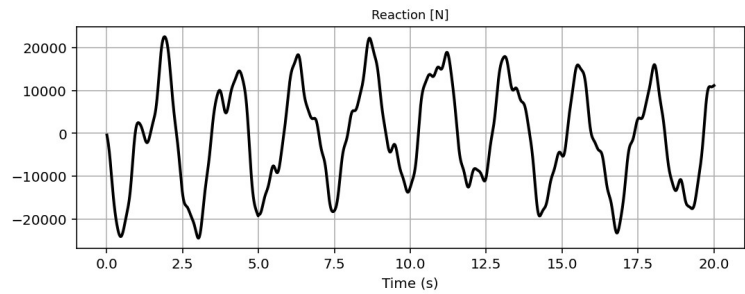


External Time-dependent Loading Over Time

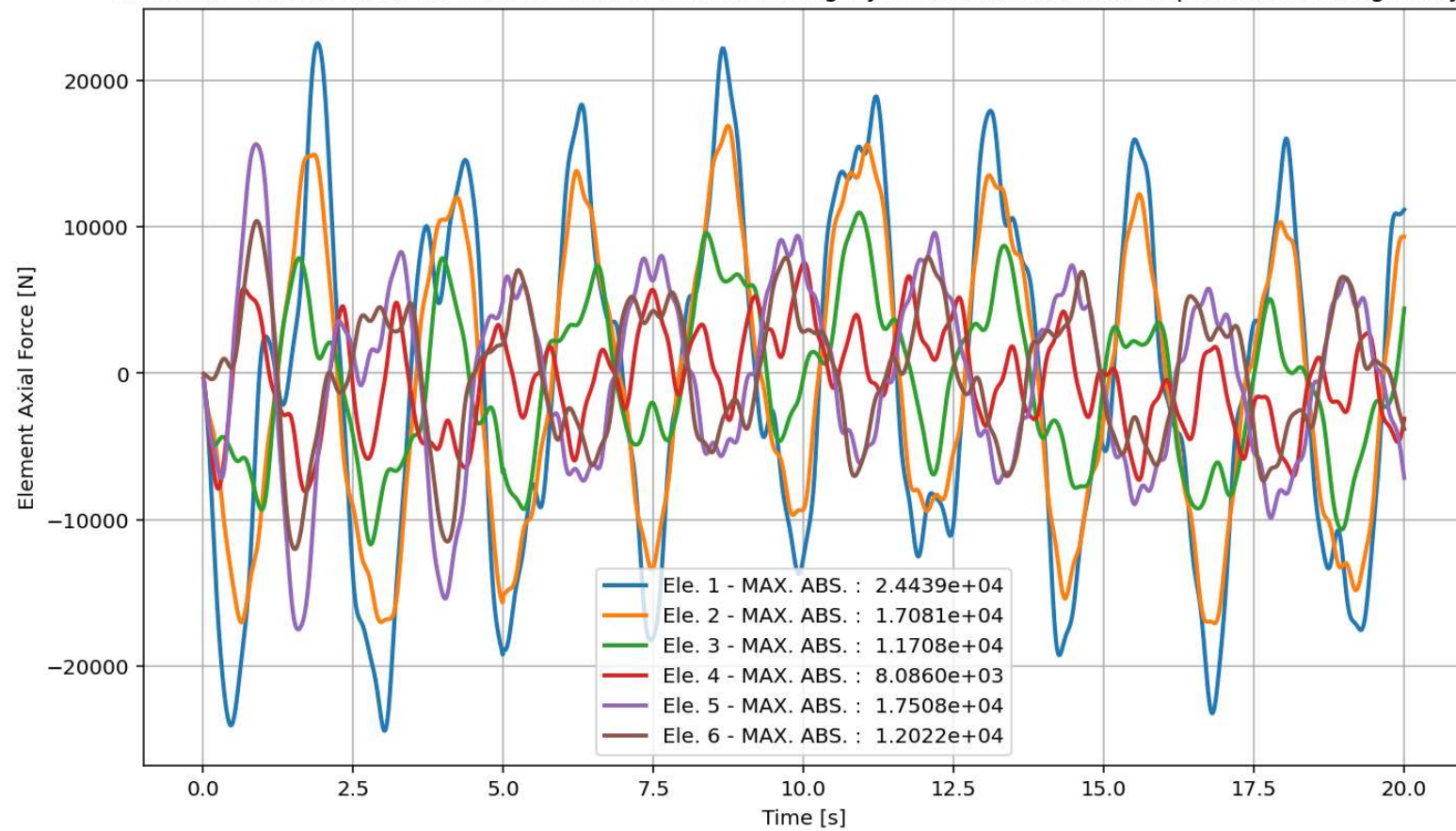


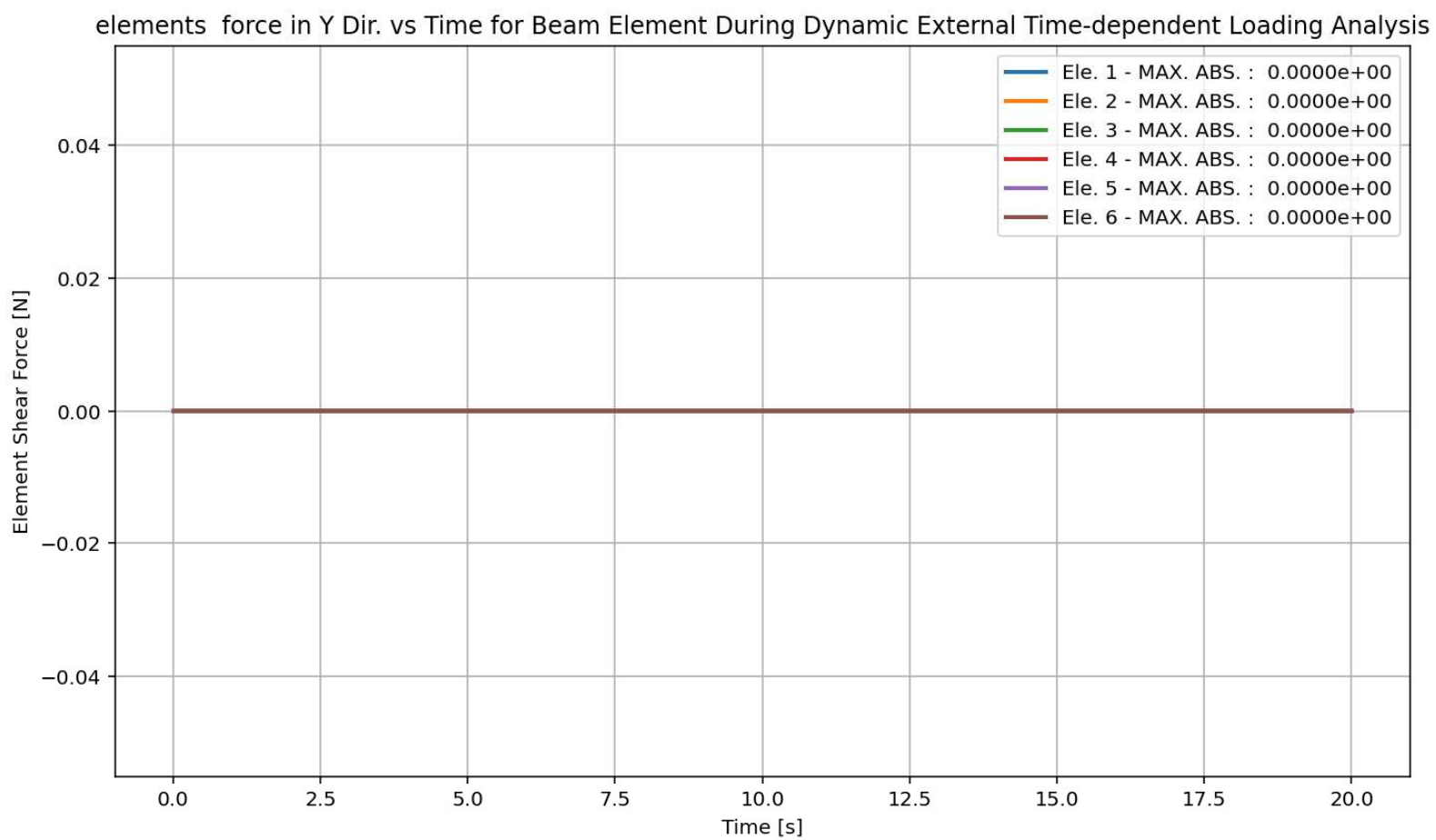
Base Reaction and Displacement of Structure During Dynamic External Time-dependent Loading Analysis



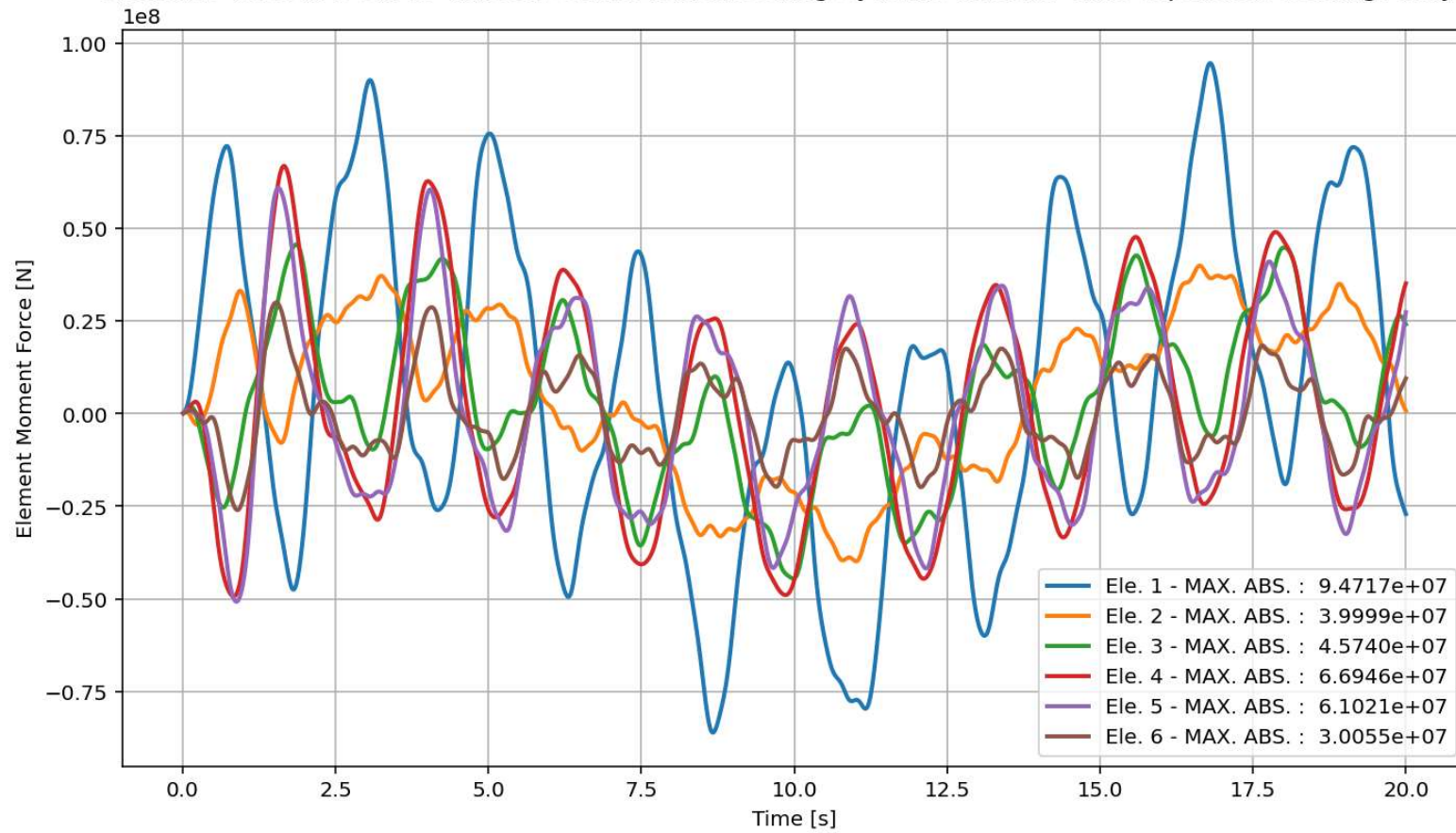


elements force in X Dir. vs Time for Beam Element During Dynamic External Time-dependent Loading Analysis

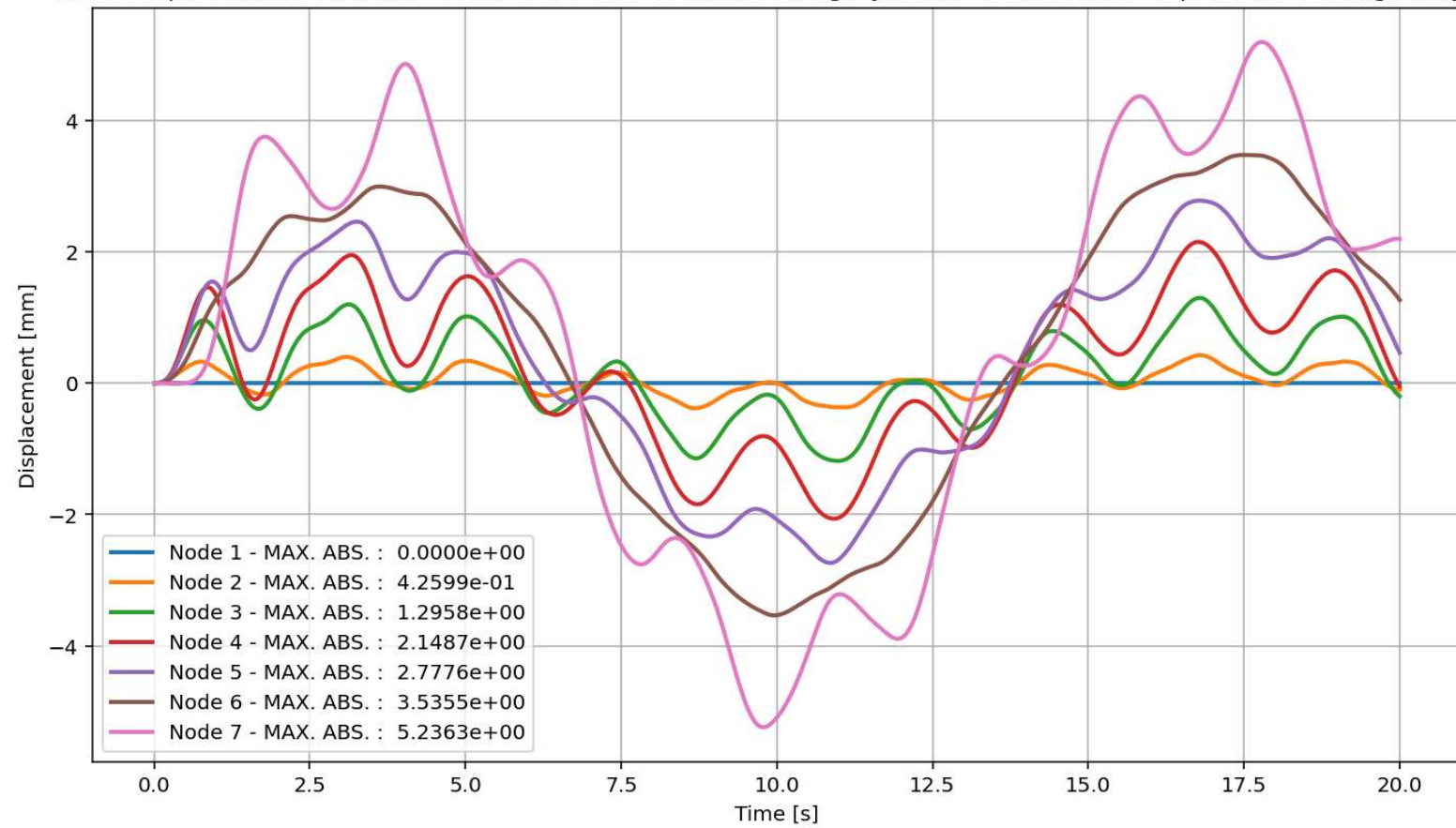




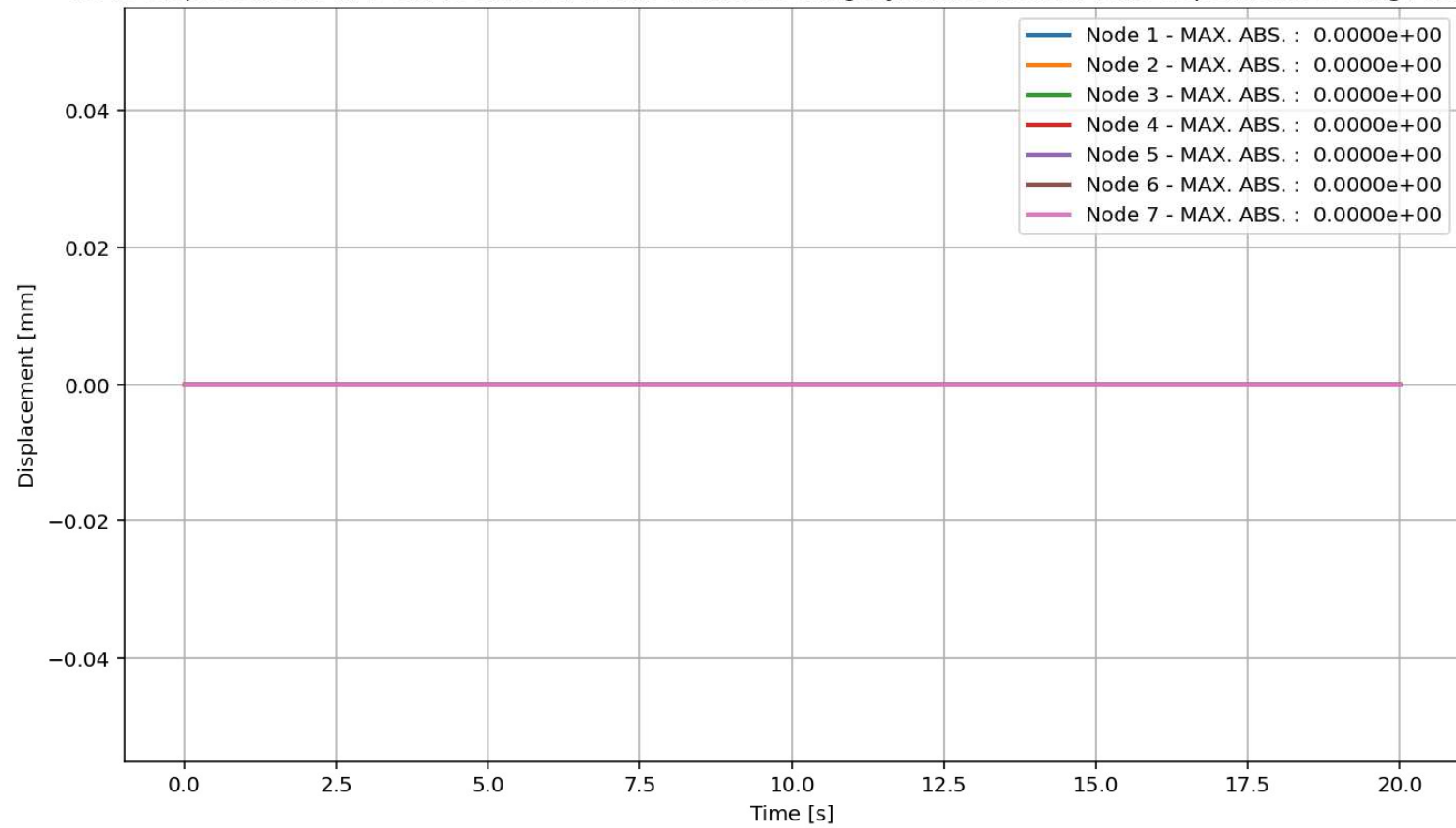
elements force in Z Dir. vs Time for Beam Element During Dynamic External Time-dependent Loading Analysis



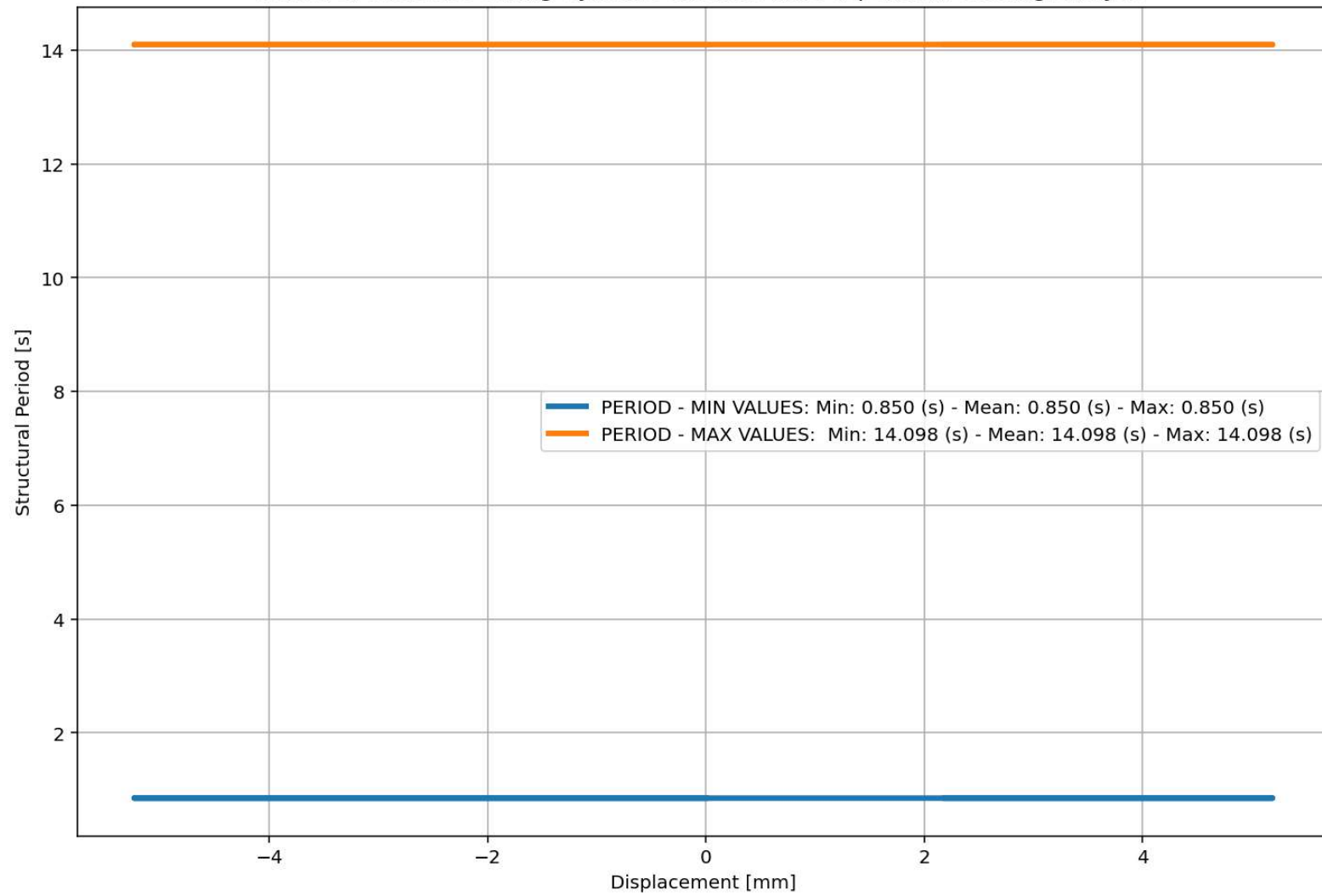
Node Displacements in X Dir. vs Time for Beam Element During Dynamic External Time-dependent Loading Analysis



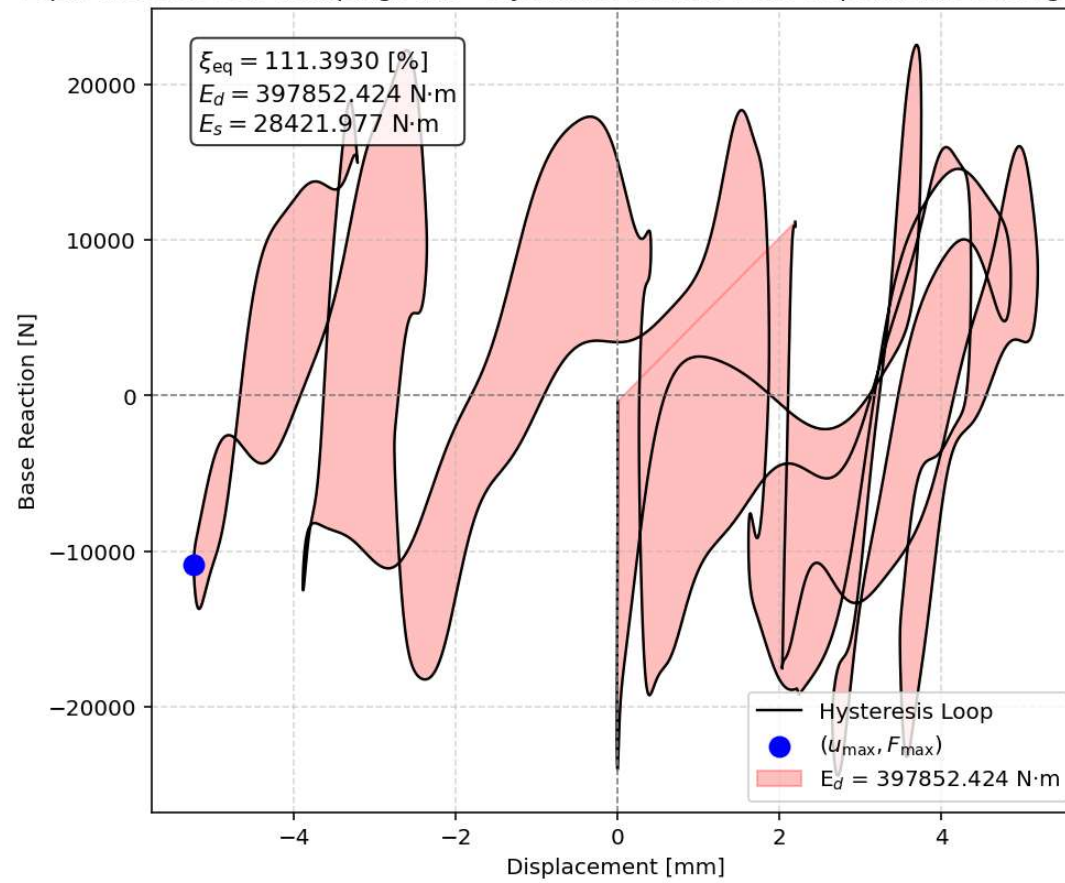
Node Displacements in Y Dir. vs Time for Beam Element During Dynamic External Time-dependent Loading Analysis

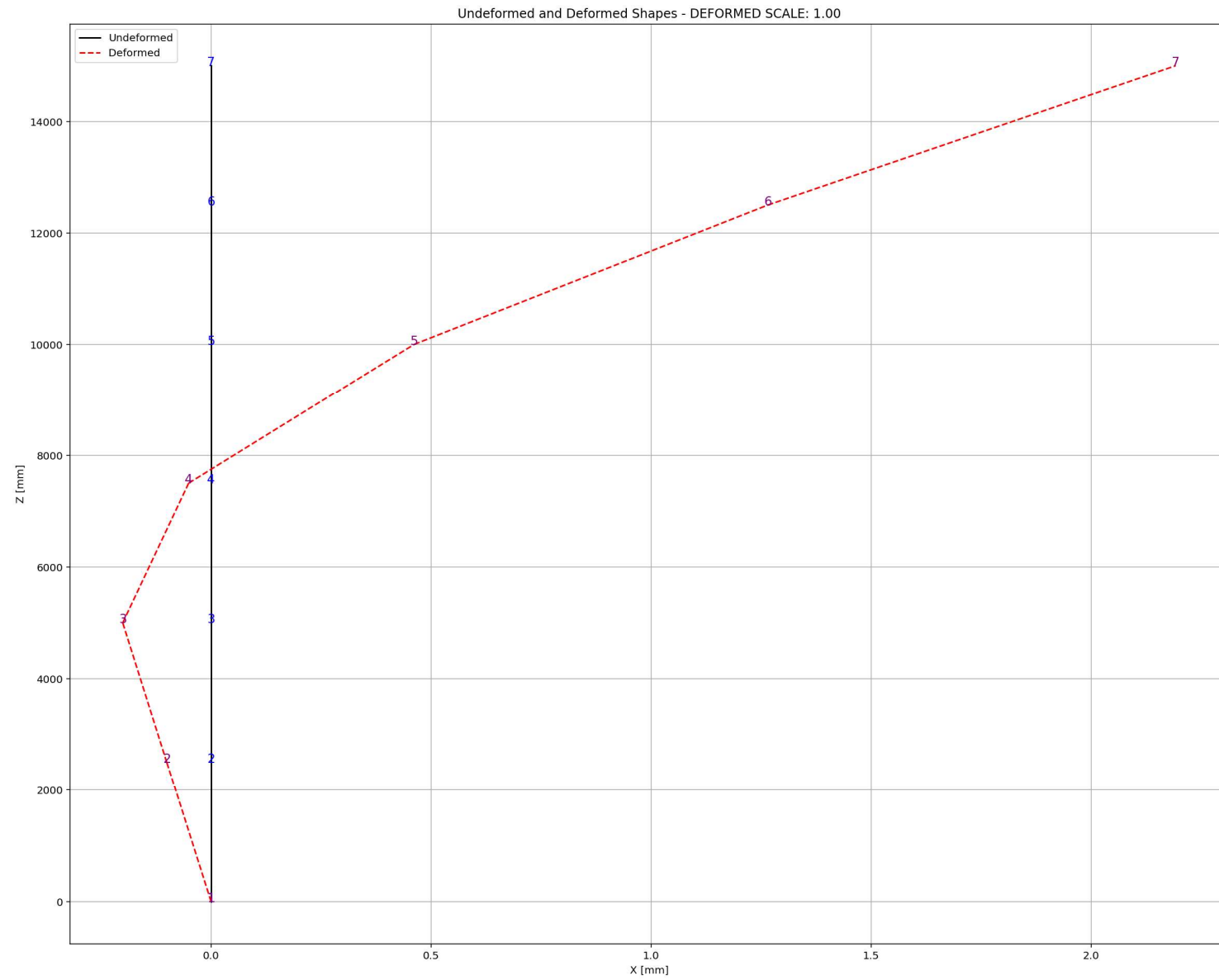


Period of Structure During Dynamic External Time-dependent Loading Analysis

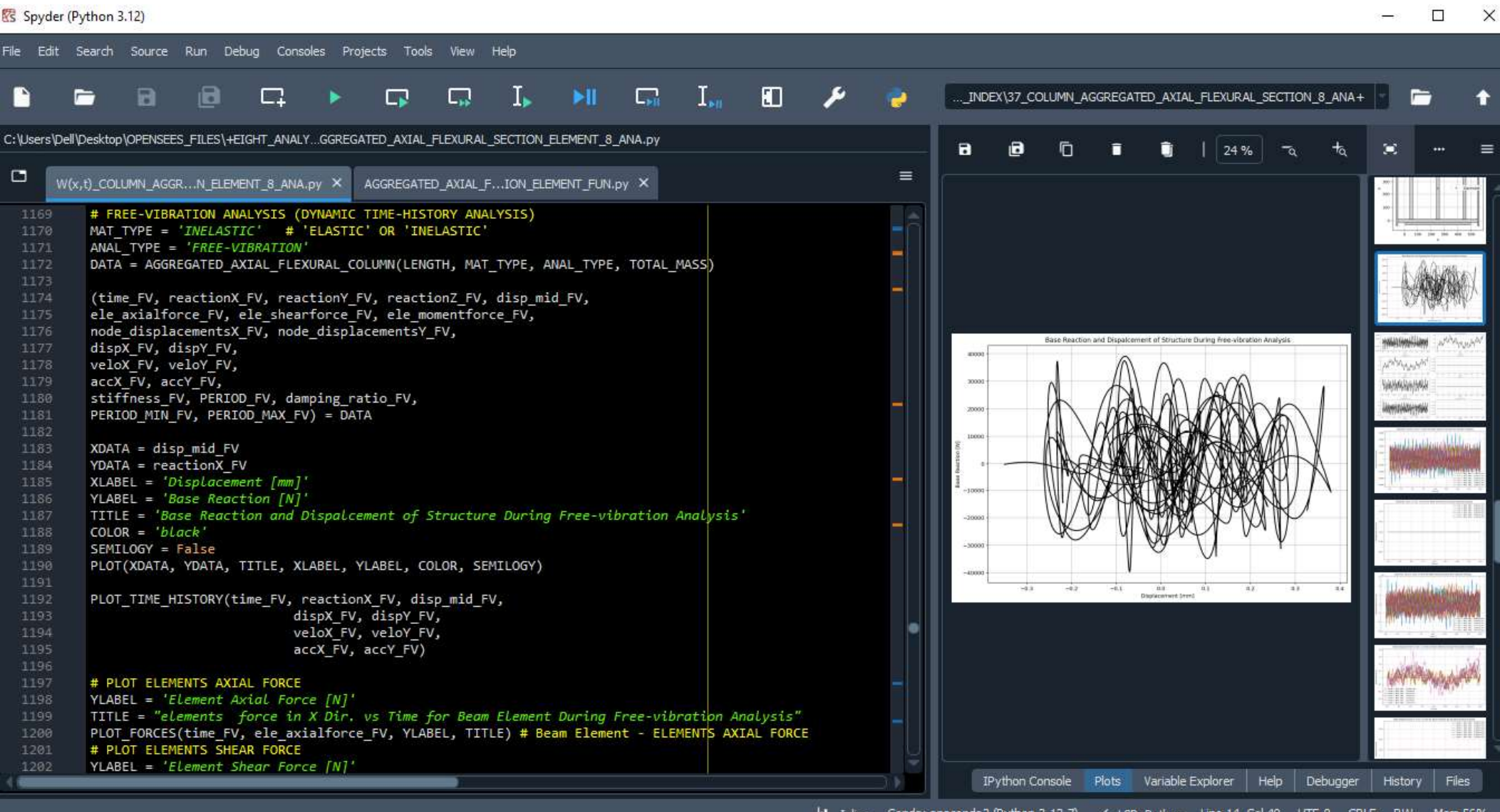


Equivalent viscous damping ratio - Dynamic External Time-dependent Loading Hysteresis

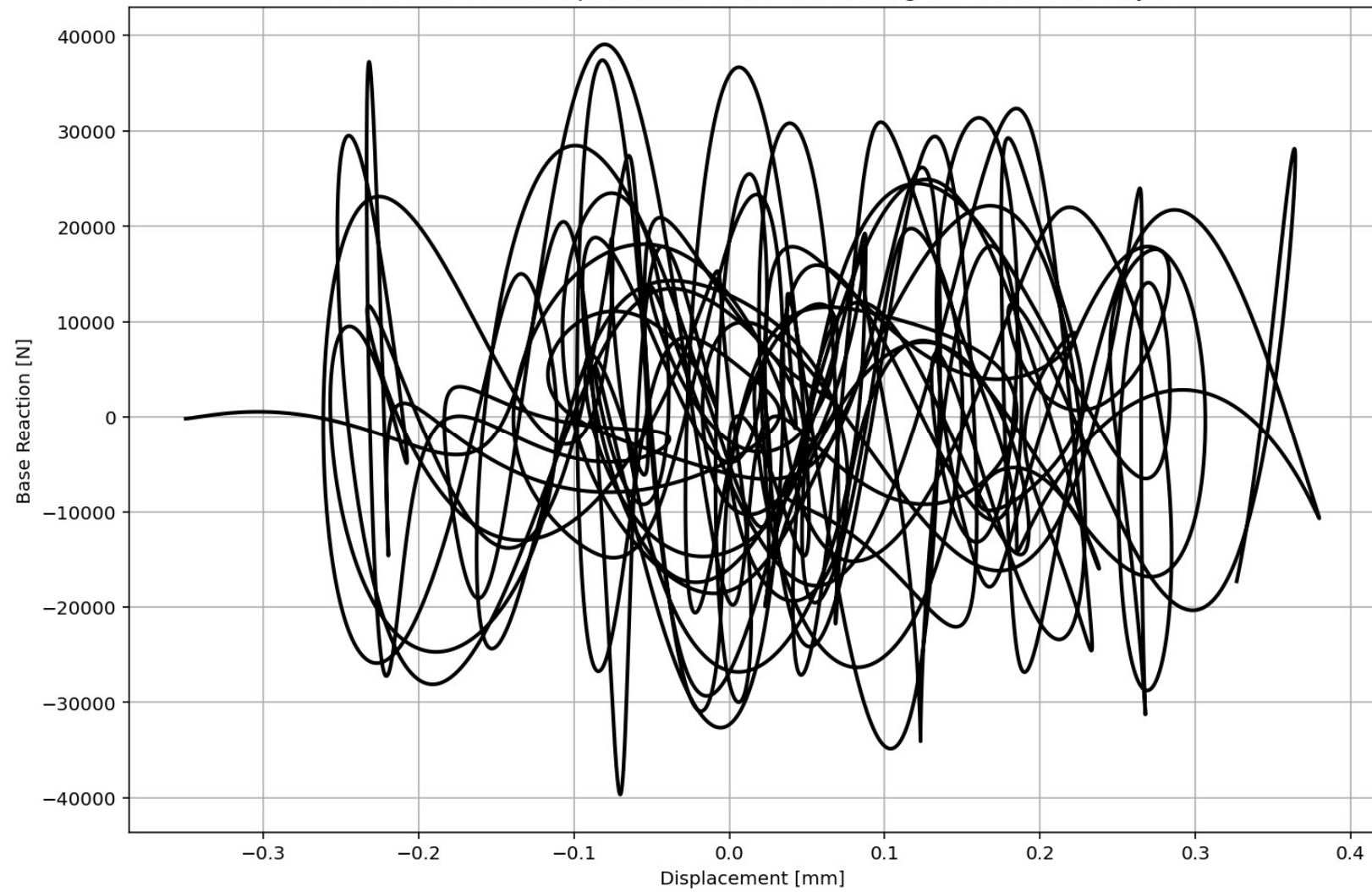


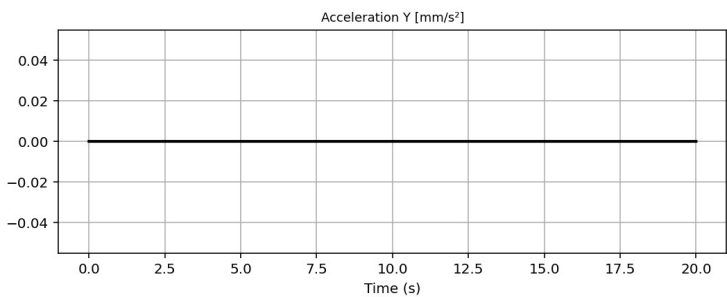
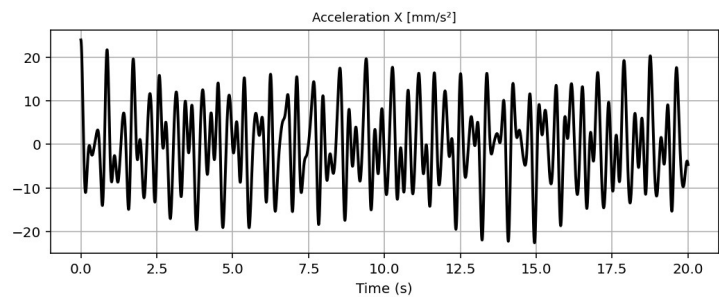
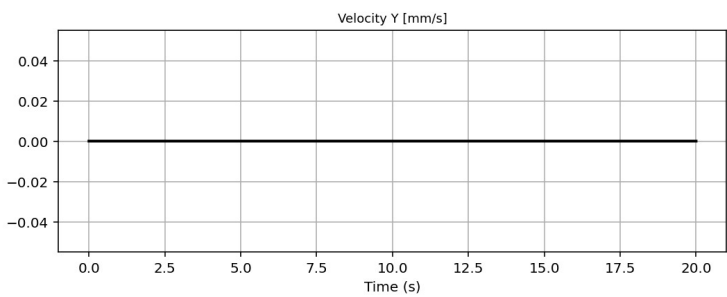
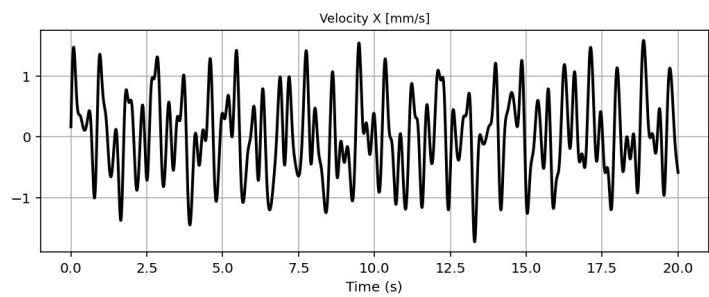
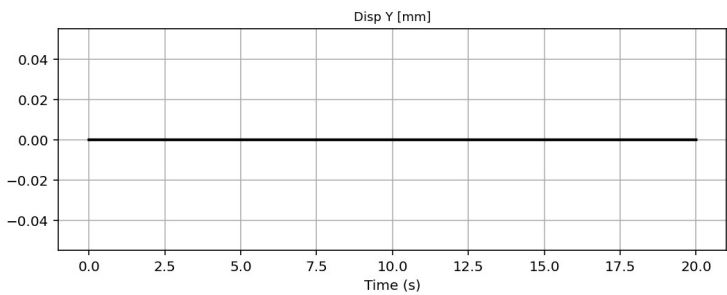
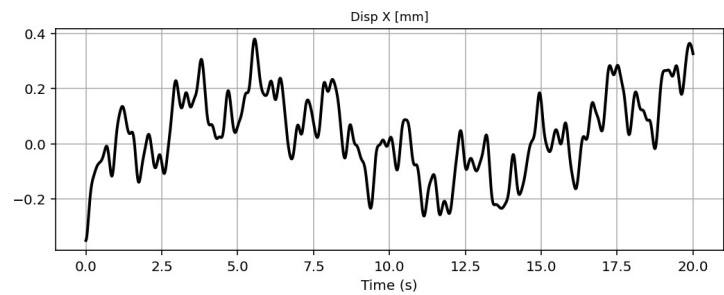
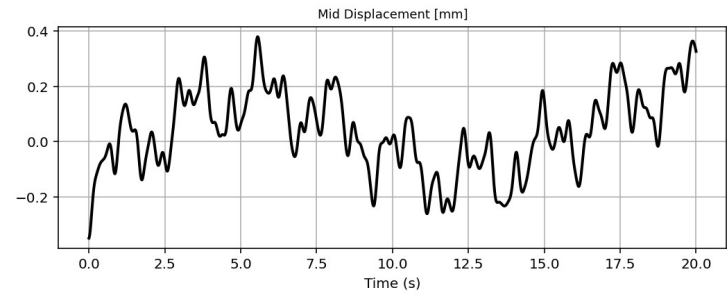
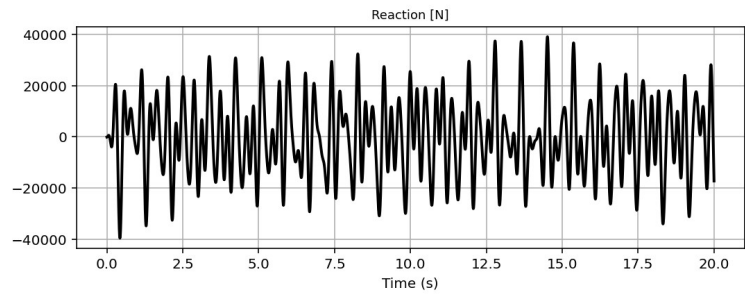


FREE-VIBRATION ANALYSIS

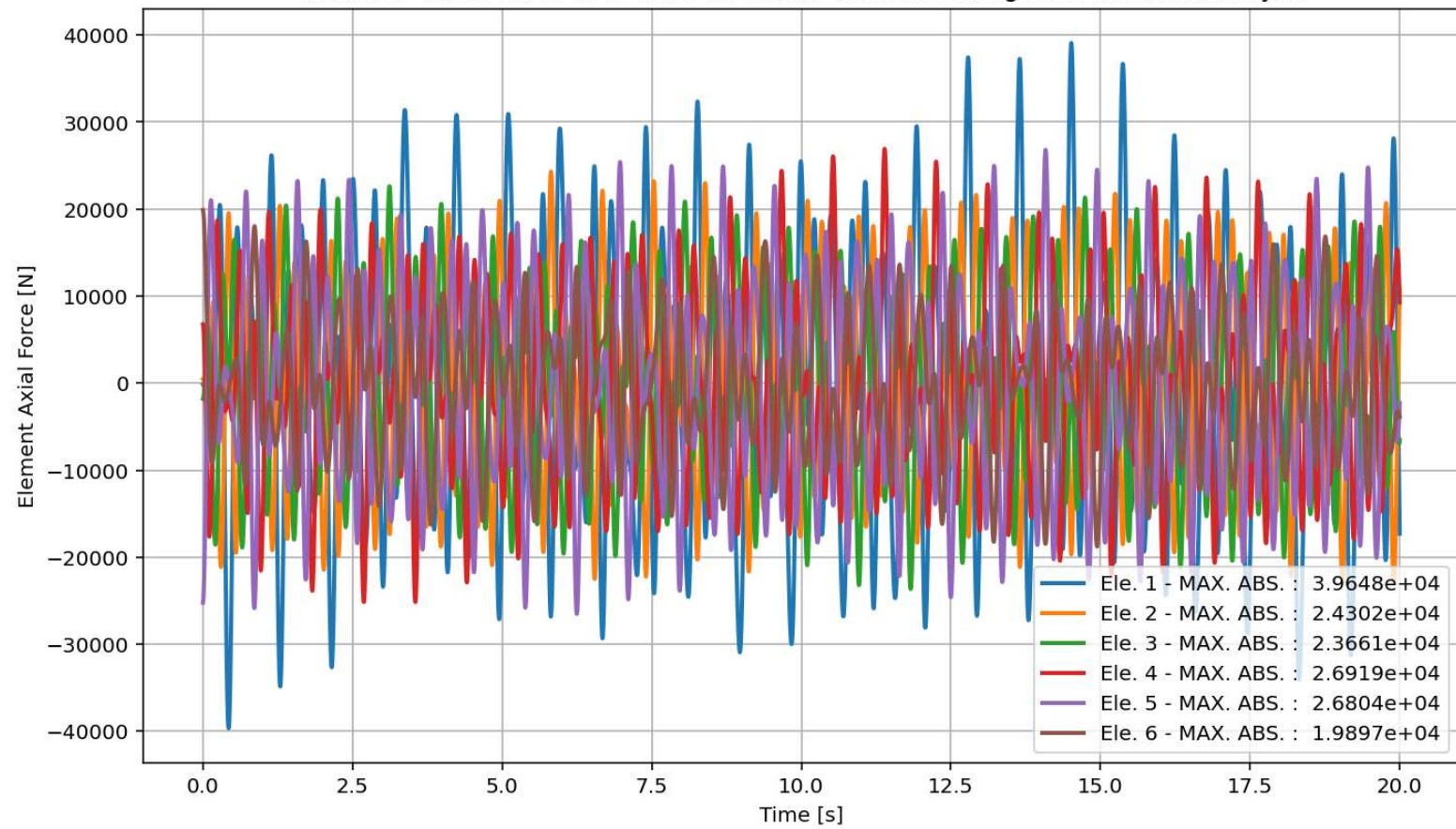


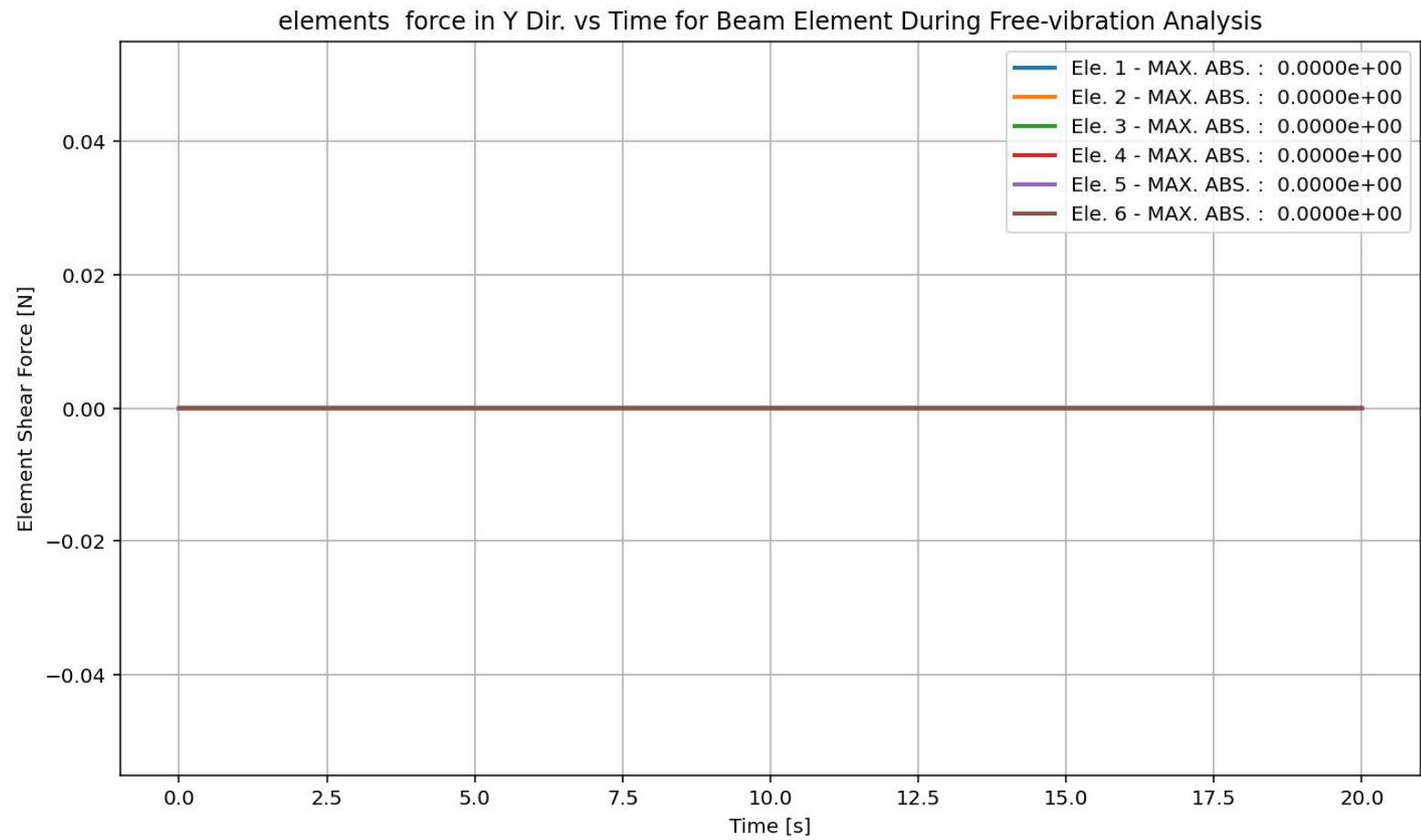
Base Reaction and Displacement of Structure During Free-vibration Analysis

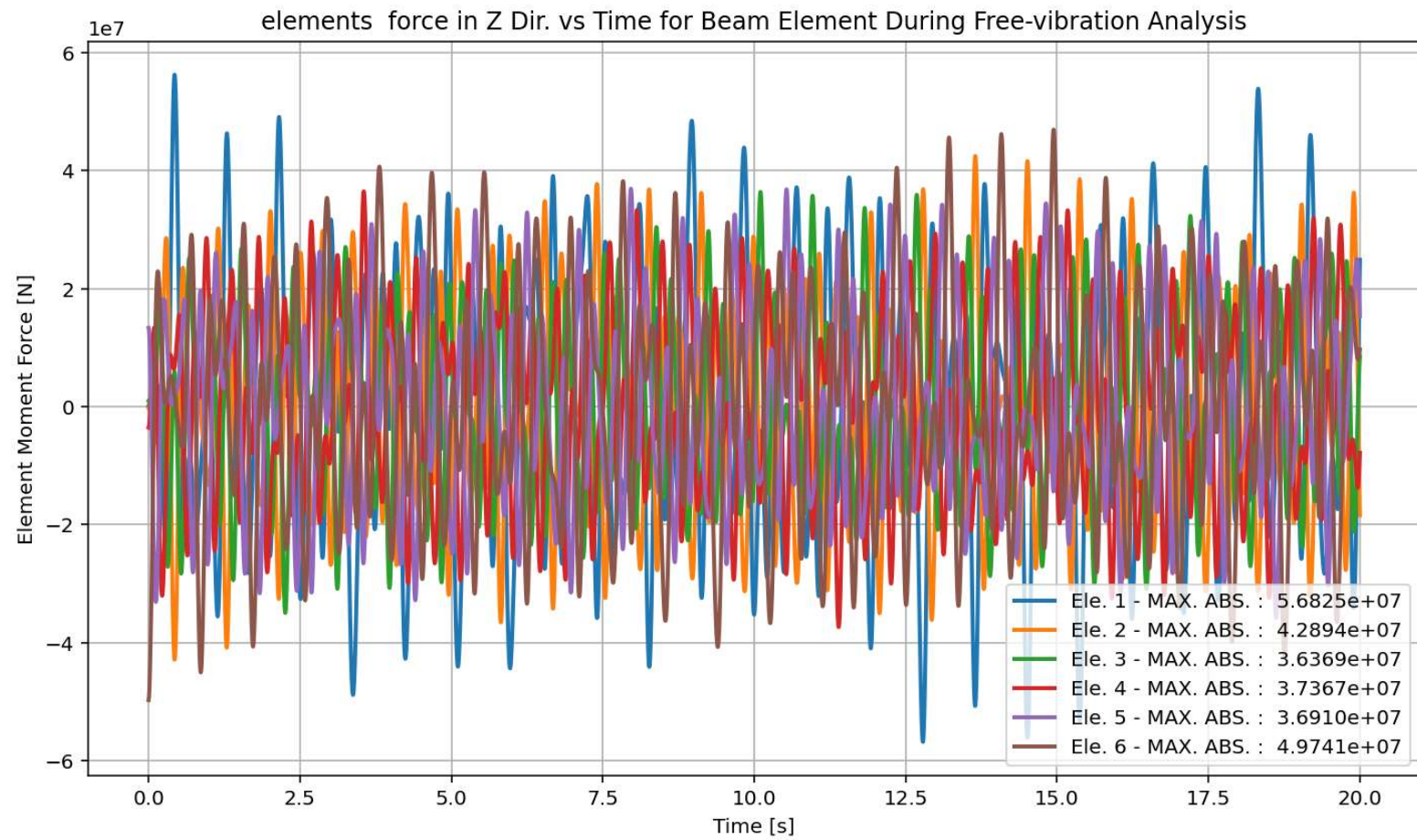




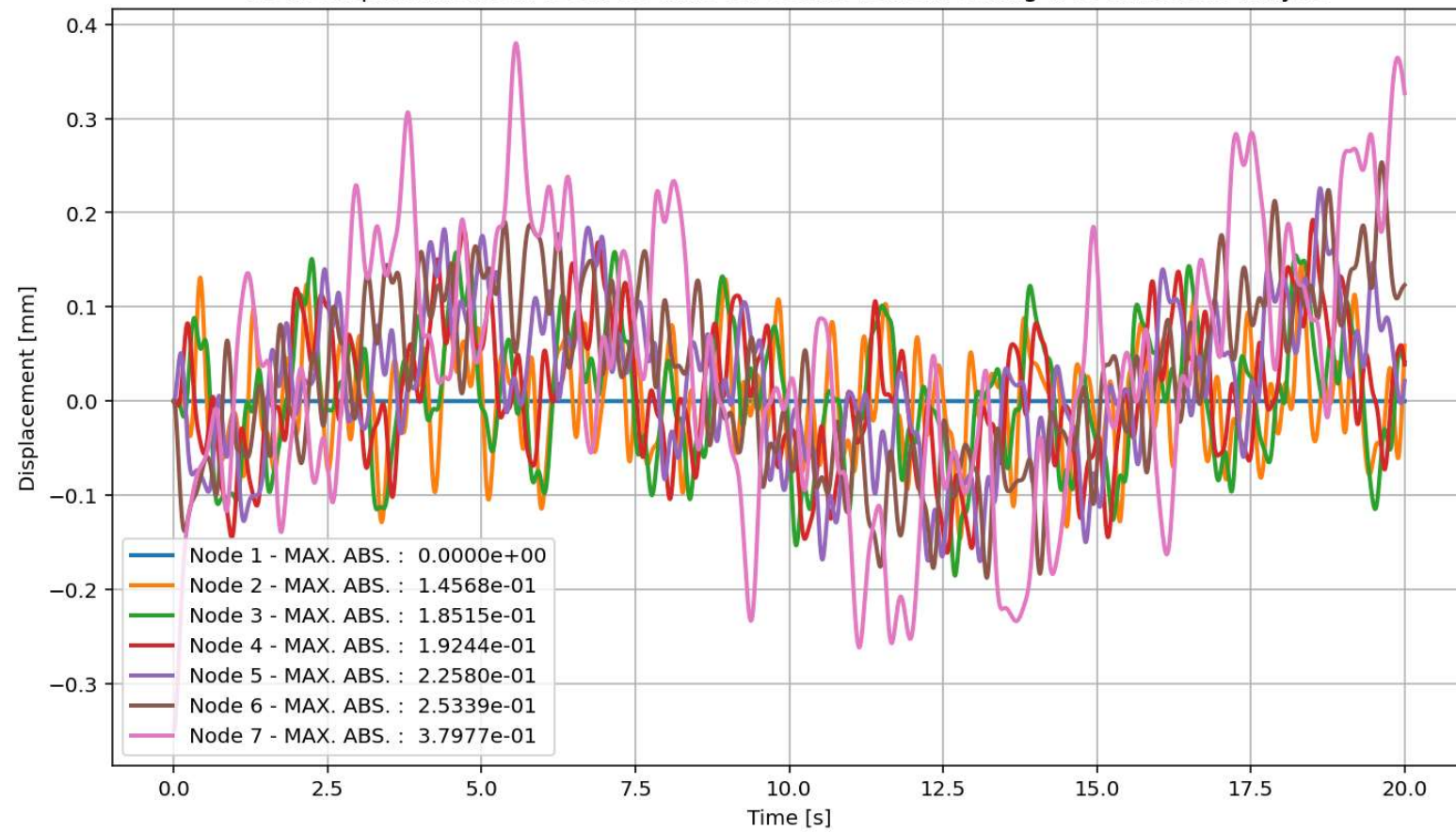
elements force in X Dir. vs Time for Beam Element During Free-vibration Analysis

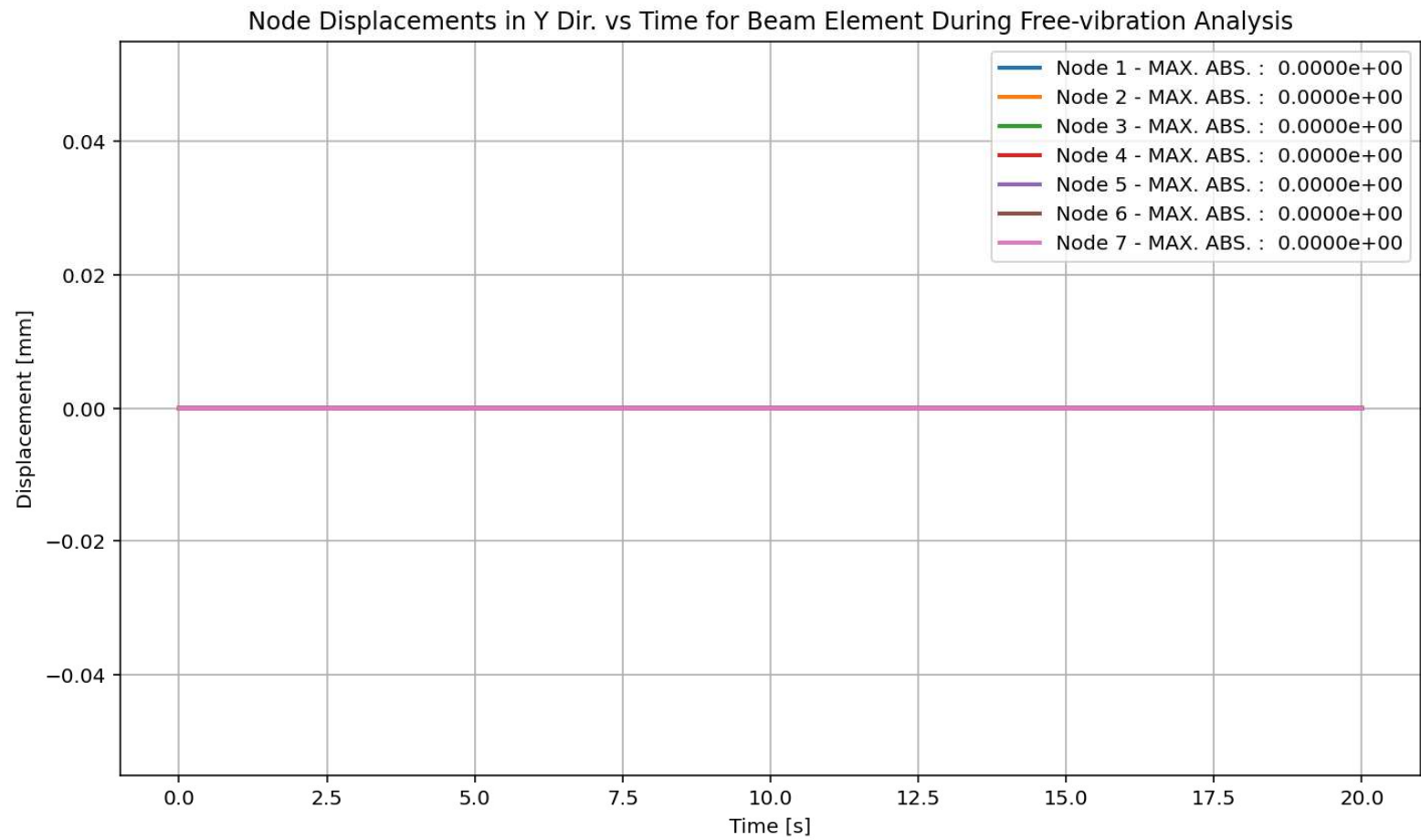


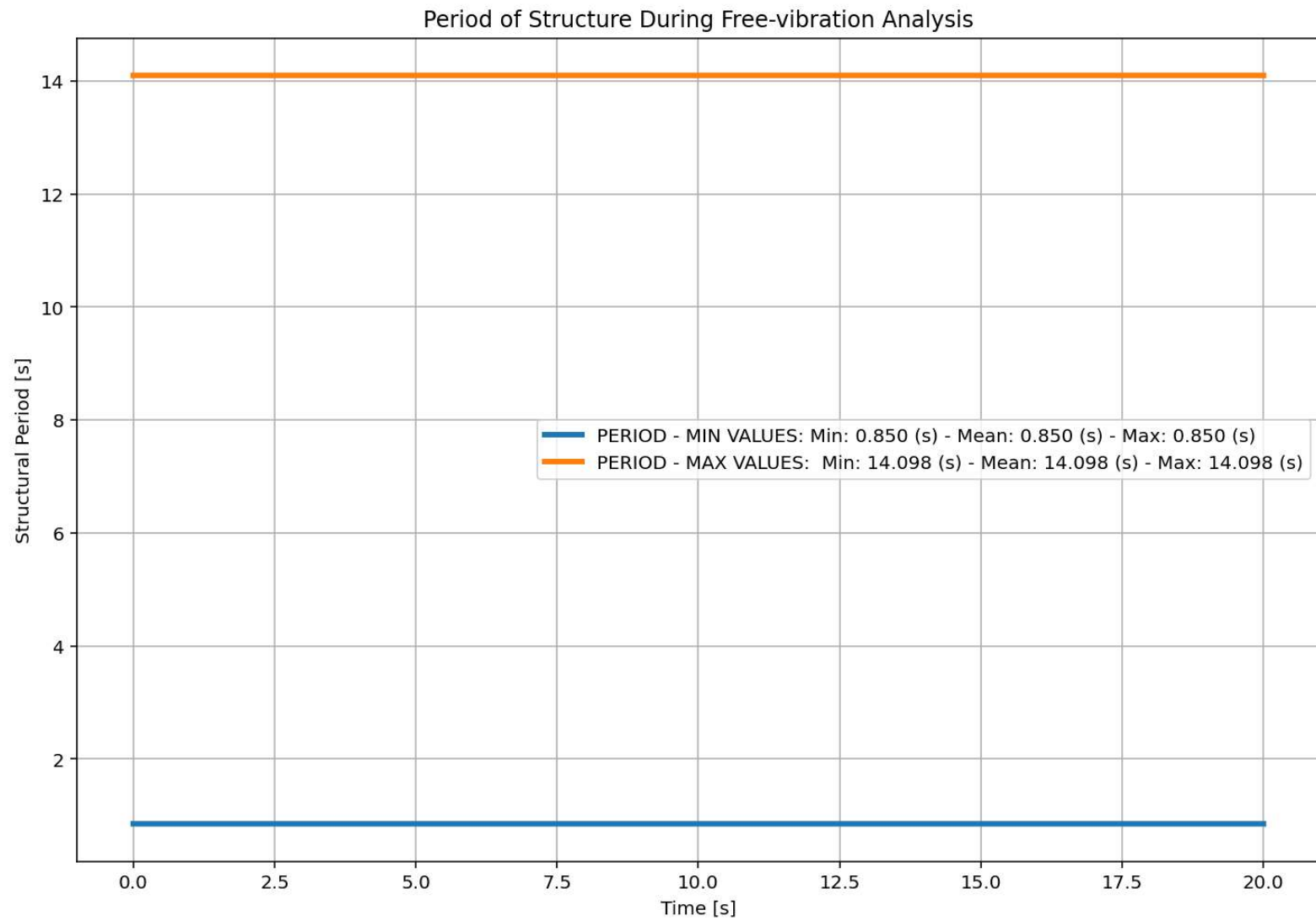


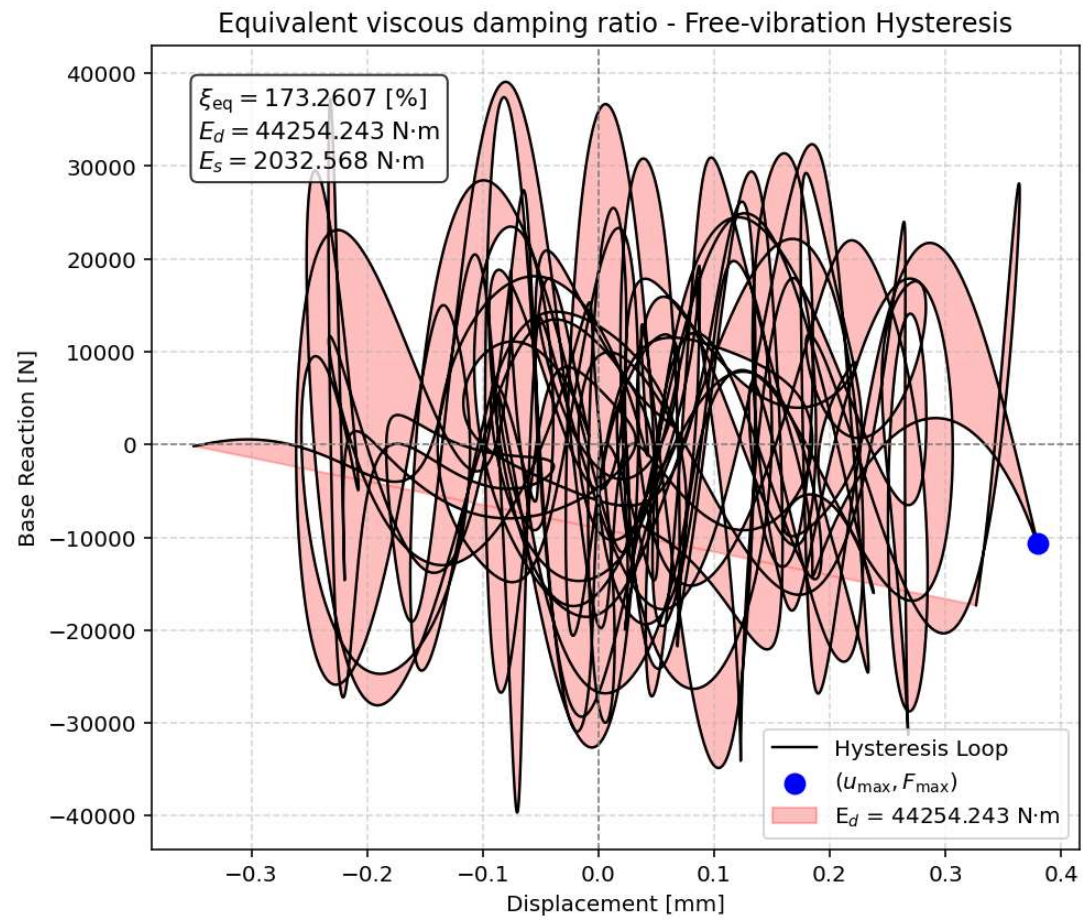


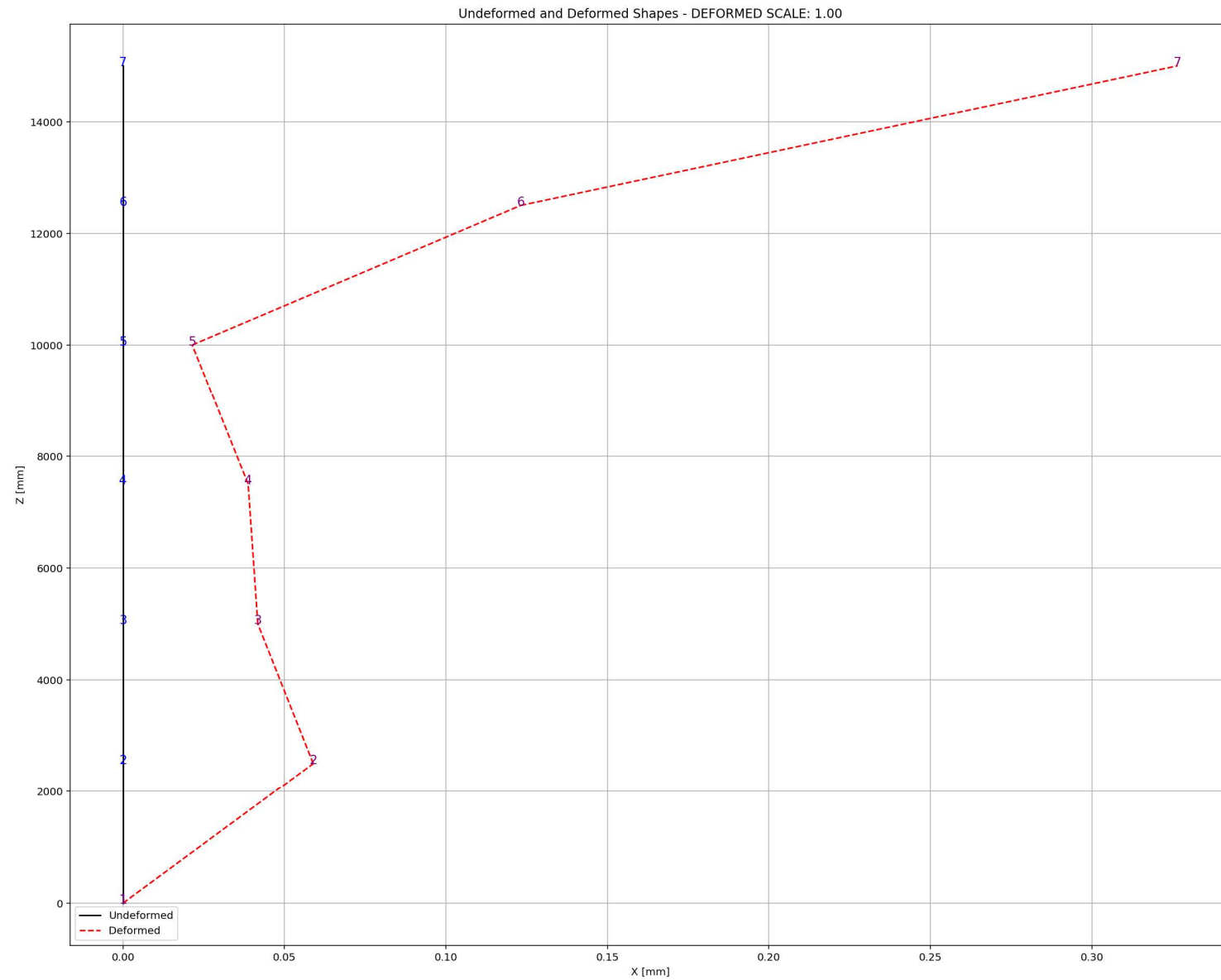
Node Displacements in X Dir. vs Time for Beam Element During Free-vibration Analysis











SEISMIC ANALYSIS

Spyder (Python 3.12)

File Edit Search Source Run Debug Consoles Projects Tools View Help

C:\Users\ DELL\Desktop\OPENSEES_FILES\+HEIGHT_ANALY...GGREGATED_AXIAL_FLEXURAL_SECTION_ELEMENT_8_ANA.py

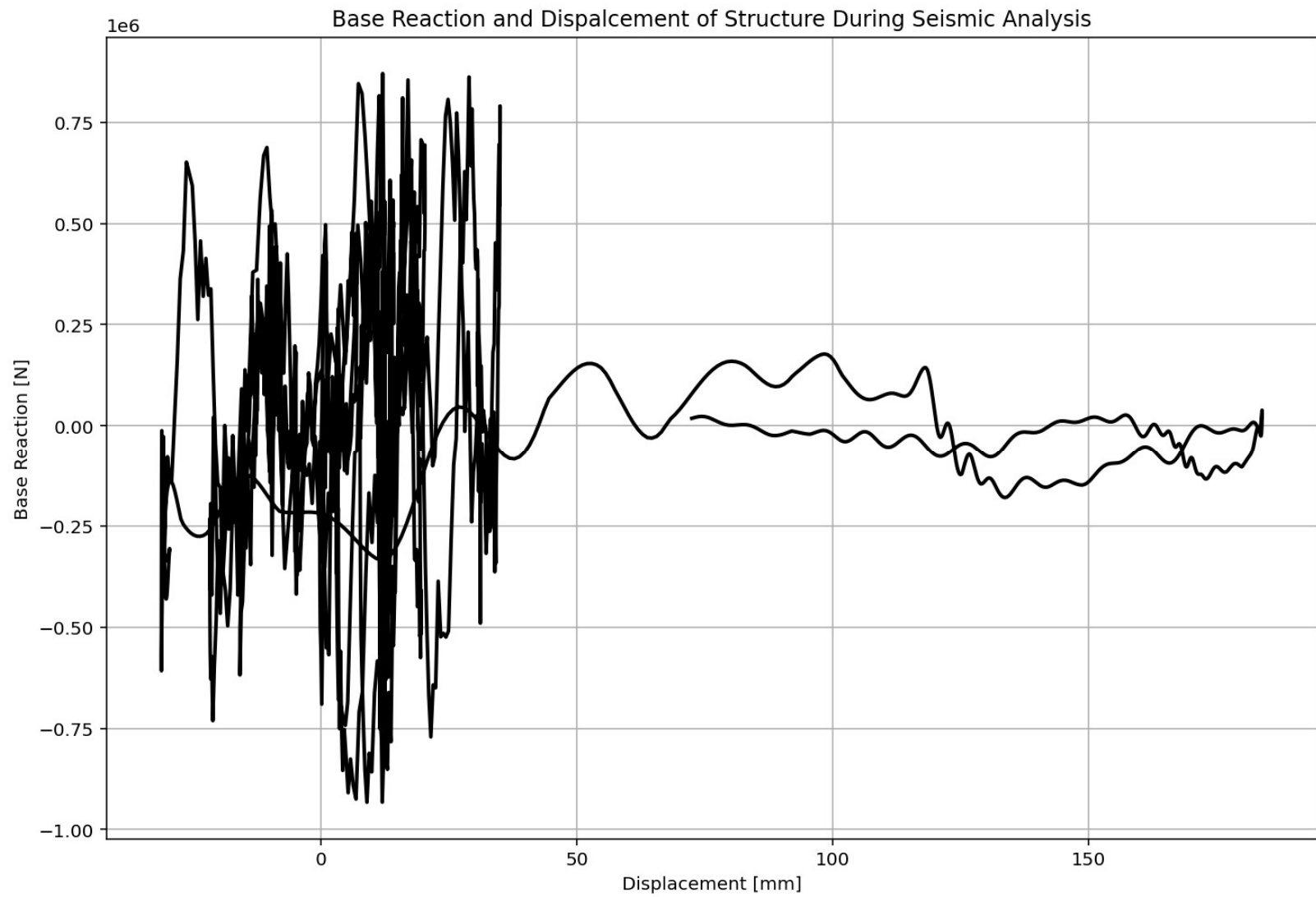
W(x,t)_COLUMN_AGGR...N_ELEMENT_8_ANA.py X AGGREGATED_AXIAL_F...ION_ELEMENT_FUN.py X

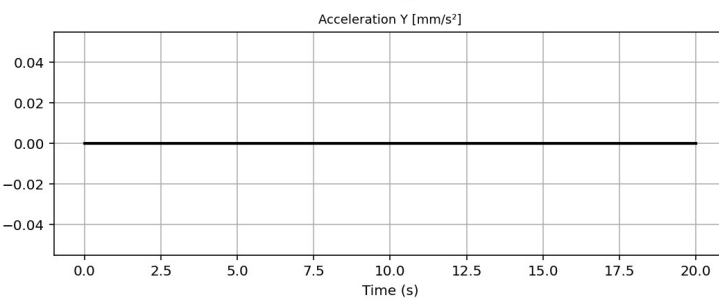
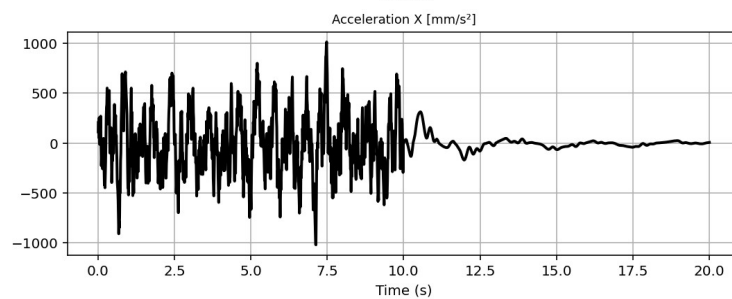
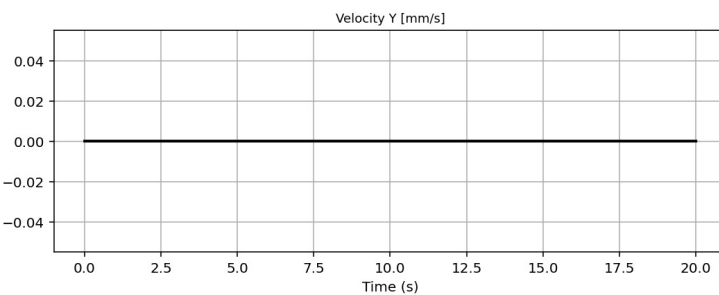
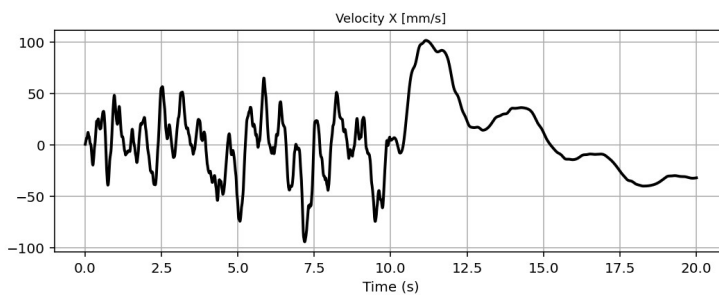
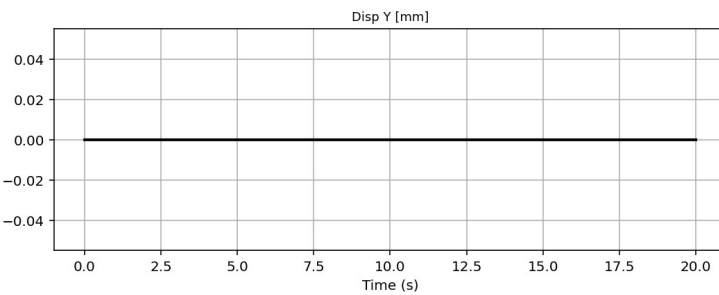
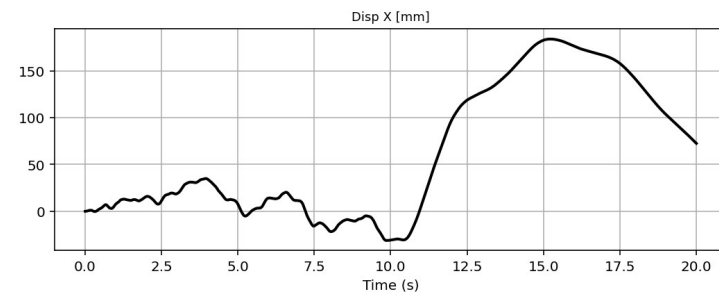
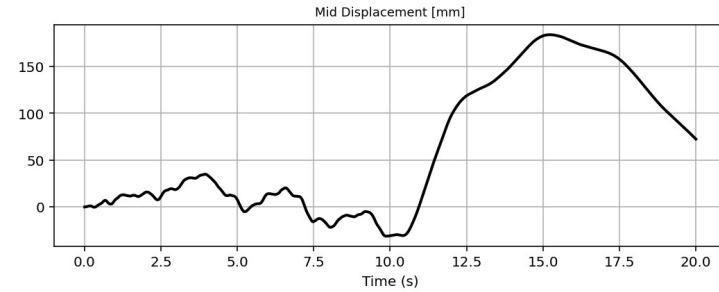
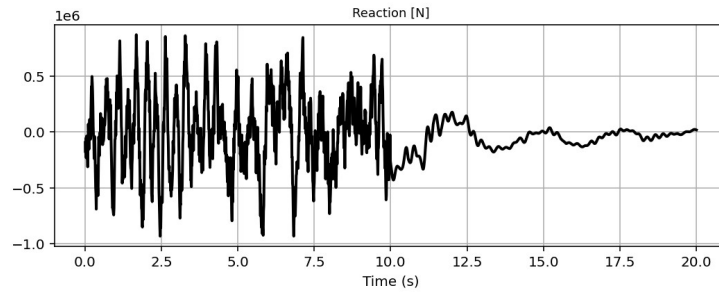
```
1237 # SEISMIC ANALYSIS (DYNAMIC TIME-HISTORY ANALYSIS)
1238 MAT_TYPE = 'INELASTIC' # 'ELASTIC' OR 'INELASTIC'
1239 ANAL_TYPE = 'SEISMIC'
1240
1241 DATA = AGGREGATED_AXIAL_FLEXURAL_COLUMN(LENGTH, MAT_TYPE, ANAL_TYPE, TOTAL_MASS)
1242
1243 (time_SEI, reactionX_SEI, reactionY_SEI, reactionZ_SEI, disp_mid_SEI,
1244  ele_axialforce_SEI, ele_shearforce_SEI, ele_momentforce_SEI,
1245  node_displacementsX_SEI, node_displacementsY_SEI,
1246  dispX_SEI, dispY_SEI,
1247  veloX_SEI, veloY_SEI,
1248  accX_SEI, accY_SEI,
1249  stiffness_SEI, PERIOD_SEI, damping_ratio_SEI,
1250  PERIOD_MIN_SEI, PERIOD_MAX_SEI) = DATA
1251
1252
1253 XDATA = disp_mid_SEI
1254 YDATA = reactionX_SEI
1255 XLABEL = 'Displacement [mm]'
1256 YLABEL = 'Base Reaction [N]'
1257 TITLE = 'Base Reaction and Displcement of Structure During Seismic Analysis'
1258 COLOR = 'black'
1259 SEMIOLOGY = False
1260 PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMIOLOGY)
1261
1262 PLOT_TIME_HISTORY(time_SEI, reactionX_SEI, disp_mid_SEI,
1263                   dispX_SEI, dispY_SEI,
1264                   veloX_SEI, veloY_SEI,
1265                   accX_SEI, accY_SEI)
1266
1267 # PLOT ELEMENTS AXIAL FORCE
1268 YLABEL = 'Element Axial Force [N]'
1269 TITLE = "elements force in X Dir. vs Time for Beam Element During Seismic Analysis"
1270 PLOT FORCES(time SEI, ele_axialforce SEI, YLABEL, TITLE) # Beam Element - ELEMENTS AXIAL FORCE
```

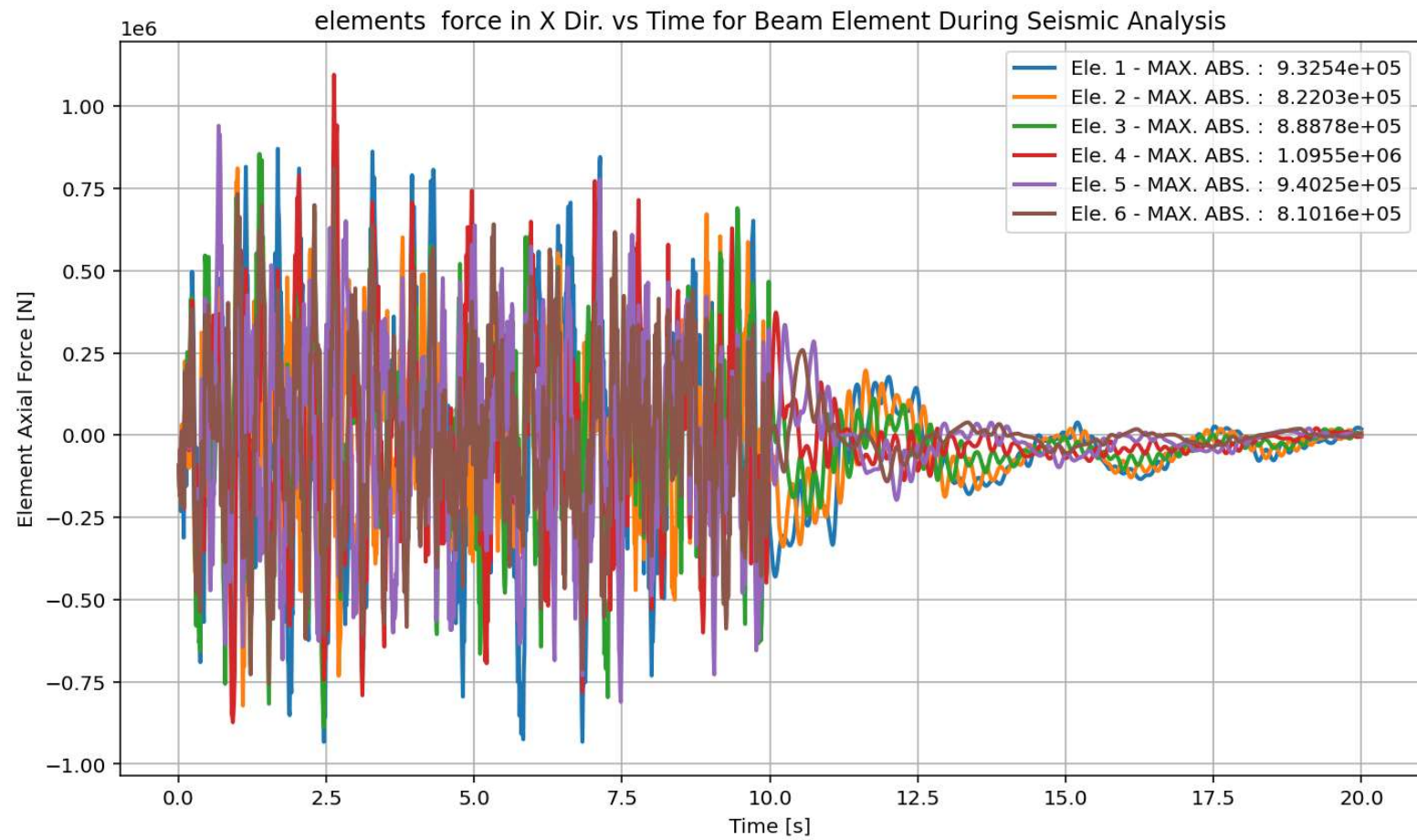
24 %

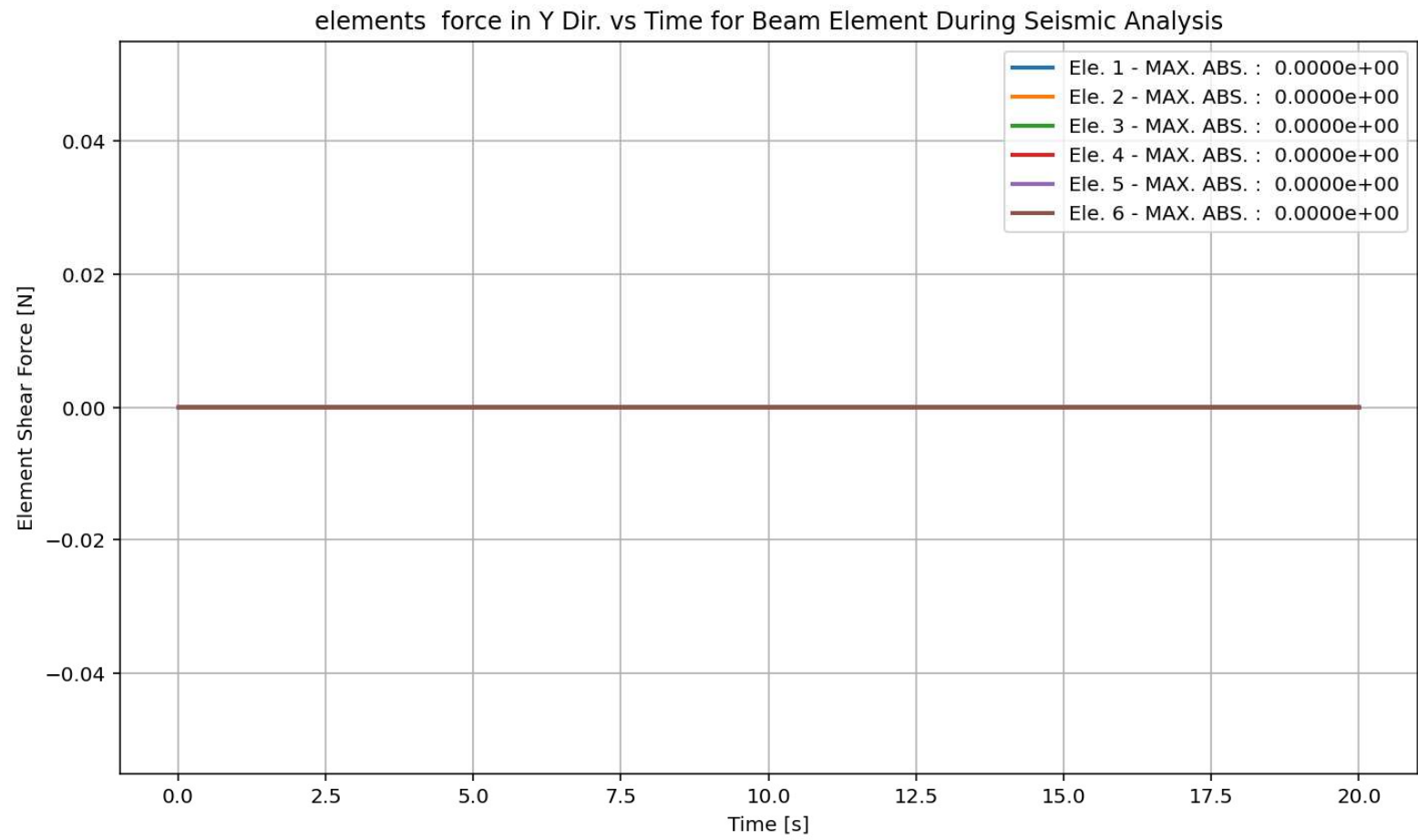
Base Reaction and Displcement of Structure During Seismic Analysis

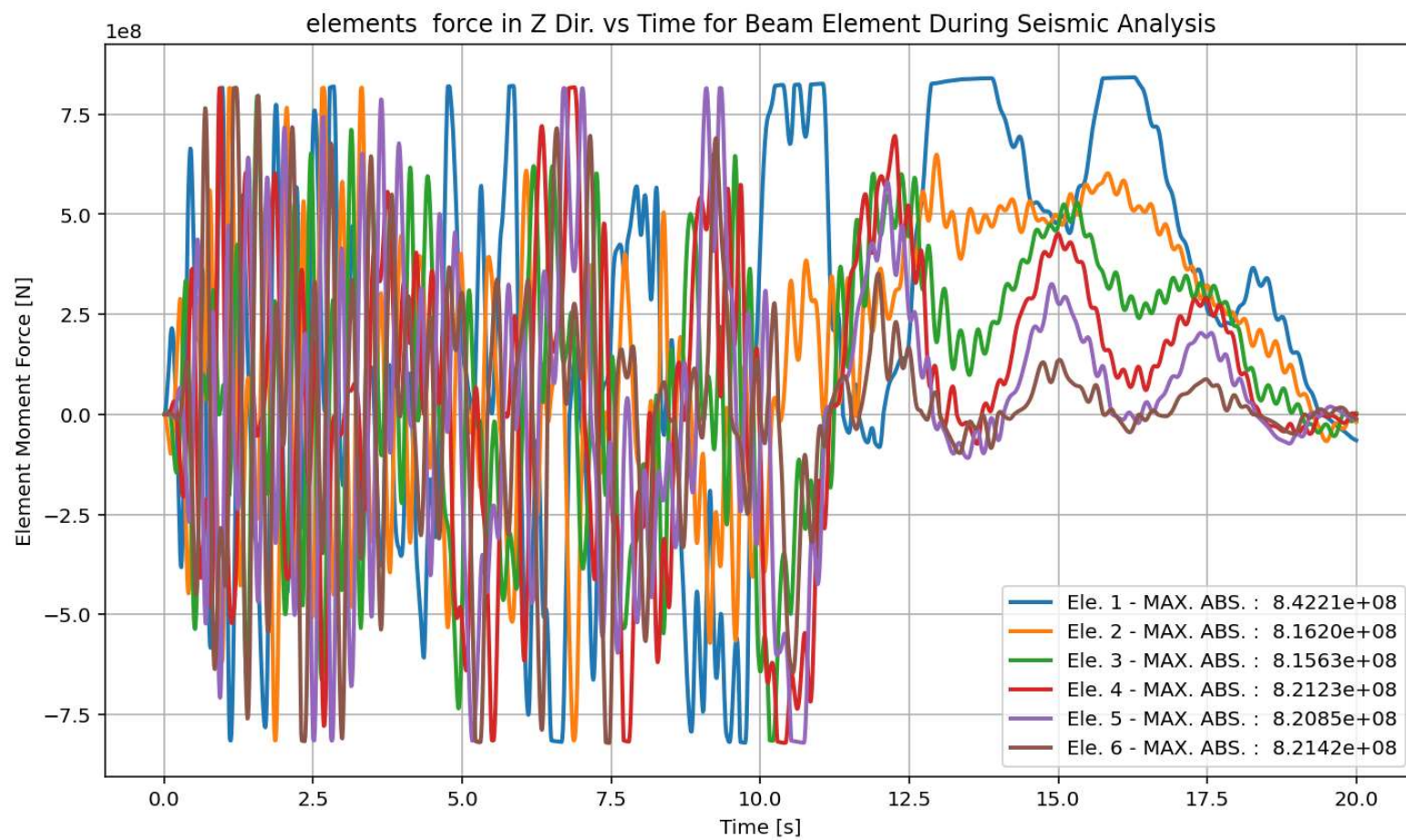
IPython Console Plots Variable Explorer Help Debugger History Files



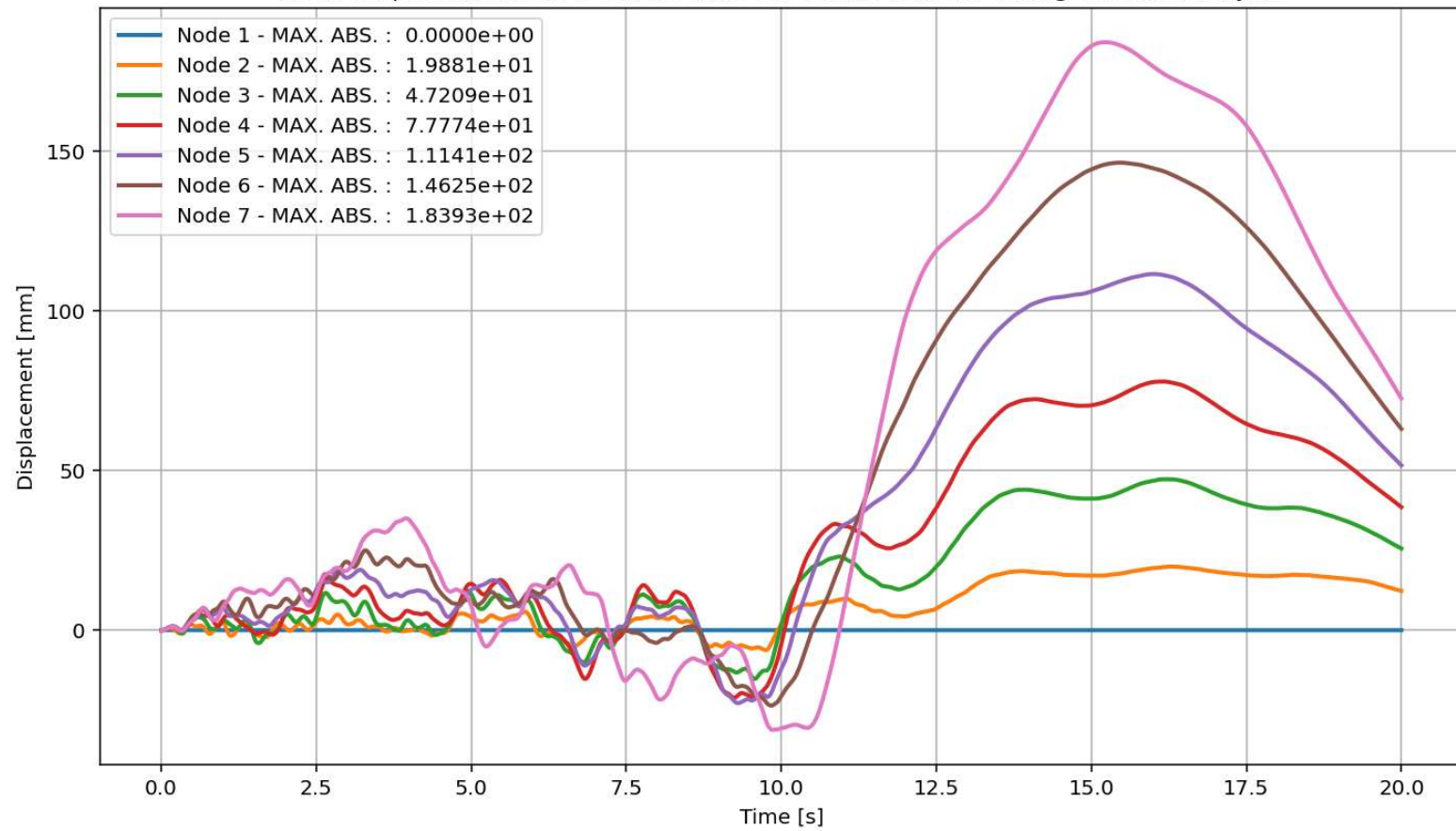




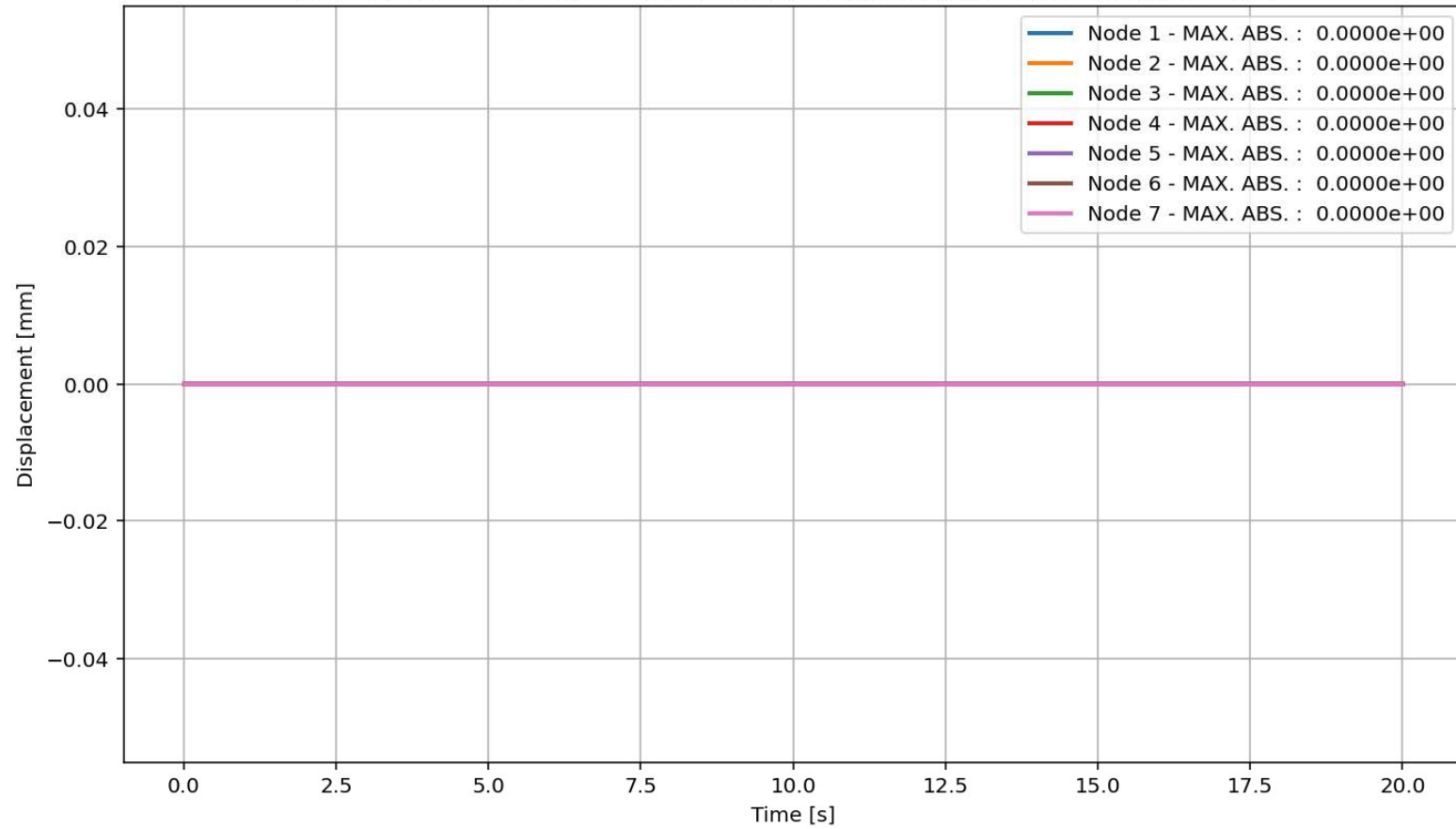




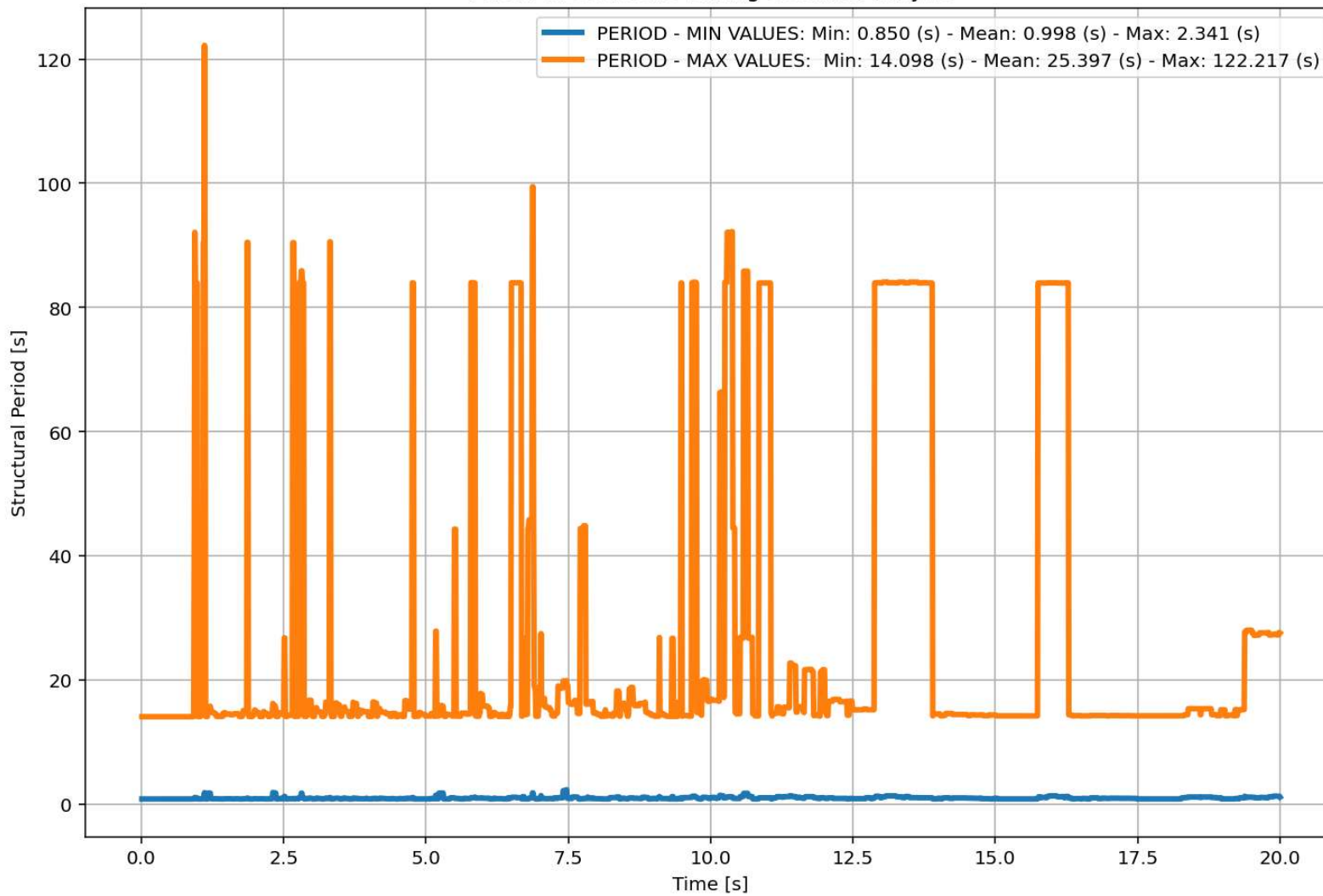
Node Displacements in X Dir. vs Time for Beam Element During Seismic Analysis

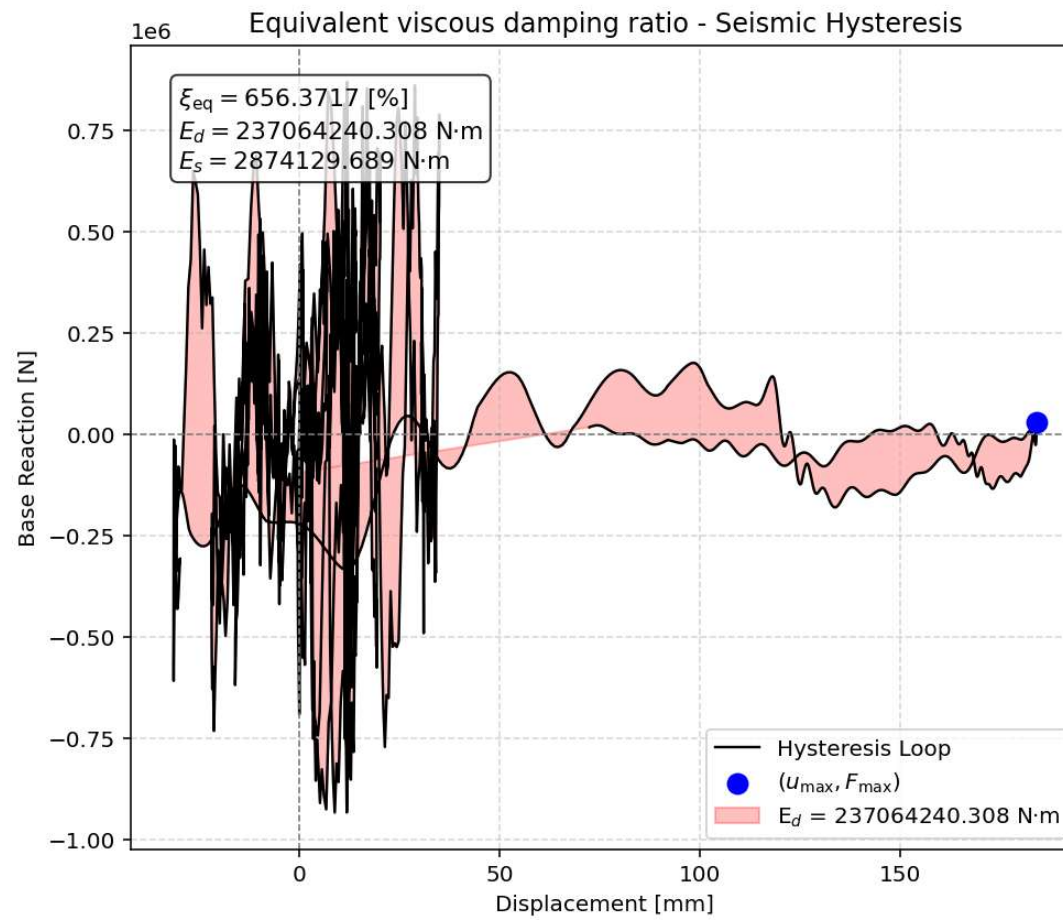


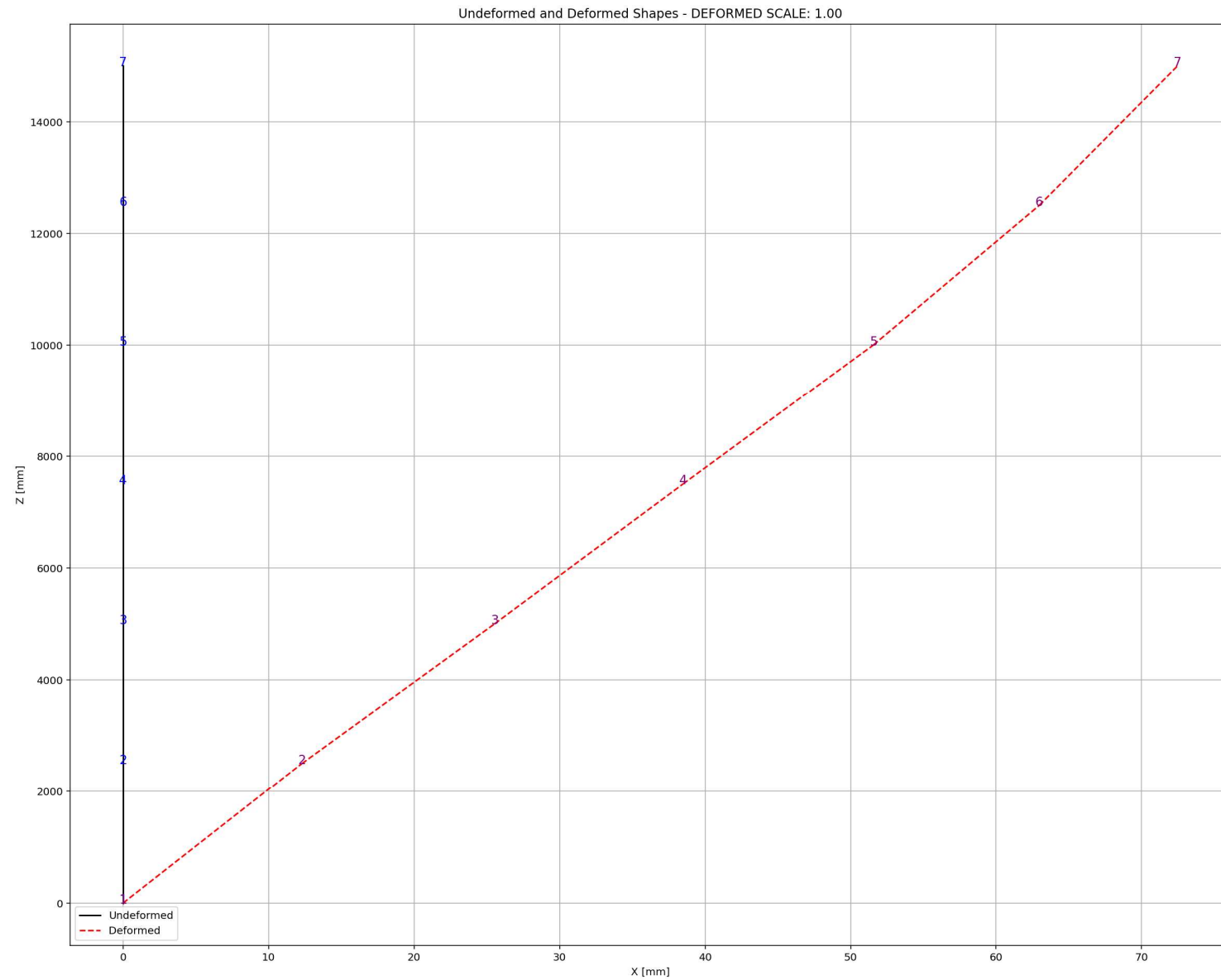
Node Displacements in Y Dir. vs Time for Beam Element During Seismic Analysis

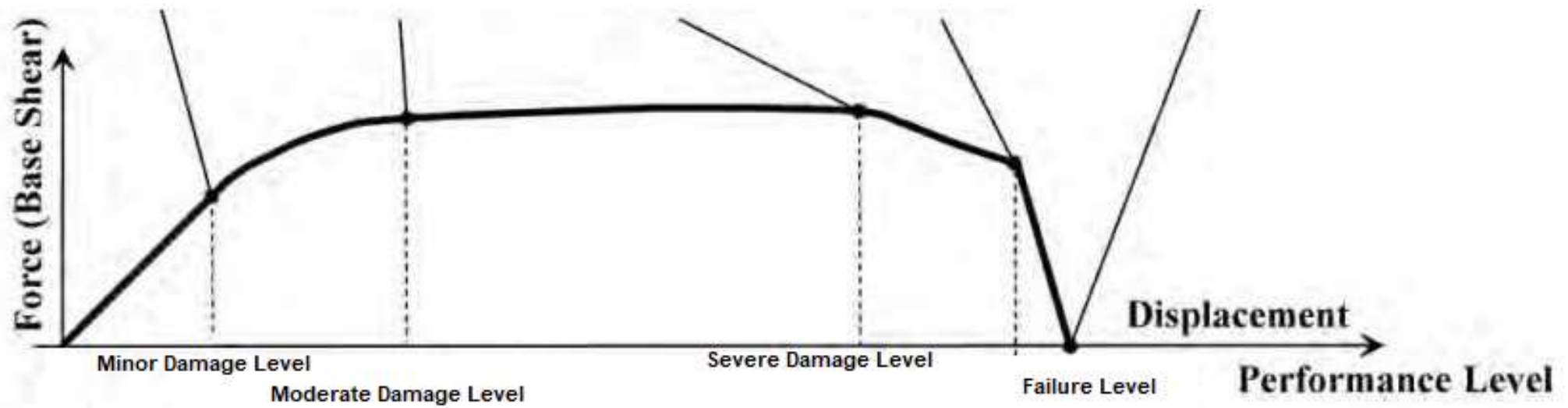
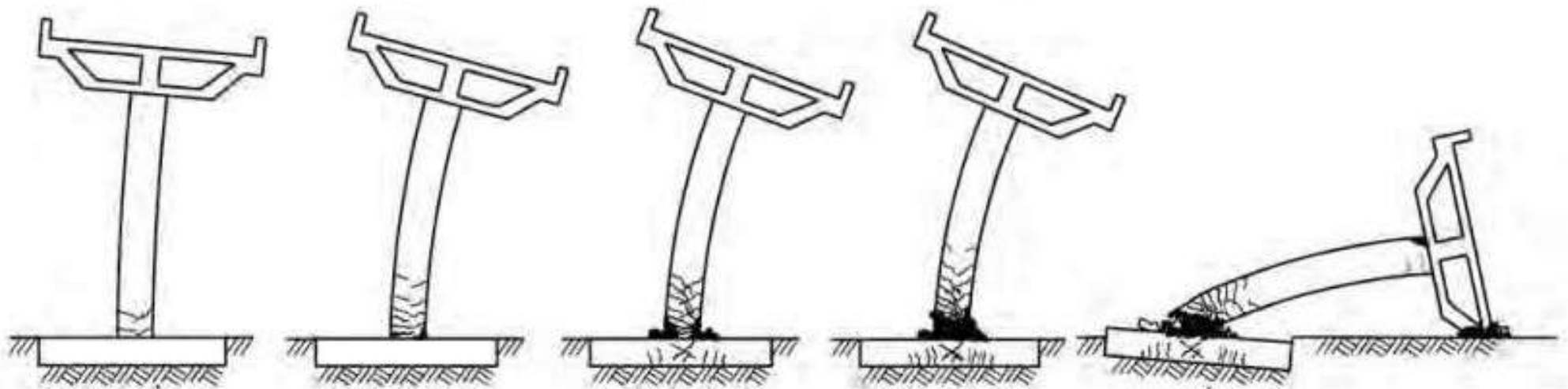


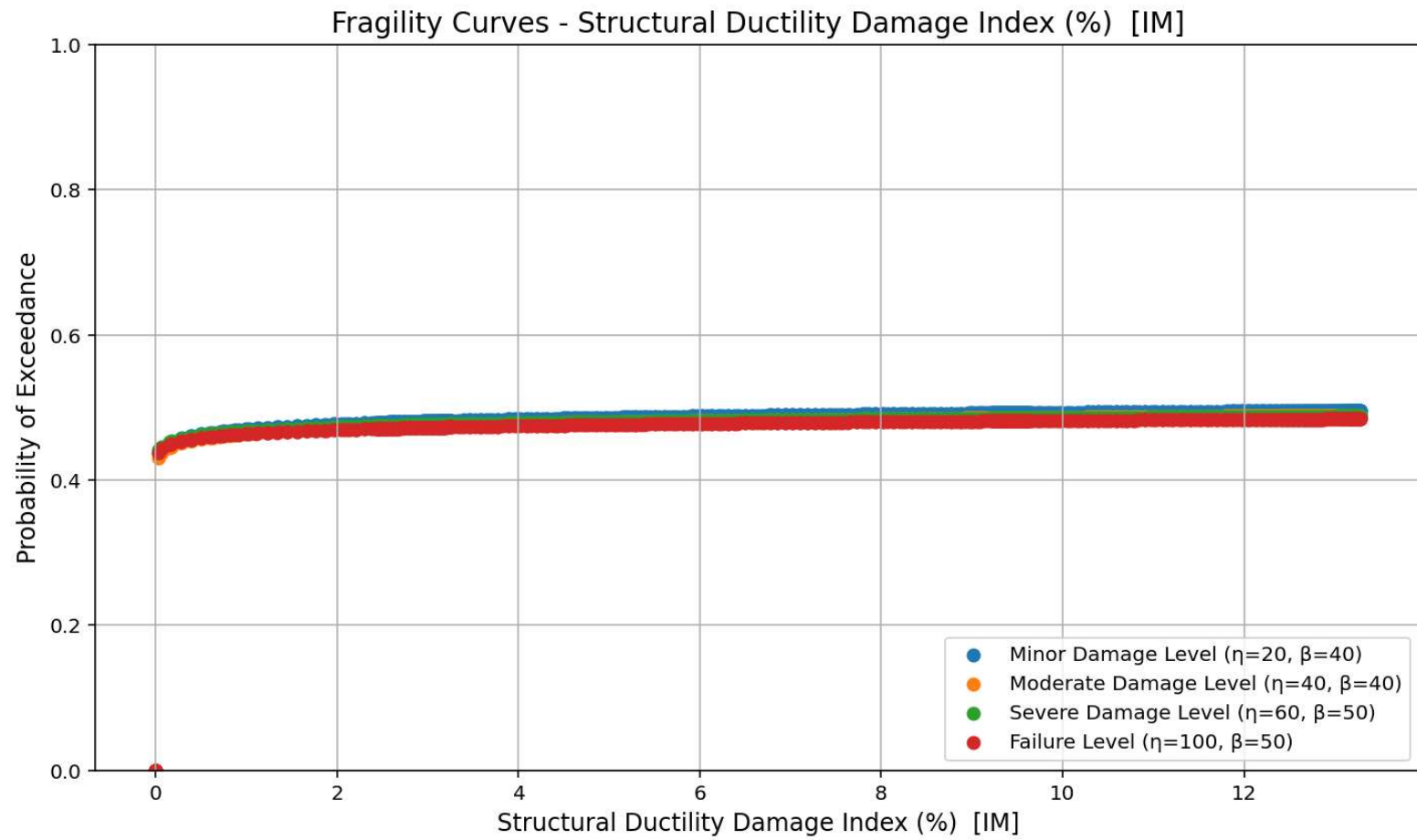
Period of Structure During Seismic Analysis

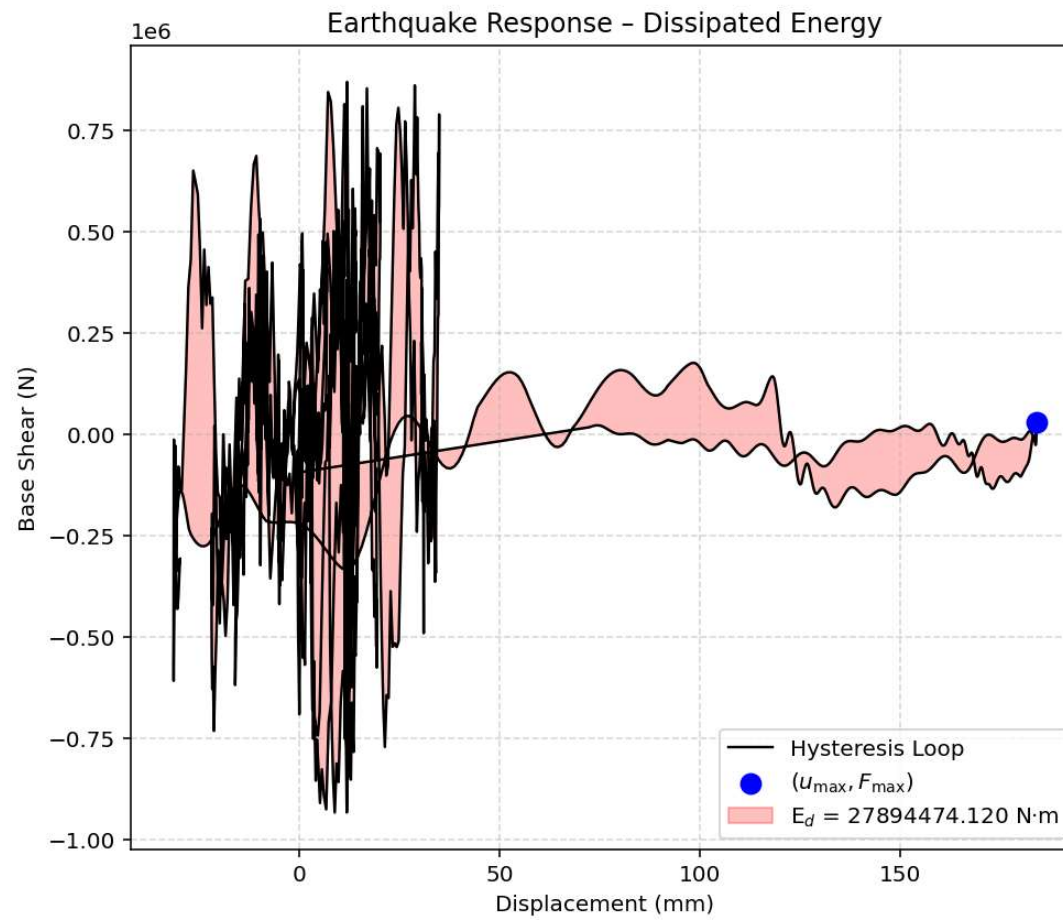


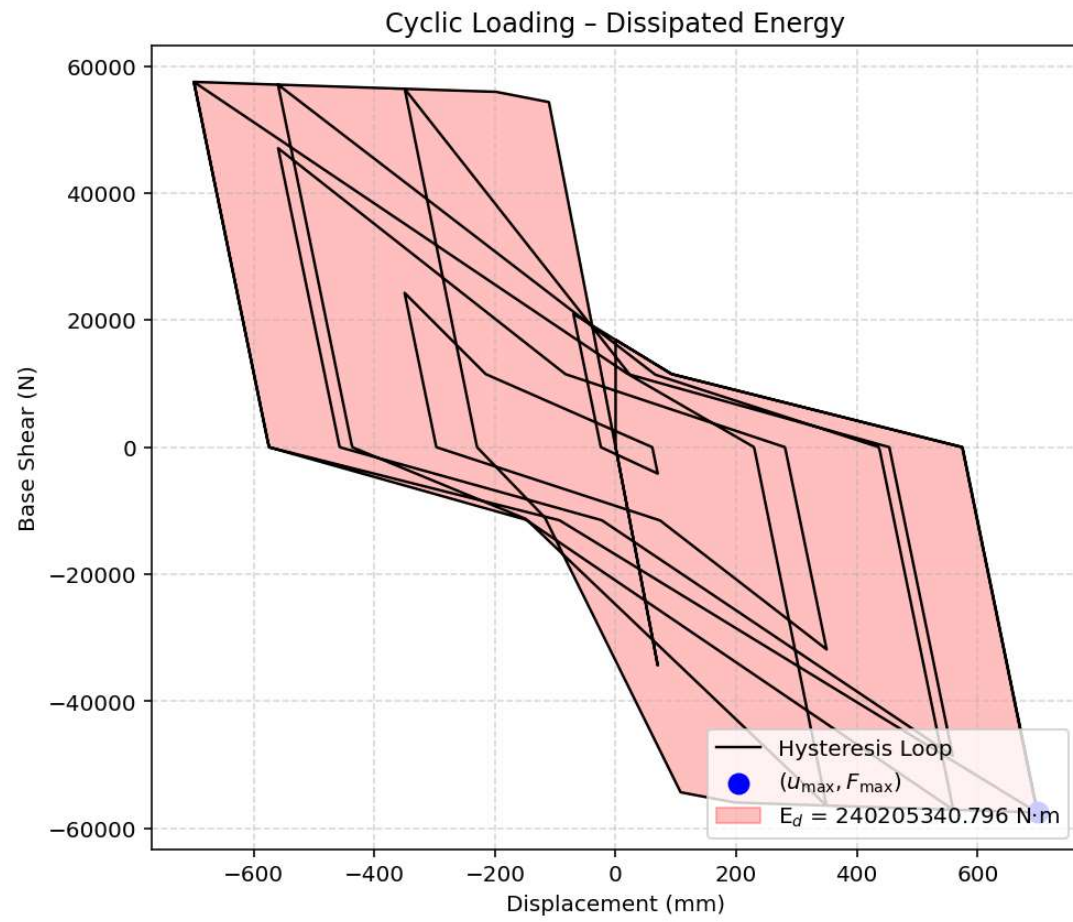










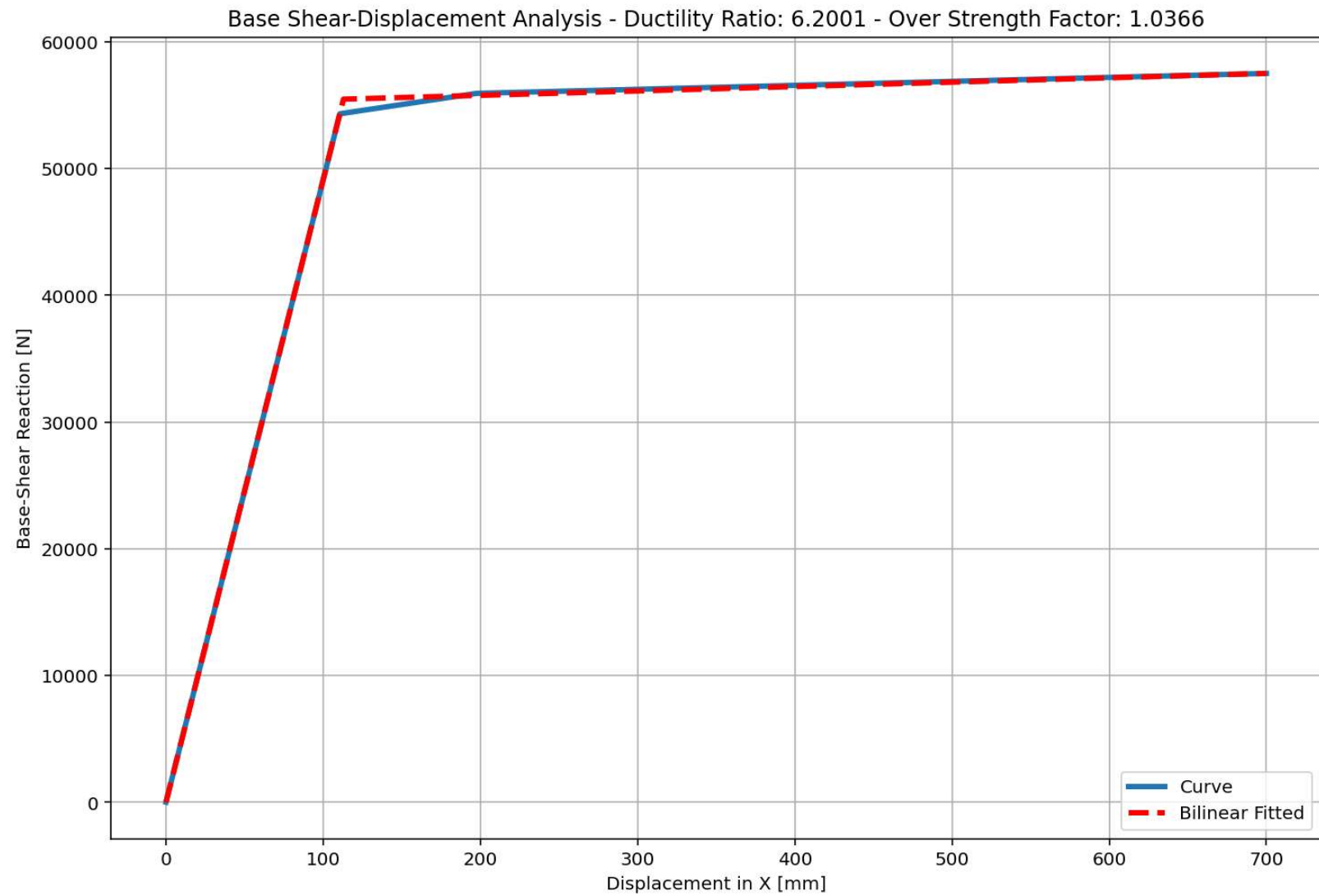


Dissipated Energy from Earthquake= 27894474.12 N·mm

Dissipated Energy from Cyclic Displacement= 240205340.80 N·mm

DISSIPATED ENERGY CAPACITY INDEX: 11.613 [%]

ZONE 2: MINOR DAMAGE



+=====+

= Analysis curve fitted =

Disp Base Shear

[[0.00000000e+00 0.00000000e+00]

[8.19232859e+01 1.88095150e+05]

[7.50000000e+02 2.23204874e+05]]

+=====+

+-----+

Structure Elastic Stiffness : 2295.99

Structure Plastic Stiffness : 297.61

Structure Tangent Stiffness : 52.55

Structure Ductility Ratio : 9.15

Structure Over Strength Factor: 1.19

+-----+

Over Strength Coefficient (Ω_0): 1.0366

Displacement Ductility Ratio (μ): 6.2001

Ductility Coefficient (R_μ): 6.2001

Structural Behavior Coefficient (R): 6.4271

Structural Ductility Damage Index in X Direction: 12.0977 (%)

